

Matrices A and B are **unitarily similar** if $B = U^*AU$ for some unitary matrix U . The term **unitarily equivalent** is sometimes used in the literature.

The **numerical range** of A is $W(A) = \{\mathbf{v}^*A\mathbf{v} \mid \mathbf{v}^*\mathbf{v} = 1\}$.

The **Frobenius (Euclidean) norm** of the matrix A is $\|A\|_F = (\sum_{i,j=1}^n |a_{ij}|^2)^{1/2} = (\text{tr}(A^*A))^{1/2}$. (See Chapter 37 for more information on norms.)

The **operator norm** of the matrix A induced by the vector 2-norm $\|\cdot\|_2$ is $\|A\|_2 = \max\{\|A\mathbf{v}\| \mid \|\mathbf{v}\| = 1\}$; this norm is also called the **spectral norm**.

Facts:

Most of the material in this section can be found in one or more of the following: [HJ85, Chap. 2] [Hal87, Chap. 3] [Gan59, Chap. IX] [MM64, I.4, III.5]. Specific references are also given for some facts.

1. A real, orthogonal matrix is unitary.
2. The following are equivalent:
 - U is unitary.
 - U is invertible and $U^{-1} = U^*$.
 - The columns of U are orthonormal.
 - The rows of U are orthonormal.
 - For any vectors $\mathbf{x}$ and $\mathbf{y}$, we have $\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$.
 - For any vector $\mathbf{x}$, we have $\|U\mathbf{x}\| = \|\mathbf{x}\|$.
3. If U is unitary, then U^* , U^T , and $\bar{U}$ are also unitary.
4. If U is unitary, then every eigenvalue of U has modulus 1 and $|\det(U)| = 1$. Also, $\|U\|_2 = 1$.
5. The product of two unitary matrices is unitary and the product of two orthogonal matrices is orthogonal.
6. The set of $n \times n$ unitary matrices, denoted $U(n)$, is a subgroup of $GL(n, \mathbb{C})$, called the unitary group. The subgroup of elements of $U(n)$ with determinant one is the special unitary group, denoted $SU(n)$. Similarly, the set of $n \times n$ real orthogonal matrices, denoted $O(n)$, is a subgroup of $GL(n, \mathbb{R})$, called the real, orthogonal group, and the subgroup of real, orthogonal matrices of determinant one is $SO(n)$, the special orthogonal group.
7. Let U be unitary. Then
 - $\|A\|_F = \|U^*AU\|_F$.
 - $\|A\|_2 = \|U^*AU\|_2$.
 - A and U^*AU have the same singular values, as well as the same eigenvalues.
 - $W(A) = W(U^*AU)$.
8. [Sch09] Any square, complex matrix A is unitarily similar to a triangular matrix. If $T = U^*AU$ is triangular, then the diagonal entries of T are the eigenvalues of A . The unitary matrix U can be chosen to get the eigenvalues in any desired order along the diagonal of T . Algorithm 1 below gives a method for finding U , assuming that one knows how to find an eigenvalue and eigenvector, e.g., by exact methods for small matrices (Section 4.3), and how to find an orthonormal basis containing the given vector, e.g., by the Gram-Schmidt process (Section 5.5). This algorithm is designed to illuminate the result, not for computation with large matrices in finite precision arithmetic; for such problems appropriate numerical methods should be used (cf. Section 43.2).

Algorithm 1: Unitary TriangularizationInput: $A \in \mathbb{C}^{n \times n}$.Output: unitary U such that $U^*AU = T$ is triangular.

1. $A_1 = A$.
2. FOR $k = 1, \dots, n - 1$
 - (a) Find an eigenvalue and normalized eigenvector $\mathbf{x}$ of the $(n + 1 - k) \times (n + 1 - k)$ matrix A_k .
 - (b) Find an orthonormal basis $\mathbf{x}, \mathbf{y}_2, \dots, \mathbf{y}_{n+1-k}$ for $\mathbb{C}^{n+1-k}$.
 - (c) $U_k = [\mathbf{x}, \mathbf{y}_2, \dots, \mathbf{y}_{n+1-k}]$.
 - (d) $\tilde{U}_k = I_{k-1} \oplus U_k$ ($\tilde{U}_1 = U_1$).
 - (e) $B_k = U_k^* A_k U_k$.
 - (f) $A_{k+1} = B_k(1)$, the $(n - k) \times (n - k)$ matrix obtained from B_k by deleting the first row and column.
3. $U = \tilde{U}_1 \tilde{U}_2, \dots, \tilde{U}_{n-1}$.

9. (A strictly real version of the Schur unitary triangularization theorem) If A is a real matrix, then there is a real, orthogonal matrix Q such that $Q^T A Q$ is block triangular, with the blocks of size 1×1 or 2×2 . Each real eigenvalue of A appears as a 1×1 block of $Q^T A Q$ and each nonreal pair of complex conjugate eigenvalues corresponds to a 2×2 diagonal block of $Q^T A Q$.
10. If $\mathcal{F}$ is a commuting family of matrices, then $\mathcal{F}$ is simultaneously unitarily triangularizable — i.e., there is a unitary matrix U such that U^*AU is triangular for every matrix A in $\mathcal{F}$. This fact has the analogous real form also.
11. [Lit53] [Mit53] [Sha91] Let $\lambda_1, \lambda_2, \dots, \lambda_t$ be the distinct eigenvalues of A with multiplicities $m_1, m_2, \dots, m_t$. Suppose U^*AU is block triangular with diagonal blocks $A_1, A_2, \dots, A_t$, where A_i is size $m_i \times m_i$ and λ_i is the only eigenvalue of A_i for each i . Then the Jordan canonical form of A is the direct sum of the Jordan canonical forms of the blocks $A_1, A_2, \dots, A_t$. *Note:* This conclusion also holds if the unitary similarity U is replaced by an ordinary similarity.
12. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of the $n \times n$ matrix A and let $T = U^*AU$ be triangular. Then $\|A\|_F^2 = \sum_{i=1}^n |\lambda_i|^2 + \sum_{i < j} |t_{ij}|^2$. Hence, $\|A\|_F^2 \geq \sum_{i=1}^n |\lambda_i|^2$ and equality holds if and only if T is diagonal, or equivalently, if and only if A is normal (see Section 7.2).
13. A 2×2 matrix A with eigenvalues λ_1, λ_2 is unitarily similar to the triangular matrix $\begin{bmatrix} \lambda_1 & r \\ 0 & \lambda_2 \end{bmatrix}$, where $r = \sqrt{\|A\|_F^2 - (|\lambda_1|^2 + |\lambda_2|^2)}$. Note that r is real and nonnegative.
14. Two 2×2 matrices, A and B , are unitarily similar if and only if they have the same eigenvalues and $\|A\|_F = \|B\|_F$.
15. Any square matrix A is unitarily similar to a matrix in which all of the diagonal entries are equal to $\frac{\text{tr}(A)}{n}$.
16. [Spe40] Two $n \times n$ matrices, A and B , are unitarily equivalent if and only if $\text{tr } \omega(A, A^*) = \text{tr } \omega(B, B^*)$ for every word $\omega(s, t)$ in two noncommuting variables.
17. [Pea62] Two $n \times n$ matrices, A and B , are unitarily equivalent if and only if $\text{tr } \omega(A, A^*) = \text{tr } \omega(B, B^*)$ for every word $\omega(s, t)$ in two noncommuting variables of degree at most $2n^2$.

Examples:

1. The matrix $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$ is unitary but not orthogonal.
2. The matrix $\frac{1}{\sqrt{1+2i}} \begin{bmatrix} 1 & 1+i \\ 1+i & -1 \end{bmatrix}$ is orthogonal but not unitary.

3. Fact 13 shows that $A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$ is unitarily similar to $A = \begin{bmatrix} 4 & 1 \\ 0 & 1 \end{bmatrix}$.
4. For any nonzero r , the matrices $\begin{bmatrix} 3 & r \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ are similar, but not unitarily similar.
5. Let $A = \begin{bmatrix} -31 & 21 & 48 \\ -4 & 4 & 6 \\ -20 & 13 & 31 \end{bmatrix}$. Apply Algorithm 1 to A :

Step 1. $A_1 = A$.

Step 2. For

$k = 1$: (a) $p_{A_1}(x) = x^3 - 4x^2 + 5x - 2 = (x - 2)(x - 1)^2$, so the eigenvalues are 1, 1, 2. From the reduced row echelon form of $A - I_3$, we see that $[3, 0, 2]^T$ is an eigenvector for 1 and, thus, $\mathbf{x} = [\frac{3}{\sqrt{13}}, 0, \frac{2}{\sqrt{13}}]^T$ is a normalized eigenvector.

(b) One expects to apply the Gram-Schmidt process to a basis that includes $\mathbf{x}$ as the first vector to produce an orthonormal basis. In this example, it is obvious how to find an orthonormal basis for $\mathbb{C}^3$:

$$(c) \quad U_1 = \begin{bmatrix} \frac{3}{\sqrt{13}} & 0 & -\frac{2}{\sqrt{13}} \\ 0 & 1 & 0 \\ \frac{2}{\sqrt{13}} & 0 & \frac{3}{\sqrt{13}} \end{bmatrix}.$$

(d) unnecessary.

$$(e) \quad B_1 = U_1^* A_1 U_1 = \begin{bmatrix} 1 & \frac{89}{\sqrt{13}} & 68 \\ 0 & 4 & 2\sqrt{13} \\ 0 & -\frac{3}{\sqrt{13}} & -1 \end{bmatrix}.$$

$$(f) \quad A_2 = \begin{bmatrix} 4 & 2\sqrt{13} \\ -\frac{3}{\sqrt{13}} & -1 \end{bmatrix}.$$

$k = 2$: (a) 1 is still an eigenvalue of A_2 . From the reduced row echelon form of $A_2 - I_2$, we see that $[-2\sqrt{13}, 3]^T$ is an eigenvector for 1 and, thus, $\mathbf{x} = [-2\sqrt{\frac{13}{61}}, \frac{3}{\sqrt{61}}]^T$ is a normalized eigenvector.

(b) Again, the orthonormal basis is obvious:

$$(c) \quad U_2 = \begin{bmatrix} -2\sqrt{\frac{13}{61}} & \frac{3}{\sqrt{61}} \\ \frac{3}{\sqrt{61}} & 2\sqrt{\frac{13}{61}} \end{bmatrix}.$$

$$(d) \quad \tilde{U}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2\sqrt{\frac{13}{61}} & \frac{3}{\sqrt{61}} \\ 0 & \frac{3}{\sqrt{61}} & 2\sqrt{\frac{13}{61}} \end{bmatrix}.$$

$$(e) \quad B_2 = \begin{bmatrix} 1 & -\frac{29}{\sqrt{13}} \\ 0 & 2 \end{bmatrix}.$$

(f) unnecessary.

$$\text{Step 3. } U = \tilde{U}_1 \tilde{U}_2 = \begin{bmatrix} \frac{3}{\sqrt{13}} & -\frac{6}{\sqrt{793}} & -\frac{4}{\sqrt{61}} \\ 0 & -2\sqrt{\frac{13}{61}} & \frac{3}{\sqrt{61}} \\ \frac{2}{\sqrt{13}} & \frac{9}{\sqrt{793}} & \frac{6}{\sqrt{61}} \end{bmatrix}, \quad T = U^* A U = \begin{bmatrix} 1 & \frac{26}{\sqrt{61}} & \frac{2035}{\sqrt{793}} \\ 0 & 1 & -\frac{29}{\sqrt{13}} \\ 0 & 0 & 2 \end{bmatrix}.$$

6. [HJ85, p. 84] Schur's theorem tells us that every complex, square matrix is unitarily similar to a triangular matrix. However, it is not true that every complex, square matrix is similar to a triangular matrix via a complex, orthogonal similarity. For, suppose $A = QTQ^T$, where Q is complex orthogonal and T is triangular. Let $\mathbf{q}$ be the first column of Q . Then $\mathbf{q}$ is an eigenvector of A and $\mathbf{q}^T \mathbf{q} = 1$. However, the matrix $A = \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}$ has no such eigenvector; A is nilpotent and any eigenvector of A is a scalar multiple of $\begin{bmatrix} 1 \\ i \end{bmatrix}$.

7.2 Normal Matrices and Spectral Theory

In this subsection, all matrices are over the complex numbers and are square. All vector spaces are finite dimensional complex inner product spaces.

Definitions:

The matrix A is **normal** if $AA^* = A^*A$.

The matrix A is **Hermitian** if $A^* = A$.

The matrix A is **skew-Hermitian** if $A^* = -A$.

The linear operator, T , on the complex inner product space V is **normal** if $TT^* = T^*T$.

Two orthogonal projections, P and Q , are **pairwise orthogonal** if $PQ = QP = 0$. (See Section 5.4 for information about orthogonal projection.)

The matrices A and B are said to have **Property L** if their eigenvalues $\alpha_k, \beta_k, (k = 1, \dots, n)$ may be ordered in such a way that the eigenvalues of $xA + yB$ are given by $x\alpha_k + y\beta_k$ for all complex numbers x and y .

Facts:

Most of the material in this section can be found in one or more of the following: [HJ85, Chap. 2] [Hal87, Chap. 3] [Gan59, Chap. IX] [MM64, I.4, III.3.5, III.5] [GJSW87]. Specific references are also given for some facts.

1. Diagonal, Hermitian, skew-Hermitian, and unitary matrices are all normal. Note that real symmetric matrices are Hermitian, real skew-symmetric matrices are skew-Hermitian, and real, orthogonal matrices are unitary, so all of these matrices are normal.
2. If U is unitary, then A is normal if and only if U^*AU is normal.
3. Let T be a linear operator on the complex inner product space V . Let $\mathcal{B}$ be an ordered orthonormal basis of V and let $A = [T]_{\mathcal{B}}$. Then T is normal if and only if A is a normal matrix.
4. (Spectral Theorem) The following three versions are equivalent.
 - A matrix is normal if and only if it is unitarily similar to a diagonal matrix. (*Note:* This is sometimes taken as the definition of normal. See Fact 6 below for a strictly real version.)
 - The matrix A is normal if and only if there is an orthonormal basis of eigenvectors of A .
 - Let $\lambda_1, \lambda_2, \dots, \lambda_t$ be the distinct eigenvalues of A with algebraic multiplicities $m_1, m_2, \dots, m_t$. Then A is normal if and only if there exist t pairwise orthogonal, orthogonal projections $P_1, P_2, \dots, P_t$ such that $\sum_{i=1}^t P_i = I$, $\text{rank}(P_i) = m_i$, and $A = \sum_{i=1}^t \lambda_i P_i$. (Note that the two orthogonal projections P and Q are pairwise orthogonal if and only if $\text{range}(P)$ and $\text{range}(Q)$ are orthogonal subspaces.)
5. (Principal Axes Theorem) A real matrix A is symmetric if and only if $A = QDQ^T$, where Q is a real, orthogonal matrix and D is a real, diagonal matrix. Equivalently, a real matrix A is symmetric

if and only if there is a real, orthonormal basis of eigenvectors of A . Note that the eigenvalues of A appear on the diagonal of D , and the columns of Q are eigenvectors of A . The Principal Axes Theorem follows from the Spectral Theorem, and the fact that all of the eigenvalues of a Hermitian matrix are real.

6. (A strictly real version of the Spectral Theorem) If A is a real, normal matrix, then there is a real, orthogonal matrix Q such that $Q^T A Q$ is block diagonal, with the blocks of size 1×1 or 2×2 . Each real eigenvalue of A appears as a 1×1 block of $Q^T A Q$ and each nonreal pair of complex conjugate eigenvalues corresponds to a 2×2 diagonal block of $Q^T A Q$.
7. The following are equivalent. See also Facts 4 and 8. See [GJSW87] and [EI98] for more equivalent conditions.

- A is normal.
- A^* can be expressed as a polynomial in A .
- For any B , $AB = BA$ implies $A^*B = BA^*$.
- Any eigenvector of A is also an eigenvector of A^* .
- Each invariant subspace of A is also an invariant subspace of A^* .
- For each invariant subspace, $\mathcal{V}$, of A , the orthogonal complement, $\mathcal{V}^\perp$, is also an invariant subspace of A .
- $\langle A\mathbf{x}, A\mathbf{y} \rangle = \langle A^*\mathbf{x}, A^*\mathbf{y} \rangle$ for all vectors $\mathbf{x}$ and $\mathbf{y}$.
- $\langle A\mathbf{x}, A\mathbf{x} \rangle = \langle A^*\mathbf{x}, A^*\mathbf{x} \rangle$ for every vector $\mathbf{x}$.
- $\|A\mathbf{x}\| = \|A^*\mathbf{x}\|$ for every vector $\mathbf{x}$.
- $A^* = UA$ for some unitary matrix U .
- $\|A\|_F^2 = \sum_{i=1}^n |\lambda_i|^2$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A .
- The singular values of A are $|\lambda_1|, |\lambda_2|, \dots, |\lambda_n|$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A .
- If $A = UP$ is a polar decomposition of A , then $UP = PU$. (See Section 8.4.)
- A commutes with a normal matrix with distinct eigenvalues.
- A commutes with a Hermitian matrix with distinct eigenvalues.
- The Hermitian matrix $AA^* - A^*A$ is semidefinite (i.e., it does not have both positive and negative eigenvalues).

8. Let $H = \frac{A + A^*}{2}$ and $K = \frac{A - A^*}{2i}$. Then H and K are Hermitian and $A = H + iK$. The matrix A is normal if and only if $HK = KH$.

9. If A is normal, then

- A is Hermitian if and only if all of the eigenvalues of A are real.
- A is skew-Hermitian if and only if all of the eigenvalues of A are pure imaginary.
- A is unitary if and only if all of the eigenvalues of A have modulus 1.

10. The matrix U is unitary if and only if $U = \exp(iH)$ where H is Hermitian.
11. If Q is a real matrix with $\det(Q) = 1$, then Q is orthogonal if and only if $Q = \exp(K)$, where K is a real, skew-symmetric matrix.
12. (Cayley's Formulas/Cayley Transform) If U is unitary and does not have -1 as an eigenvalue, then $U = (I + iH)(I - iH)^{-1}$, where $H = i(I - U)(I + U)^{-1}$ is Hermitian.
13. (Cayley's Formulas/Cayley Transform, real version) If Q is a real, orthogonal matrix which does not have -1 as an eigenvalue, then $Q = (I - K)(I + K)^{-1}$, where $K = (I - Q)(I + Q)^{-1}$ is a real, skew-symmetric matrix.
14. A triangular matrix is normal if and only if it is diagonal. More generally, if the block triangular

matrix, $\begin{bmatrix} B_{11} & B_{12} \\ 0 & B_{22} \end{bmatrix}$ (where the diagonal blocks, B_{ii} , $i = 1, 2$, are square), is normal, then $B_{12} = 0$.

15. Let A be a normal matrix. Then the diagonal entries of A are the eigenvalues of A if and only if A is diagonal.
16. If A and B are normal and commute, then AB is normal. However, the product of two noncommuting normal matrices need not be normal. (See Example 3 below.)
17. If A is normal, then $\rho(A) = \|A\|_2$. Consequently, if A is normal, then $\rho(A) \geq |a_{ij}|$ for all i and j . The converses of both of these facts are false (see Example 4 below).
18. [MM64, p. 168] [MM55] [ST80] If A is normal, then $W(A)$ is the convex hull of the eigenvalues of A . The converse of this statement holds when $n \leq 4$, but not for $n \geq 5$.
19. [WW49] [MM64, page 162] Let A be a normal matrix and suppose x is a vector such that $(Ax)_i = 0$ whenever $x_i = 0$. For each nonzero component, x_j , of x , define $\mu_j = \frac{(Ax)_j}{x_j}$. Note that μ_j is a complex number, which we regard as a point in the plane. Then any closed disk that contains all of the points μ_j must contain an eigenvalue of A .
20. [HW53] Let A and B be normal matrices with eigenvalues $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$. Then

$$\min_{\sigma \in S_n} \sum_{i=1}^n |\alpha_i - \beta_{\sigma(i)}|^2 \leq \|A - B\|_F^2 \leq \max_{\sigma \in S_n} \sum_{i=1}^n |\alpha_i - \beta_{\sigma(i)}|^2,$$

where the minimum and maximum are over all permutations σ in the symmetric group S_n (i.e., the group of all permutations of $1, \dots, n$).

21. [Sun82] [Bha82] Let A and B be $n \times n$ normal matrices with eigenvalues $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$. Let Λ_A, Λ_B be the diagonal matrices with diagonal entries $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$, respectively. Let $\|\cdot\|$ be any unitarily invariant norm. Then, if $A - B$ is normal, we have

$$\min_P \| \Lambda_A - P^{-1} \Lambda_B P \| \leq \|A - B\| \leq \max_P \| \Lambda_A - P^{-1} \Lambda_B P \|,$$

where the maximum and minimum are over all $n \times n$ permutation matrices P .

Observe that if A and B are Hermitian, then $A - B$ is also Hermitian and, hence, normal, so this inequality holds for all pairs of Hermitian matrices. However, Example 6 gives a pair of 2×2 normal matrices (with $A - B$ not normal) for which the inequality does not hold. Note that for the Frobenius norm, we get the Hoffman–Wielandt inequality (20), which does hold for all pairs of normal matrices.

For the operator norm, $\|\cdot\|_2$, this gives the inequality

$$\min_{\sigma \in S_n} \max_j |\alpha_j - \beta_{\sigma(j)}| \leq \|A - B\|_2 \leq \max_{\sigma \in S_n} \max_j |\alpha_j - \beta_{\sigma(j)}|$$

(assuming $A - B$ is normal), which, for the case of Hermitian A and B , is a classical result of Weyl [Wey12].

22. [OS90] [BEK97] [BDM83] [BDK89] [Hol92] [AN86] Let A and B be normal matrices with eigenvalues $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$, respectively. Using $\|A\|_2 \leq \|A\|_F \leq \sqrt{n} \|A\|_2$ together with the Hoffman–Wielandt inequality (20) yields

$$\frac{1}{\sqrt{n}} \min_{\sigma \in S_n} \max_j |\alpha_j - \beta_{\sigma(j)}| \leq \|A - B\|_2 \leq \sqrt{n} \max_{\sigma \in S_n} \max_j |\alpha_j - \beta_{\sigma(j)}|.$$

On the right-hand side, the factor $\sqrt{n}$ may be replaced by $\sqrt{2}$ and it is known that this constant is the best possible. On the left-hand side, the factor $\frac{1}{\sqrt{n}}$ may be replaced by the constant $\frac{1}{2.91}$, but the best possible value for this constant is still unknown. Thus, we have

$$\frac{1}{2.91} \min_{\sigma \in S_n} \max_j |\alpha_j - \beta_{\sigma(j)}| \leq \|A - B\|_2 \leq \sqrt{2} \max_{\sigma \in S_n} \max_j |\alpha_j - \beta_{\sigma(j)}|.$$

See also [Bha82], [Bha87], [BH85], [Sun82], [Sund82].

23. If A and B are normal matrices, then $AB = BA$ if and only if A and B have Property L. This was established for Hermitian matrices by Motzkin and Taussky [MT52] and then generalized to the normal case by Wiegmann [Wieg53]. For a stronger generalization see [Wiel53].
24. [Fri02] Let $a_{ij}, i = 1, \dots, n, j = 1, \dots, n$, be any set of $\frac{n(n+1)}{2}$ complex numbers. Then there exists an $n \times n$ normal matrix, N , such that $n_{ij} = a_{ij}$ for $i \leq j$. Thus, any upper triangular matrix A can be completed to a normal matrix.
25. [Bha87, p. 54] Let A be a normal $n \times n$ matrix and let B be an arbitrary $n \times n$ matrix such that $\|A - B\|_2 < \epsilon$. Then every eigenvalue of B is within distance ϵ of an eigenvalue of A . Example 7 below shows that this need not hold for an arbitrary pair of matrices.
26. There are various ways to measure the “nonnormality” of a matrix. For example, if A has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, the quantity $\sqrt{\|A\|_F^2 - \sum_{i=1}^n |\lambda_i|^2}$ is a natural measure of nonnormality, as is $\|A^*A - AA^*\|_2$. One could also consider $\|A^*A - AA^*\|$ for other choices of norm, or look at $\min\{\|A - N\| : N \text{ is normal}\}$. Fact 8 above suggests $\|HK - KH\|$ as a possible measure of nonnormality, while the polar decomposition (see Fact 7 above) $A = UP$ of A suggests $\|UP - PU\|$. See [EP87] for more measures of nonnormality and comparisons between them.
27. [Lin97] [FR96] For any $\epsilon > 0$ there is a $\delta > 0$ such that, for any $n \times n$ complex matrix A with $\|AA^* - A^*A\|_2 < \delta$, there is a normal matrix N with $\|N - A\|_2 < \epsilon$. Thus, a matrix which is approximately normal is close to a normal matrix.

Examples:

- Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Then $U^*AU = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ and $A = 4P_1 + 2P_2$, where the P_i 's are the pairwise orthogonal, orthogonal projection matrices

$$P_1 = U \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} U^* = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad P_2 = U \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} U^* = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$
- $A = \begin{bmatrix} 1 & 4+2i & 6 \\ 0 & 8+2i & 0 \\ 2 & -2i & 4i \end{bmatrix} = H + iK$, where $H = \begin{bmatrix} 1 & 2+i & 4 \\ 2-i & 8 & i \\ 4 & -i & 0 \end{bmatrix}$ and $K = \begin{bmatrix} 0 & 1-2i & -2i \\ 1+2i & 2 & -1 \\ 2i & -1 & 4 \end{bmatrix}$ are Hermitian.
- $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ are both normal matrices, but the product $AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is not normal.
- Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Then $\rho(A) = 2 = \|A\|_2$, but A is not normal.
- Let $Q = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$. Put $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$ and $D = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$. Then $Q = UDU^* = U \left(\exp i \begin{bmatrix} \theta & 0 \\ 0 & -\theta \end{bmatrix} \right) U^* = \exp i \left(U \begin{bmatrix} \theta & 0 \\ 0 & -\theta \end{bmatrix} U^* \right)$. Put $H = U \begin{bmatrix} \theta & 0 \\ 0 & -\theta \end{bmatrix} U^* = \begin{bmatrix} 0 & -i\theta \\ i\theta & 0 \end{bmatrix}$. Then H is Hermitian and $Q = \exp(iH)$. Also, $K = iH = \begin{bmatrix} 0 & \theta \\ -\theta & 0 \end{bmatrix}$ is a real, skew-symmetric matrix and $Q = \exp(K)$.
- Here is an example from [Sund82] showing that the condition that $A - B$ be normal cannot be dropped from 21. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then A is Hermitian with eigenvalues ± 1

and B is skew-Hermitian with eigenvalues $\pm i$. So, we have $\|\Lambda_A - P^{-1}\Lambda_B P\|_2 = \sqrt{2}$, regardless of the permutation P . However, $A - B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ and $\|A - B\|_2 = 2$.

7. This example shows that Fact 25 above does not hold for general pairs of matrices. Let $\alpha > \beta > 0$ and put $A = \begin{bmatrix} 0 & \alpha \\ \beta & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & \alpha - \beta \\ 0 & 0 \end{bmatrix}$. Then the eigenvalues of A are $\pm\sqrt{\alpha\beta}$ and both eigenvalues of B are zero. We have $A - B = \begin{bmatrix} 0 & \beta \\ \beta & 0 \end{bmatrix}$ and $\|A - B\|_2 = \beta$. But, since $\alpha > \beta$, we have $\sqrt{\alpha\beta} > \beta = \|A - B\|_2$.

References

- [AN86] T. Ando and Y. Nakamura. "Bounds for the antidistance." Technical Report, Hokkaido University, Japan, 1986.
- [BDK89] R. Bhatia, C. Davis, and P. Koosis. An extremal problem in Fourier analysis with applications to operator theory. *J. Funct. Anal.*, 82:138–150, 1989.
- [BDM83] R. Bhatia, C. Davis, and A. McIntosh. Perturbation of spectral subspaces and solution of linear operator equations. *Linear Algebra Appl.*, 52/53:45–67, 1983.
- [BEK97] R. Bhatia, L. Elsner, and G.M. Krause. Spectral variation bounds for diagonalisable matrices. *Aequationes Mathematicae*, 54:102–107, 1997.
- [Bha82] R. Bhatia. Analysis of spectral variation and some inequalities. *Transactions of the American Mathematical Society*, 272:323–331, 1982.
- [Bha87] R. Bhatia. *Perturbation Bounds for Matrix Eigenvalues*. Longman Scientific & Technical, Essex, U.K. (copublished in the United States with John Wiley & Sons, New York), 1987.
- [BH85] R. Bhatia and J. A. R. Holbrook. Short normal paths and spectral variation. *Proc. Amer. Math. Soc.*, 94:377–382, 1985.
- [EI98] L. Elsner and Kh.D. Ikramov. Normal matrices: an update. *Linear Algebra Appl.*, 285:291–303, 1998.
- [EP87] L. Elsner and M.H.C. Paardekooper. On measures of nonnormality of matrices. *Linear Algebra Appl.*, 92:107–124, 1987.
- [Fri02] S. Friedland. Normal matrices and the completion problem. *SIAM J. Matrix Anal. Appl.*, 23:896–902, 2002.
- [FR96] P. Friis and M. Rørdam. Almost commuting self-adjoint matrices — a short proof of Huaxin Lin's theorem. *J. Reine Angew. Math.*, 479:121–131, 1996.
- [Gan59] F.R. Gantmacher. *Matrix Theory, Vol. I*. Chelsea Publishing, New York, 1959.
- [GJSW87] R. Grone, C.R. Johnson, E.M. Sa, and H. Wolkowicz. Normal matrices. *Linear Algebra Appl.*, 87:213–225, 1987.
- [Hal87] P.R. Halmos. *Finite-Dimensional Vector Spaces*. Springer-Verlag, New York, 1987.
- [HJ85] R.A. Horn and C.R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 1985.
- [Hol92] J.A. Holbrook. Spectral variation of normal matrices. *Linear Algebra Appl.*, 174:131–144, 1992.
- [HOS96] J. Holbrook, M. Omladič, and P. Šemrl. Maximal spectral distance. *Linear Algebra Appl.*, 249:197–205, 1996.
- [HW53] A.J. Hoffman and H.W. Wielandt. The variation of the spectrum of a normal matrix. *Duke Math. J.*, 20:37–39, 1953.
- [Lin97] H. Lin. Almost commuting self-adjoint matrices and applications. *Operator algebras and their applications (Waterloo, ON, 1994/95), Fields Inst. Commun.*, 13, Amer. Math. Soc., Providence, RI, 193–233, 1997.
- [Lit53] D.E. Littlewood. On unitary equivalence. *J. London Math. Soc.*, 28:314–322, 1953.

- [Mir60] L. Mirsky. Symmetric guage functions and unitarily invariant norms. *Quart. J. Math. Oxford* (2), 11:50–59, 1960.
- [Mit53] B.E. Mitchell. Unitary transformations. *Can. J. Math*, 6:69–72, 1954.
- [MM55] B.N. Moysl and M.D. Marcus. Field convexity of a square matrix. *Proc. Amer. Math. Soc.*, 6:981–983, 1955.
- [MM64] M. Marcus and H. Minc. *A Survey of Matrix Theory and Matrix Inequalities*. Allyn and Bacon, Boston, 1964.
- [MT52] T.S. Motzkin and O. Taussky Todd. Pairs of matrices with property L. *Trans. Amer. Math. Soc.*, 73:108–114, 1952.
- [OS90] M. Omladič and P. Šemrl. On the distance between normal matrices. *Proc. Amer. Math. Soc.*, 110:591–596, 1990.
- [Par48] W.V. Parker. Sets of numbers associated with a matrix. *Duke Math. J.*, 15:711–715, 1948.
- [Pea62] C. Pearcy. A complete set of unitary invariants for operators generating finite W^* -algebras of type I. *Pacific J. Math.*, 12:1405–1416, 1962.
- [Sch09] I. Schur. Über die charakteristischen Wurzeln einer linearen Substitutionen mit einer Anwendung auf die Theorie der Intergralgleichungen. *Math. Ann.*, 66:488–510, 1909.
- [Sha91] H. Shapiro. A survey of canonical forms and invariants for unitary similarity. *Linear Algebra Appl.*, 147:101–167, 1991.
- [Spe40] W. Specht. Zur Theorie der Matrizen, II. *Jahresber. Deutsch. Math.-Verein.*, 50:19–23, 1940.
- [ST80] H. Shapiro and O. Taussky. Alternative proofs of a theorem of Moysl and Marcus on the numerical range of a square matrix. *Linear Multilinear Algebra*, 8:337–340, 1980.
- [Sun82] V.S. Sunder. On permutations, convex hulls, and normal operators. *Linear Algebra Appl.*, 48:403–411, 1982.
- [Sund82] V.S. Sunder. Distance between normal operators. *Proc. Amer. Math. Soc.*, 84:483–484, 1982.
- [Wey12] H. Weyl. Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Difffferentialgleichungen. *Math. Ann.*, 71:441–479, 1912.
- [Wieg53] N. Wiegmann. Pairs of normal matrices with property L. *Proc. Am. Math. Soc.*, 4: 35–36, 1953.
- [Wiel53] H. Wielandt. Pairs of normal matrices with property L. *J. Res. Nat. Bur. Standards*, 51:89–90, 1953.
- [WW49] A.G. Walker and J.D. Weston. Inclusion theorems for the eigenvalues of a normal matrix. *J. London Math. Soc.*, 24:28–31, 1949.

8

Hermitian and Positive Definite Matrices

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8.1 Hermitian Matrices

All matrices in this section are either real or complex, unless explicitly stated otherwise.

Definitions:

A matrix $A \in \mathbb{C}^{n \times n}$ is **Hermitian** or **self-adjoint** if $A^* = A$, or element-wise, $\bar{a}_{ij} = a_{ji}$, for $i, j = 1, \dots, n$. The set of Hermitian matrices of order n is denoted by $\mathcal{H}_n$. Note that a matrix $A \in \mathbb{R}^{n \times n}$ is Hermitian if and only if $A^T = A$.

A matrix $A \in \mathbb{C}^{n \times n}$ is **symmetric** if $A^T = A$, or element-wise, $a_{ij} = a_{ji}$, for $i, j = 1, \dots, n$. The set of real symmetric matrices of order n is denoted by $\mathcal{S}_n$. Since $\mathcal{S}_n$ is a subset of $\mathcal{H}_n$, all theorems for matrices in $\mathcal{H}_n$ apply to $\mathcal{S}_n$ as well.

Let V be a complex inner product space with inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in V$. The matrix $G = [g_{ij}] \in \mathbb{C}^{n \times n}$ defined by $g_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$, $i, j \in \{1, 2, \dots, n\}$ is called the **Gram matrix** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

The inner product $\langle \mathbf{x}, \mathbf{y} \rangle$ of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ will mean the standard inner product, i.e., $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x}$, unless stated otherwise. The term orthogonal will mean orthogonal with respect to this inner product, unless stated otherwise.

Facts:

For facts without a specific reference, see [HJ85, pp. 38, 101–104, 169–171, 175], [Lax96, pp. 80–83], and [GR01, pp. 169–171]. Many are an immediate consequence of the definition.

1. A real symmetric matrix is Hermitian, and a real Hermitian matrix is symmetric.
2. Let A, B be Hermitian.

- (a) Then $A + B$ is Hermitian.
 - (b) If $AB = BA$, then AB is Hermitian.
 - (c) If $c \in \mathbb{R}$, then cA is Hermitian.
3. $A + A^*$, $A^* + A$, AA^* , and A^*A are Hermitian for all $A \in \mathbb{C}^{n \times n}$.
 4. If $A \in \mathcal{H}_n$, then $\langle A\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A\mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$.
 5. If $A \in \mathcal{H}_n$, then $A^k \in \mathcal{H}_n$ for all $k \in \mathbb{N}$.
 6. If $A \in \mathcal{H}_n$ is invertible, then $A^{-1} \in \mathcal{H}_n$.
 7. The main diagonal entries of a Hermitian matrix are real.
 8. All eigenvalues of a Hermitian matrix are real.
 9. Eigenvectors corresponding to distinct eigenvalues of a Hermitian matrix are orthogonal.
 10. *Spectral Theorem — Diagonalization version:* If $A \in \mathcal{H}_n$, there is a unitary matrix $U \in \mathbb{C}^{n \times n}$ such that $U^*AU = D$, where D is a real diagonal matrix whose diagonal entries are the eigenvalues of A . If $A \in \mathcal{S}_n$, the same conclusion holds with an orthogonal matrix $Q \in \mathbb{R}^{n \times n}$, i.e., $Q^TAQ = D$.
 11. *Spectral Theorem — Orthonormal basis version:* If $A \in \mathcal{H}_n$, there is an orthonormal basis of $\mathbb{C}^n$ consisting of eigenvectors of A . If $A \in \mathcal{S}_n$, the same conclusion holds with $\mathbb{C}^n$ replaced by $\mathbb{R}^n$.
 12. [Lay97, p. 447] *Spectral Theorem — Sum of rank one projections version:* Let $A \in \mathcal{H}_n$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, and corresponding orthonormal eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$. Then

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^* + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^* + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^*.$$

If $A \in \mathcal{S}_n$, then

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T.$$

13. If $A \in \mathcal{H}_n$, then $\text{rank } A$ equals the number of nonzero eigenvalues of A .
14. Each $A \in \mathbb{C}^{n \times n}$ can be written uniquely as $A = H + iK$, where $H, K \in \mathcal{H}_n$.
15. Given $A \in \mathbb{C}^{n \times n}$, then $A \in \mathcal{H}_n$ if and only if $\mathbf{x}^* A \mathbf{x}$ is real for all $\mathbf{x} \in \mathbb{C}^n$.
16. Any Gram matrix is Hermitian. Some examples of how Gram matrices arise are given in Chapter 66 and [Lax96, p. 124].
17. The properties given above for $\mathcal{H}_n$ and $\mathcal{S}_n$ are generally not true for symmetric matrices in $\mathbb{C}^{n \times n}$, but there is a substantial theory associated with them. (See [HJ85, sections 4.4 and 4.6].)

Examples:

1. The matrix $\begin{bmatrix} 3 & 2-i \\ 2+i & -5 \end{bmatrix} \in \mathcal{H}_2$ and $\begin{bmatrix} 6 & 0 & 2 \\ 0 & -1 & 5 \\ 2 & 5 & 3 \end{bmatrix} \in \mathcal{S}_3$.
2. Let D be an open set in $\mathbb{R}^n$ containing the point $\mathbf{x}_0$, and let $f : D \rightarrow \mathbb{R}$ be a twice continuously differentiable function on D . Define $H \in \mathbb{R}^{n \times n}$ by $h_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0)$. Then H is a real symmetric matrix, and is called the **Hessian** of f .
3. Let $G = (V, E)$ be a simple undirected graph with vertex set $V = \{1, 2, 3, \dots, n\}$. The $n \times n$ **adjacency matrix** $A(G) = [a_{ij}]$ (see Section 28.3) is defined by

$$a_{ij} = \begin{cases} 1 & \text{if } ij \in E \\ 0 & \text{otherwise.} \end{cases}$$

In particular, all diagonal entries of $A(G)$ are 0. Since ij is an edge of G if and only if ji is, the adjacency matrix is real symmetric. Observe that for each $i \in V$, $\sum_{j=1}^n a_{ij} = \delta(i)$, i.e., the sum of the i^{th} row is the degree of vertex i .

8.2 Order Properties of Eigenvalues of Hermitian Matrices

Definitions:

Given $A \in \mathcal{H}_n$, the **Rayleigh quotient** $R_A : \mathbb{C}^n \setminus \{0\} \rightarrow \mathbb{R}$ is $R_A(x) = \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \frac{\langle A \mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle}$.

Facts:

For facts without a specific reference, see [HJ85, Sections 4.2, 4.3]; however, in that source the eigenvalues are labeled from smallest to greatest and the definition of majorizes (see Preliminaries) has a similar reversal of notation.

1. *Rayleigh–Ritz Theorem:* Let $A \in \mathcal{H}_n$, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Then

$$\lambda_n \leq \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} \leq \lambda_1, \quad \text{for all nonzero } \mathbf{x} \in \mathbb{C}^n,$$

$$\lambda_1 = \max_{\mathbf{x} \neq 0} \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \max_{\|\mathbf{x}\|_2=1} \mathbf{x}^* A \mathbf{x},$$

and

$$\lambda_n = \min_{\mathbf{x} \neq 0} \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \min_{\|\mathbf{x}\|_2=1} \mathbf{x}^* A \mathbf{x}.$$

2. *Courant–Fischer Theorem:* Let $A \in M_n$ be a Hermitian matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, and let k be a given integer with $1 \leq k \leq n$. Then

$$\max_{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{n-k} \in \mathbb{C}^n} \min_{\substack{\mathbf{x} \neq 0, \mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \perp \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{n-k}}} \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \lambda_k$$

and

$$\min_{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{k-1} \in \mathbb{C}^n} \max_{\substack{\mathbf{x} \neq 0, \mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \perp \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{k-1}}} \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \lambda_k.$$

3. (Also [Bha01, p. 291]) *Weyl Inequalities:* Let $A, B \in \mathcal{H}_n$ and assume that the eigenvalues of A, B and $A+B$ are arranged in decreasing order. Then for every pair of integers j, k such that $1 \leq j, k \leq n$ and $j+k \leq n+1$,

$$\lambda_{j+k-1}(A+B) \leq \lambda_j(A) + \lambda_k(B)$$

and for every pair of integers j, k such that $1 \leq j, k \leq n$ and $j+k \geq n+1$,

$$\lambda_{j+k-n}(A+B) \geq \lambda_j(A) + \lambda_k(B).$$

4. *Weyl Inequalities:* These inequalities are a prominent special case of Fact 3. Let $A, B \in \mathcal{H}_n$ and assume that the eigenvalues of A, B and $A+B$ are arranged in decreasing order. Then for each $j \in \{1, 2, \dots, n\}$,

$$\lambda_j(A) + \lambda_n(B) \leq \lambda_j(A+B) \leq \lambda_j(A) + \lambda_1(B).$$

5. *Interlacing Inequalities:* Let $A \in \mathcal{H}_n$, let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be the eigenvalues of A , and for any $i \in \{1, 2, \dots, n\}$, let $\mu_1 \geq \mu_2 \geq \dots \geq \mu_{n-1}$ be the eigenvalues of $A(i)$, where $A(i)$ is the principal

submatrix of A obtained by deleting its i^{th} row and column. Then

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \lambda_3 \geq \dots \geq \lambda_{n-1} \geq \mu_{n-1} \geq \lambda_n.$$

6. Let $A \in \mathcal{H}_n$ and let B be any principal submatrix of A . If λ_k is the k^{th} largest eigenvalue of A and μ_k is the k^{th} largest eigenvalue of B , then $\lambda_k \geq \mu_k$.
7. Let $A \in \mathcal{H}_n$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Let S be a k -dimensional subspace of $\mathbb{C}^n$ with $k \in \{1, 2, \dots, n\}$. Then
 - (a) If there is a constant c such that $\mathbf{x}^* A \mathbf{x} \geq c \mathbf{x}^* \mathbf{x}$ for all $\mathbf{x} \in S$, then $\lambda_k \geq c$.
 - (b) If there is a constant c such that $\mathbf{x}^* A \mathbf{x} \leq c \mathbf{x}^* \mathbf{x}$ for all $\mathbf{x} \in S$, then $\lambda_{n-k+1} \leq c$.
8. Let $A \in \mathcal{H}_n$.
 - (a) If $\mathbf{x}^* A \mathbf{x} \geq 0$ for all $\mathbf{x}$ in a k -dimensional subspace of $\mathbb{C}^n$, then A has at least k nonnegative eigenvalues.
 - (b) If $\mathbf{x}^* A \mathbf{x} > 0$ for all nonzero $\mathbf{x}$ in a k -dimensional subspace of $\mathbb{C}^n$, then A has at least k positive eigenvalues.
9. Let $A \in \mathcal{H}_n$, let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n)$ be the vector of eigenvalues of A arranged in decreasing order, and let $\alpha = (a_1, a_2, \dots, a_n)$ be the vector consisting of the diagonal entries of A arranged in decreasing order. Then $\lambda \succeq \alpha$. (See Preliminaries for the definition of $\succeq$.)
10. Let $\alpha = (a_1, a_2, \dots, a_n)$, $\beta = (b_1, b_2, \dots, b_n)$ be decreasing sequences of real numbers such that $\alpha \succeq \beta$. Then there exists an $A \in \mathcal{H}_n$ such that the eigenvalues of A are $a_1, a_2, \dots, a_n$, and the diagonal entries of A are $b_1, b_2, \dots, b_n$.
11. [Lax96, pp. 133–6] or [Bha01, p. 291] (See also Chapter 15.) Let $A, B \in \mathcal{H}_n$ with eigenvalues $\lambda_1(A) \geq \lambda_2(A) \geq \dots \geq \lambda_n(A)$ and $\lambda_1(B) \geq \lambda_2(B) \geq \dots \geq \lambda_n(B)$. Then
 - (a) $|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_2$, $i = 1, \dots, n$.
 - (b) $\sum_{i=1}^n [\lambda_i(A) - \lambda_i(B)]^2 \leq \|A - B\|_F^2$.

Examples:

1. Setting $\mathbf{x} = \mathbf{e}_i$ in the Rayleigh-Ritz theorem, we obtain $\lambda_n \leq a_{ii} \leq \lambda_1$. Thus, for any $A \in \mathcal{H}_n$, we have $\lambda_1 \geq \max\{a_{ii} | i \in \{1, 2, \dots, n\}\}$ and $\lambda_n \leq \min\{a_{ii} | i \in \{1, 2, \dots, n\}\}$.
2. Setting $\mathbf{x} = [1, 1, \dots, 1]^T$ in the Rayleigh-Ritz theorem, we find that $\lambda_n \leq \frac{1}{n} \sum_{i,j=1}^n a_{ij} \leq \lambda_1$. If we take A to be the adjacency matrix of a graph, then this inequality implies that the largest eigenvalue of the graph is greater than or equal to its average degree.
3. The Weyl inequalities in Fact 3 above are a special case of the following general class of inequalities:

$$\sum_{k \in K} \lambda_k(A + B) \leq \sum_{i \in I} \lambda_i(A) + \sum_{j \in J} \lambda_j(B),$$

where I, J, K are certain subsets of $\{1, 2, \dots, n\}$. In 1962, A. Horn conjectured which inequalities of this form are valid for all Hermitian A, B , and this conjecture was proved correct in papers by A. Klyachko in 1998 and by A. Knutson and T. Tao in 1999. Two detailed accounts of the problem and its solution are given in [Bha01] and [Ful00].

4. Let $A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ have eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$. Since $A(4) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

has eigenvalues 3, 0, 0, by the interlacing inequalities, $\lambda_1 \geq 3 \geq \lambda_2 \geq 0 \geq \lambda_3 \geq 0 \geq \lambda_4$. In particular, $\lambda_3 = 0$.

Applications:

1. To use the Rayleigh–Ritz theorem effectively to estimate the largest or smallest eigenvalue of a Hermitian matrix, one needs to take into account the relative magnitudes of the entries of the

matrix. For example, let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. In order to estimate λ_1 , we should try to maximize the

Rayleigh quotient. A vector $\mathbf{x} \in \mathbb{R}^3$ is needed for which no component is zero, but such that each component is weighted more than the last. In a few trials, one is led to $\mathbf{x} = [1, 2, 3]^T$, which gives a Rayleigh quotient of 5. So $\lambda_1 \geq 5$. This is close to the actual value of λ_1 , which is $\frac{1}{4} \csc^2 \frac{\pi}{14} \approx 5.049$.

This example is only meant to illustrate the method; its primary importance is as a tool for estimating the largest (smallest) eigenvalue of a large Hermitian matrix when it can neither be found exactly nor be computed numerically.

2. The interlacing inequalities can sometimes be used to efficiently find all the eigenvalues of a Hermitian matrix. The Laplacian matrix (from spectral graph theory, see Section 28.4) of a star is

$$L = \begin{bmatrix} n-1 & -1 & -1 & \cdots & -1 & -1 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ -1 & 0 & 1 & & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ -1 & 0 & 0 & & 1 & 0 \\ -1 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}.$$

Since $L(1)$ is an identity matrix, the interlacing inequalities relative to $L(1)$ are: $\lambda_1 \geq 1 \geq \lambda_2 \geq 1 \geq \cdots \geq \lambda_{n-1} \geq 1 \geq \lambda_n$. Therefore, $n-2$ of the eigenvalues of L are equal to 1. Since the columns sum to 0, another eigenvalue is 0. Finally, since $\text{tr } L = 2n - 2$, the remaining eigenvalue is n .

3. The sixth fact above is applied in spectral graph theory to establish the useful fact that the k^{th} largest eigenvalue of a graph is greater than or equal to the k^{th} largest eigenvalue of any induced subgraph.

8.3 Congruence

Definitions:

Two matrices $A, B \in \mathcal{H}_n$ are ***congruent** if there is an invertible matrix $C \in \mathbb{C}^{n \times n}$ such that $B = C^*AC$, denoted $A \stackrel{*}{\sim} B$. If C is real, then A and B are also called **congruent**.

Let $A \in \mathcal{H}_n$. The **inertia** of A is the ordered triple $\text{in}(A) = (\pi(A), \nu(A), \delta(A))$, where $\pi(A)$ is the number of positive eigenvalues of A , $\nu(A)$ is the number of negative eigenvalues of A , and $\delta(A)$ is the number of zero eigenvalues of A .

In the event that $A \in \mathbb{C}^{n \times n}$ has all real eigenvalues, we adopt the same definition for $\text{in}(A)$.

Facts:

The following can be found in [HJ85, pp. 221–223] and a variation of the last in [Lax96, pp. 77–78].

1. Unitary similarity is a special case of *congruence.
2. *Congruence is an equivalence relation.
3. For $A \in \mathcal{H}_n$, $\pi(A) + \nu(A) + \delta(A) = n$.
4. For $A \in \mathcal{H}_n$, $\text{rank } A = \pi(A) + \nu(A)$.
5. Let $A \in \mathcal{H}_n$ with inertia (r, s, t) . Then A is *congruent to $I_r \oplus (-I_s) \oplus 0_t$. A matrix C that implements this *congruence is found as follows. Let U be a unitary matrix for which $U^*AU = D$

is a diagonal matrix with $d_{11}, \dots, d_{rr}$ the positive eigenvalues, $d_{r+1,r+1}, \dots, d_{r+s,r+s}$ the negative eigenvalues, and $d_{ii} = 0$, $k > r + s$. Let

$$s_i = \begin{cases} 1/\sqrt{d_{ii}}, & i = 1, \dots, r \\ 1/\sqrt{-d_{ii}}, & i = r + 1, \dots, s \\ 1, & i > r + s \end{cases}$$

and let $S = \text{diag}(s_1, s_2, \dots, s_n)$. Then $C = US$.

6. *Sylvester's Law of Inertia*: Two matrices $A, B \in \mathcal{H}_n$ are *congruent if and only if they have the same inertia.

Examples:

1. Let $A = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 4 \\ 3 & 4 & 0 \end{bmatrix}$. Since $\text{rank } A = 2$, $\pi(A) + \nu(A) = 2$, so $\delta(A) = 1$. Since $\text{tr } A = 0$, we have $\pi(A) = \nu(A) = 1$, and $\text{in}(A) = (1, 1, 1)$. Letting

$$C = \begin{bmatrix} \frac{3}{5\sqrt{10}} & \frac{3}{5\sqrt{10}} & \frac{4}{5} \\ \frac{4}{5\sqrt{10}} & \frac{4}{5\sqrt{10}} & -\frac{3}{5} \\ \frac{1}{\sqrt{10}} & -\frac{1}{\sqrt{10}} & 0 \end{bmatrix}$$

we have

$$C^*AC = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Now suppose

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Clearly $\text{in}(B) = (1, 1, 1)$ also. By Sylvester's law of inertia, B must be *congruent to A .

8.4 Positive Definite Matrices

Definitions:

A matrix $A \in \mathcal{H}_n$ is **positive definite** if $\mathbf{x}^*A\mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{C}^n$. It is **positive semidefinite** if $\mathbf{x}^*A\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{C}^n$. It is **indefinite** if neither A nor $-A$ is positive semidefinite. The set of positive definite matrices of order n is denoted by PD_n , and the set of positive semidefinite matrices of order n by PSD_n . If the dependence on n is not significant, these can be abbreviated as PD and PSD. Finally, PD (PSD) are also used to abbreviate "positive definite" ("positive semidefinite").

Let k be a positive integer. If A, B are PSD and $B^k = A$, then B is called a **PSD k^{th} root** of A and is denoted $A^{1/k}$.

A **correlation matrix** is a PSD matrix in which every main diagonal entry is 1.

Facts:

For facts without a specific reference, see [HJ85, Sections 7.1 and 7.2] and [Fie86, pp. 51–57].

1. $A \in \mathcal{S}_n$ is PD if $\mathbf{x}^T A \mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{R}^n$, and is PSD if $\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
2. Let $A, B \in \text{PSD}_n$.
 - (a) Then $A + B \in \text{PSD}_n$.
 - (b) If, in addition, $A \in \text{PD}_n$, then $A + B \in \text{PD}_n$.
 - (c) If $c \geq 0$, then $cA \in \text{PSD}_n$.
 - (d) If, in addition, $A \in \text{PD}_n$ and $c > 0$, then $cA \in \text{PD}_n$.
3. If $A_1, A_2, \dots, A_k \in \text{PSD}_n$, then so is $A_1 + A_2 + \dots + A_k$. If, in addition, there is an $i \in \{1, 2, \dots, k\}$ such that $A_i \in \text{PD}_n$, then $A_1 + A_2 + \dots + A_k \in \text{PD}_n$.
4. Let $A \in \mathcal{H}_n$. Then A is PD if and only if every eigenvalue of A is positive, and A is PSD if and only if every eigenvalue of A is nonnegative.
5. If A is PD, then $\text{tr } A > 0$ and $\det A > 0$. If A is PSD, then $\text{tr } A \geq 0$ and $\det A \geq 0$.
6. A PSD matrix is PD if and only if it is invertible.
7. *Inheritance Principle:* Any principal submatrix of a PD (PSD) matrix is PD (PSD).
8. All principal minors of a PD (PSD) matrix are positive (nonnegative).
9. Each diagonal entry of a PD (PSD) matrix is positive (nonnegative). If a diagonal entry of a PSD matrix is 0, then every entry in the row and column containing it is also 0.
10. Let $A \in \mathcal{H}_n$. Then A is PD if and only if every leading principal minor of A is positive. A is PSD if and only if every principal minor of A is nonnegative. (The matrix $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$ shows that it is not sufficient that every leading principal minor be nonnegative in order for A to be PSD.)
11. Let A be PD (PSD). Then A^k is PD (PSD) for all $k \in \mathbb{N}$.
12. Let $A \in \text{PSD}_n$ and express A as $A = UDU^*$, where U is unitary and D is the diagonal matrix of eigenvalues. Given any positive integer k , there exists a unique **PSD** k^{th} **root** of A given by $A^{1/k} = UD^{1/k}U^*$. If A is real so is $A^{1/k}$. (See also Chapter 11.2.)
13. If A is PD, then A^{-1} is PD.
14. Let $A \in \text{PSD}_n$ and let $C \in \mathbb{C}^{n \times m}$. Then C^*AC is PSD.
15. Let $A \in \text{PD}_n$ and let $C \in \mathbb{C}^{n \times m}$, $n \geq m$. Then C^*AC is PD if and only if $\text{rank } C = m$; i.e., if and only if C has linearly independent columns.
16. Let $A \in \text{PD}_n$ and $C \in \mathbb{C}^{n \times n}$. Then C^*AC is PD if and only if C is invertible.
17. Let $A \in \mathcal{H}_n$. Then A is PD if and only if there is an invertible $B \in \mathbb{C}^{n \times n}$ such that $A = B^*B$.
18. *Cholesky Factorization:* Let $A \in \mathcal{H}_n$. Then A is PD if and only if there is an invertible lower triangular matrix L with positive diagonal entries such that $A = LL^*$. (See Chapter 38 for information on the computation of the Cholesky factorization.)
19. Let $A \in \text{PSD}_n$ with $\text{rank } A = r < n$. Then A can be factored as $A = B^*B$ with $B \in \mathbb{C}^{r \times n}$. If A is a real matrix, then B can be taken to be real and $A = B^T B$. Equivalently, there exist vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in \mathbb{C}^r$ (or $\mathbb{R}^r$) such that $a_{ij} = \mathbf{v}_i^* \mathbf{v}_j$ (or $\mathbf{v}_i^T \mathbf{v}_j$). Note that A is the Gram matrix (see section 8.1) of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$. In particular, any rank 1 PSD matrix has the form $\mathbf{x}\mathbf{x}^*$ for some nonzero vector $\mathbf{x} \in \mathbb{C}^n$.
20. [Lax96, p. 123]; see also [HJ85, p. 407] The Gram matrix G of a set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is PSD. If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent, then G is PD.
21. [HJ85, p. 412] *Polar Form:* Let $A \in \mathbb{C}^{m \times n}$, $m \geq n$. Then A can be factored $A = UP$, where $P \in \text{PSD}_n$, $\text{rank } P = \text{rank } A$, and $U \in \mathbb{C}^{m \times n}$ has orthonormal columns. Moreover, P is uniquely determined by A and equals $(A^*A)^{1/2}$. If A is real, then P and U are real. (See also Section 17.1.)
22. [HJ85, p. 400] Any matrix $A \in \text{PD}_n$ is diagonally congruent to a correlation matrix via the diagonal matrix $D = (1/\sqrt{a_{11}}, \dots, 1/\sqrt{a_{nn}})$.

23. [BJT93] *Parameterization of Correlation Matrices in S_3* : Let $0 \leq \alpha, \beta, \gamma \leq \pi$. Then the matrix

$$C = \begin{bmatrix} 1 & \cos \alpha & \cos \gamma \\ \cos \alpha & 1 & \cos \beta \\ \cos \gamma & \cos \beta & 1 \end{bmatrix}$$

is PSD if and only if $\alpha \leq \beta + \gamma$, $\beta \leq \alpha + \gamma$, $\gamma \leq \alpha + \beta$, $\alpha + \beta + \gamma \leq 2\pi$. Furthermore, C is PD if and only if all of these inequalities are strict.

24. [HJ85, p. 472] and [Fie86, p. 55] Let $A = \begin{bmatrix} B & C \\ C^* & D \end{bmatrix} \in \mathcal{H}_n$, and assume that B is invertible. Then

A is PD if and only if the matrices B and its Schur complement $S = D - C^*B^{-1}C$ are PD.

25. [Joh92] and [LB96, pp. 93–94] Let $A = \begin{bmatrix} B & C \\ C^* & D \end{bmatrix}$ be PSD. Then any column of C lies in the span of the columns of B .

26. [HJ85, p. 465] Let $A \in \text{PD}_n$ and $B \in \mathcal{H}_n$. Then

- (a) AB is diagonalizable.
- (b) All eigenvalues of AB are real.
- (c) $\text{in}(AB) = \text{in}(B)$.

27. Any diagonalizable matrix A with real eigenvalues can be factored as $A = BC$, where B is PSD and C is Hermitian.

28. If $A, B \in \text{PD}_n$, then every eigenvalue of AB is positive.

29. [Lax96, p. 120] Let $A, B \in \mathcal{H}_n$. If A is PD and $AB + BA$ is PD, then B is PD. It is not true that if

A, B are both PD, then $AB + BA$ is PD as can be seen by the example $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$.

30. [HJ85, pp. 466–467] and [Lax96, pp. 125–126] The real valued function $f(X) = \log(\det X)$ is concave on the set PD_n ; i.e., $f((1-t)X + tY) \geq (1-t)f(X) + tf(Y)$ for all $t \in [0, 1]$ and all $X, Y \in \text{PD}_n$.

31. [Lax96, p. 129] If $A \in \text{PD}_n$ is real, $\int_{\mathbb{R}^n} e^{-\mathbf{x}^T A \mathbf{x}} d\mathbf{x} = \frac{\pi^{n/2}}{\sqrt{\det A}}$.

32. [Fie60] Let $A = [a_{ij}]$, $B = [b_{ij}] \in \text{PD}_n$, with $A^{-1} = [\alpha_{ij}]$, $B^{-1} = [\beta_{ij}]$. Then

$$\sum_{i,j=1}^n (a_{ij} - b_{ij})(\alpha_{ij} - \beta_{ij}) \leq 0,$$

with equality if and only if $A = B$.

33. [Ber73, p. 55] Consider PD_n to be a subset of $\mathbb{C}^{n^2}$ (or for real matrices of $\mathbb{R}^{n^2}$). Then the (topological) boundary of PD_n is PSD $_n$.

Examples:

1. If $A = [a]$ is 1×1 , then A is PD if and only if $a > 0$, and is PSD if and only if $a \geq 0$; so PD and PSD matrices are a generalization of positive numbers and nonnegative numbers.
2. If one attempts to define PD (or PSD) for nonsymmetric real matrices according to the usual definition, many of the facts above for (Hermitian) PD matrices no longer hold. For example,

suppose $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Then $\mathbf{x}^T A \mathbf{x} = 0$ for all $\mathbf{x} \in \mathbb{R}^2$. But $\sigma(A) = \{i, -i\}$, which does not agree

with Fact 4 above.

3. The matrix $A = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$ factors as $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 25 & 0 \\ 0 & 9 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, so $A^{1/2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$.
4. A self-adjoint linear operator on a complex inner product space V (see Section 5.3) is called **positive** if $\langle A\mathbf{x}, \mathbf{x} \rangle > 0$ for all nonzero $\mathbf{x} \in V$. For the usual inner product in $\mathbb{C}^n$ we have $\langle A\mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^* A\mathbf{x}$, in which case the definition of positive operator and positive definite matrix coincide.
5. Let $X_1, X_2, \dots, X_n$ be real-valued random variables on a probability space, each with mean zero and finite second moment. Define the matrix

$$a_{ij} = E(X_i X_j), \quad i, j \in \{1, 2, \dots, n\}.$$

The real symmetric matrix A is called the **covariance matrix** of $X_1, X_2, \dots, X_n$, and is necessarily PSD. If we let $X = (X_1, X_2, \dots, X_n)^T$, then we may abbreviate the definition to $A = E(XX^T)$.

Applications:

- [HFKLMO95, p. 181] or [MT88, p. 253] *Test for Maxima and Minima in Several Variables*: Let D be an open set in $\mathbb{R}^n$ containing the point $\mathbf{x}_0$, let $f : D \rightarrow \mathbb{R}$ be a twice continuously differentiable function on D , and assume that all first derivatives of f vanish at $\mathbf{x}_0$. Let H be the Hessian matrix of f (Example 2 of Section 8.1). Then
 - f has a relative minimum at $\mathbf{x}_0$ if $H(\mathbf{x}_0)$ is PD.
 - f has a relative maximum at $\mathbf{x}_0$ if $-H(\mathbf{x}_0)$ is PD.
 - f has a saddle point at $\mathbf{x}_0$ if $H(\mathbf{x}_0)$ is indefinite.

Otherwise, the test is inconclusive.

- Section 1.3 of the textbook [Str86] is an elementary introduction to real PD matrices emphasizing the significance of the Cholesky-like factorization LDL^T of a PD matrix. This representation is then used as a framework for many applications throughout the first three chapters of this text.
- Let A be a real matrix in PD_n . A **multivariate normal distribution** is one whose probability density function in $\mathbb{R}^n$ is given by

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det A}} e^{-\frac{1}{2} \mathbf{x}^T A^{-1} \mathbf{x}}.$$

It follows from Fact 31 above that $\int_{\mathbb{R}^n} f(\mathbf{x}) d\mathbf{x} = 1$. A **Gaussian family** $X_1, X_2, \dots, X_n$, where each X_i has mean zero, is a set of random variables that have a multivariate normal distribution. The entries of the matrix A satisfy the identity $a_{ij} = E(X_i X_j)$, so the distribution is completely determined by its covariance matrix.

8.5 Further Topics in Positive Definite Matrices

Definitions:

Let $A, B \in F^{n \times n}$, where F is a field. The **Hadamard product** or **Schur product** of A and B , denoted $A \circ B$, is the matrix in $F^{n \times n}$ whose $(i, j)^{\text{th}}$ entry is $a_{ij} b_{ij}$.

A function $f : \mathbb{R} \rightarrow \mathbb{C}$ is called **positive semidefinite** if for each $n \in \mathbb{N}$ and all $x_1, x_2, \dots, x_n \in \mathbb{R}$, the $n \times n$ matrix $[f(x_i - x_j)]$ is PSD.

Let $A, B \in \mathcal{H}_n$. We write $A \succ B$ if $A - B$ is PD, and $A \succeq B$ if $A - B$ is PSD. The partial ordering on $\mathcal{H}_n$ induced by $\succeq$ is called the **partial semidefinite ordering** or the **Loewner ordering**.

Let V be an n -dimensional inner product space over $\mathbb{C}$ or $\mathbb{R}$. A set $K \subseteq V$ is called a **cone** if

- (a) For each $\mathbf{x}, \mathbf{y} \in K$, $\mathbf{x} + \mathbf{y} \in K$.
- (b) If $\mathbf{x} \in K$ and $c \geq 0$, then $c\mathbf{x} \in K$.

A cone is frequently referred to as a convex cone. A cone K is **closed** if K is a closed subset of V , is **pointed** if $K \cap -K = \{0\}$, and is **full** if it has a nonempty interior. The set

$$K^* = \{\mathbf{y} \in V \mid \langle \mathbf{x}, \mathbf{y} \rangle \geq 0 \quad \forall \mathbf{x} \in K\}$$

is called the **dual space**.

Facts:

1. [HJ91, pp. 308–309]; also see [HJ85, p. 458] or [Lax96, pp. 124, 234] *Schur Product Theorem*: If $A, B \in \text{PSD}_n$, then so is $A \circ B$. If $A \in \text{PSD}_n$, $a_{ii} > 0$, $i = 1, \dots, n$, and $B \in \text{PD}_n$, then $A \circ B \in \text{PD}_n$. In particular, if A and B are both PD, then so is $A \circ B$.
2. [HJ85, p. 459] *Fejer's Theorem*: Let $A = [a_{ij}] \in \mathcal{H}_n$. Then A is PSD if and only if

$$\sum_{i,j=1}^n a_{ij} b_{ij} \geq 0$$

for all matrices $B \in \text{PSD}_n$.

3. [HJ91, pp. 245–246] If $A \in \text{PD}_m$ and $B \in \text{PD}_n$, then the Kronecker (tensor) product (see Section 10.4) $A \otimes B \in \text{PD}_{mn}$. If $A \in \text{PSD}_m$ and $B \in \text{PSD}_n$, then $A \otimes B \in \text{PSD}_{mn}$.
 4. [HJ85, p. 477] or [Lax96, pp. 126–127, 131–132] *Hadamard's Determinantal Inequality*: If $A \in \text{PD}_n$, then $\det A \leq \prod_{i=1}^n a_{ii}$. Equality holds if and only if A is a diagonal matrix.
 5. [FJ00, pp. 199–200] or [HJ85, p. 478] *Fischer's Determinantal Inequality*: If $A \in \text{PD}_n$ and α is any subset of $\{1, 2, \dots, n\}$, then $\det A \leq \det A[\alpha] \det A[\alpha^c]$ (where $\det A[\emptyset] = 1$). Equality occurs if and only if $A[\alpha, \alpha^c]$ is a zero matrix. (See Chapter 1.2 for the definition of $A[\alpha]$ and $A[\alpha, \beta]$.)
 6. [FJ00, pp. 199–200] or [HJ85, p. 485] *Koteljanskii's Determinantal Inequality*: Let $A \in \text{PD}_n$ and let α, β be any subsets of $\{1, 2, \dots, n\}$. Then $\det A[\alpha \cup \beta] \det A[\alpha \cap \beta] \leq \det A[\alpha] \det A[\beta]$. Note that if $\alpha \cap \beta = \emptyset$, Koteljanskii's inequality reduces to Fischer's inequality. Koteljanskii's inequality is also called the Hadamard–Fischer inequality.
- For other determinantal inequalities for PD matrices, see [FJ00] and [HJ85, §7.8].
7. [Fel71, pp. 620–623] and [Rud62, pp. 19–21] *Bochner's Theorem*: A continuous function from $\mathbb{R}$ into $\mathbb{C}$ is positive semidefinite if and only if it is the Fourier transform of a finite positive measure.
 8. [Lax96, p. 118] and [HJ85, p. 475, 470] Let $A, B, C, D \in \mathcal{H}_n$.
 - (a) If $A \prec B$ and $C \prec D$, then $A + C \prec B + D$.
 - (b) If $A \prec B$ and $B \prec C$, then $A \prec C$.
 - (c) If $A \prec B$ and $S \in \mathbb{C}^{n \times n}$ is invertible, then $S^*AS \prec S^*BS$.

The three statements obtained by replacing each occurrence of $\prec$ by $\leq$ are also valid.

9. [Lax96, pp. 118–119, 121–122] and [HJ85, pp. 471–472] Let $A, B \in \text{PD}_n$ with $A \prec B$. Then
 - (a) $A^{-1} \succ B^{-1}$.
 - (b) $A^{1/2} \prec B^{1/2}$.
 - (c) $\det A < \det B$.
 - (d) $\text{tr } A < \text{tr } B$.

If $A \leq B$, then statement (a) holds with $\succ$ replaced by $\geq$, statement (b) holds with $\prec$ replaced by $\leq$, and statements (c) and (d) hold with $<$ replaced by $\leq$.

10. [HJ85, pp. 182, 471–472] Let $A, B \in \mathcal{H}_n$ with eigenvalues $\lambda_1(A) \geq \lambda_2(A) \geq \cdots \geq \lambda_n(A)$ and $\lambda_1(B) \geq \lambda_2(B) \geq \cdots \geq \lambda_n(B)$. If $A \prec B$, then $\lambda_k(A) < \lambda_k(B)$, $k = 1, \dots, n$. If $A \preceq B$, then $\lambda_k(A) \leq \lambda_k(B)$, $k = 1, \dots, n$.
11. [HJ85, p. 474] Let A be PD and let $\alpha \subseteq \{1, 2, \dots, n\}$. Then $A^{-1}[\alpha] \succeq (A[\alpha])^{-1}$.
12. [HJ85, p. 475] If A is PD, then $A^{-1} \circ A \succeq I \succeq (A^{-1} \circ A)^{-1}$.
13. [Hal83, p. 89] If K is a cone in an inner product space V , its dual space is a closed cone and is called the **dual cone** of K . If K is a closed cone, then $(K^*)^* = K$.
14. [Ber73, pp. 49–50, 55] and [HW87, p. 82] For each pair $A, B \in \mathcal{H}_n$, define $\langle A, B \rangle = \text{tr}(AB)$.
 - (a) $\mathcal{H}_n$ is an inner product space over the real numbers with respect to $\langle \cdot, \cdot \rangle$.
 - (b) PSD_n is a closed, pointed, full cone in $\mathcal{H}_n$.
 - (c) $(\text{PSD}_n)^* = \text{PSD}_n$.

Examples:

1. The matrix $C = [\cos|i - j|] \in \mathcal{S}_n$ is PSD, as can be verified with Fact 19 of section 8.4 and the addition formula for the cosine. But a quick way to see it is to consider the measure $\mu(x) = \frac{1}{2}[\delta(x+1) + \delta(x-1)]$; i.e., $\mu(E) = 0$ if $-1, 1 \notin E$, $\mu(E) = 1$ if $-1, 1 \in E$, and $\mu(E) = 1/2$ if exactly one of $-1, 1 \in E$. Since the Fourier transform of μ is $\cos t$, if we let $x_1, x_2, \dots, x_n$ be $1, 2, \dots, n$ in the definition of positive definite function, we see immediately by Bochner's Theorem that the matrix $[\cos(i - j)] = [\cos|i - j|] = C$ is PSD. By Hadamard's determinantal inequality $\det C \leq \prod_{i=1}^n c_{ii} = 1$.
2. Since $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \prec \begin{bmatrix} 2 & 2 \\ 2 & 7 \end{bmatrix}$, taking inverses we have $\begin{bmatrix} .7 & -.2 \\ -.2 & .2 \end{bmatrix} \prec \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$.
3. The matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ is PD with inverse $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$. Then $(A[\{1, 3\}])^{-1} = \begin{bmatrix} 1.5 & -.5 \\ -.5 & .5 \end{bmatrix} \preceq \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = A^{-1}[\{1, 3\}]$. Also, $A^{-1} \circ A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 4 & -2 \\ 0 & -2 & 3 \end{bmatrix} \succeq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \succeq \frac{1}{13} \begin{bmatrix} 8 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 7 \end{bmatrix} = (A^{-1} \circ A)^{-1}$.
4. If $A \succeq B \succeq 0$, it does not follow that $A^2 \succeq B^2$. For example, if $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, then B and $A - B$ are PSD, but $A^2 - B^2$ is not.

Applications:

1. Hadamard's determinantal inequality can be used to obtain a sharp bound on the determinant of a matrix in $\mathbb{C}^{n \times n}$ if only the magnitudes of the entries are known. [HJ85, pp. 477–478] or [Lax96, p. 127].

Hadamard's Determinantal Inequality for Matrices in $\mathbb{C}^{n \times n}$: Let $B \in \mathbb{C}^{n \times n}$. Then $|\det B| \leq \prod_{i=1}^n (\sum_{j=1}^n |b_{ij}|^2)^{1/2}$ with equality holding if and only if the rows of B are orthogonal.

In the case that B is invertible, the inequality follows from Hadamard's determinantal inequality for positive definite matrices by using $A = BB^*$; if B is singular, the inequality is obvious.

The inequality can be alternatively expressed as $|\det B| \leq \prod_{i=1}^n \|\mathbf{b}_i\|_2$, where $\mathbf{b}_i$ are the rows of B . If B is a real matrix, it has the geometric meaning that among all parallelepipeds with given side lengths $\|\mathbf{b}_i\|_2$, $i = 1, \dots, n$, the one with the largest volume is rectangular.

There is a corresponding inequality in which the right-hand side is the product of the lengths of the columns of B .

2. [Fel71, pp. 620–623] A special case of Bochner’s theorem, important in probability theory, is: A continuous function ϕ is the characteristic function of a probability distribution if and only if it is positive semidefinite and $\phi(0) = 1$.
3. Understanding the cone PSD_n is important in semidefinite programming. (See Chapter 51.)

References

- [BJT93] W. Barrett, C. Johnson, and P. Tarazaga. The real positive definite completion problem for a simple cycle. *Linear Algebra and Its Applications*, 192: 3–31 (1993).
- [Ber73] A. Berman. *Cones, Matrices, and Mathematical Programming*. Springer-Verlag, Berlin, 1973.
- [Bha97] R. Bhatia. *Matrix Analysis*. Springer-Verlag, New York, 1997.
- [Bha01] R. Bhatia. Linear algebra to quantum cohomology: The story of Alfred Horn’s inequalities. *The American Mathematical Monthly*, 108 (4): 289–318, 2001.
- [FJ00] S. M. Fallat and C. R. Johnson. Determinantal inequalities: ancient history and recent advances. D. Huynh, S. Jain, and S. López-Permouth, Eds., *Algebra and Its Applications*, Contemporary Mathematics and Its Applications, American Mathematical Society, 259: 199–212, 2000.
- [Fel71] W. Feller. *An Introduction to Probability Theory and Its Applications*, 2nd ed., Vol. II. John Wiley & Sons, New York, 1996.
- [Fie60] M. Fiedler. A remark on positive definite matrices (Czech, English summary). *Casopis pro pest. mat.*, 85: 75–77, 1960.
- [Fie86] M. Fiedler. *Special Matrices and Their Applications in Numerical Mathematics*. Martinus Nijhoff Publishers, Dordrecht, The Netherlands, 1986.
- [Ful00] W. Fulton. Eigenvalues, invariant factors, highest weights, and Schubert calculus. *Bulletin of the American Mathematical Society*, 37: 209–249, 2000.
- [GR01] C. Godsil and G. Royle. *Algebraic Graph Theory*. Springer-Verlag, New York, 2001.
- [Hal83] M. Hall, Jr. *Combinatorial Theory*. John Wiley & Sons, New York, 1983.
- [HFKLMO95] K. Heuvers, W. Francis, J. Kursti, D. Lockhart, D. Mak, and G. Ortner. *Linear Algebra for Calculus*. Brooks/Cole Publishing Company, Pacific Grove, CA, 1995.
- [HJ85] R. A. Horn and C. R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 1985.
- [HJ91] R. A. Horn and C. R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, Cambridge, 1991.
- [HW87] R. Hill and S. Waters. On the cone of positive semidefinite matrices. *Linear Algebra and Its Applications*, 90: 81–88, 1987.
- [Joh92] C. R. Johnson. Personal communication.
- [Lax96] P. D. Lax. *Linear Algebra*. John Wiley & Sons, New York, 1996.
- [Lay97] D. Lay. *Linear Algebra and Its Applications*, 2nd ed., Addison-Wesley, Reading, MA., 1997.
- [LB96] M. Lundquist and W. Barrett. Rank inequalities for positive semidefinite matrices. *Linear Algebra and Its Applications*, 248: 91–100, 1996.
- [MT88] J. Marsden and A. Tromba. *Vector Calculus*, 3rd ed., W. H. Freeman and Company, New York, 1988.
- [Rud62] W. Rudin. *Fourier Analysis on Groups*. Interscience Publishers, a division of John Wiley & Sons, New York, 1962.
- [Str86] G. Strang. *Introduction to Applied Mathematics*. Wellesley-Cambridge Press, Wellesley, MA, 1986.

Nonnegative Matrices and Stochastic Matrices

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Technion

Nonnegativity is a natural property of many measured quantities (physical and virtual). Consequently, nonnegative matrices arise in modelling transformations in numerous branches of science and engineering — these include probability theory (Markov chains), population models, iterative methods in numerical analysis, economics (input–output models), epidemiology, statistical mechanics, stability analysis, and physics. This section is concerned with properties of such matrices. The theory of the subject was originated in the pioneering work of Perron and Frobenius in [Per07a, Per07b, Fro08, Fro09, and Fro12]. There have been books, chapters in books, and hundreds of papers on the subject (e.g., [BNS89], [BP94], [Gan59, Chap. XIII], [Har02] [HJ85, Chap. 8], [LT85, Chap. 15], [Min88], [Sen81], [Var62, Chap. 1]). A brief outline of proofs of the classic result of Perron and a description of several applications of the theory can be found in the survey paper [Mac00]. Generalizations of many facts reported herein to cone-invariant matrices can be found in Chapter 26.

9.1 Notation, Terminology, and Preliminaries

Definitions:

For a positive integer n , $\langle n \rangle = \{1, \dots, n\}$.

For a matrix $A \in \mathbb{C}^{m \times n}$:

A is **nonnegative** (**positive**), written $A \geq 0$ ($A > 0$), if all of A 's elements are nonnegative (positive).

A is **semipositive**, written $A \succeq 0$ if $A \geq 0$ and $A \neq 0$.

$|A|$ will denote the nonnegative matrix obtained by taking element-wise absolute values of A 's coordinates.

For a square matrix $A = [a_{ij}] \in \mathbb{C}^{n \times n}$:

The k -**eigenspace** of A at a complex number λ , denoted $N_\lambda^k(A)$, is $\ker(A - \lambda I)^k$; a **generalized eigenvector** of P at λ is a vector in $\cup_{k=0}^\infty N_\lambda^k(A)$.

The **index** of A at λ , denoted $\nu_A(\lambda)$, is the smallest integer k with $N_\lambda^k(A) = N_\lambda^{k+1}(A)$.

The **ergodicity coefficient** of A , denoted $\tau(A)$, is $\max\{|\lambda| : \lambda \in \sigma(A) \text{ and } |\lambda| \neq \rho(A)\}$ (with the maximum over the empty set defined to be 0 and $\rho(A)$ being the spectral radius of A).

A **group inverse** of a square matrix A , denoted $A^\#$, is a matrix X satisfying $AXA = A$, $XAX = X$, and $AX = XA$ (whenever there exists such an X , it is unique).

The **digraph** of A , denoted $\Gamma(A)$, is the graph with vertex-set $V(A) = \langle n \rangle$ and arc-set $E(A) = \{(i, j) : i, j \in \langle n \rangle \text{ and } a_{ij} \neq 0\}$; in particular, $i = 1, \dots, n$ are called **vertices**.

Vertex $i \in \langle n \rangle$ **has access** to vertex $j \in \langle n \rangle$, written $i \mapsto j$, if either $i = j$ or $\Gamma(A)$ contains a simple walk (path) from i to j ; we say that i and j **communicate**, written $i \sim j$, if each has access to the other.

A subset C of $\langle n \rangle$ is **final** if no vertex in C has access to a vertex not in C .

Vertex-communication is an equivalence relation. It partitions $\langle n \rangle$ into equivalence classes, called the **access equivalence classes** of A .

$\Gamma(A)$ is **strongly connected** if there is only one access equivalence class.

An access equivalence class C **has access** to an access equivalence class C' , written $C \mapsto C'$ if some, or equivalently every, vertex in C has access to some, or equivalently every, vertex in C' ; in this case we also write $i \mapsto C'$ and $C \mapsto i'$ when $i \in C$ and $i' \in C'$.

An access equivalence class C of A is **final** if its final as a subset of $\langle n \rangle$, that is, it does not have access to any access equivalence class but itself.

The **reduced digraph** of $\Gamma(A)$, denoted $R[\Gamma(A)]$, is the digraph whose vertex-set is the set of access equivalence classes of A and whose arcs are the pairs (C, C') with C and C' as distinct classes satisfying $C \mapsto C'$.

For a sequence $\{a_m\}_{m=0,1,\dots}$ of complex numbers and a complex number a :

a is a $(C, 0)$ -**limit** of $\{a_m\}_{m=0,1,\dots}$, written $\lim_{m \rightarrow \infty} a_m = a (C, 0)$, if $\lim_{m \rightarrow \infty} a_m = a$ (in the sense of a regular limit).

a is the $(C, 1)$ -**limit** of $\{a_m\}_{m=0,1,\dots}$, written $\lim_{m \rightarrow \infty} a_m = a (C, 1)$, if $\lim_{m \rightarrow \infty} m^{-1} \sum_{s=0}^{m-1} a_s = a$.

Inductively for $k = 2, 3, \dots$, a is a (C, k) -**limit** of $\{a_m\}_{m=0,1,\dots}$, written $\lim_{m \rightarrow \infty} a_m = a (C, k)$, if $\lim_{m \rightarrow \infty} m^{-1} \sum_{s=0}^{m-1} a_s = a (C, k-1)$.

For $0 \leq \beta < 1$, $\{a_m\}_{m=0,1,\dots}$ **converges geometrically to a with (geometric) rate β** if for each $\beta < \gamma < 1$, the set of real numbers $\{\frac{a_m - a}{\gamma^m} : m = 0, 1, \dots\}$ is bounded. (For simplicity, we avoid the reference of geometric convergence for (C, k) -limits.)

For a square nonnegative matrix P :

$\rho(P)$ (the spectral radius of P) is called the **Perron value** of P (see Facts 9.2–1(b) and 9.2–5(a) and 9.3–2(a)).

A **distinguished eigenvalue** of P is a (necessarily nonnegative) eigenvalue of P that is associated with a semipositive (right) eigenvector.

For more information about generalized eigenvectors, see Chapter 6.1. An example illustrating the digraph definitions is given in Figure 9.1; additional information about digraphs can be found in Chapter 29.

9.2 Irreducible Matrices

(See Chapter 27.3, Chapter 29.5, and Chapter 29.6 for additional information.)

Definitions:

A nonnegative square matrix P is **irreducible** if it is not permutation similar to any matrix having the (nontrivial) block-partition

$$\begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$$

with A and C square.

The **period** of an irreducible nonnegative square matrix P (also known as the **index of imprimitivity** of P) is the greatest common divisor of lengths of the cycles of $\Gamma(P)$, the digraph of P .

An irreducible nonnegative square matrix P is **aperiodic** if its period is 1.

Note: We exclude from further consideration the (irreducible) trivial 0 matrix of dimension 1×1 .

Facts:

Facts requiring proofs for which no specific reference is given can be found in [BP94, Chap. 2].

1. (*Positive Matrices — Perron's Theorem*) [Per07a, Per07b] Let P be a positive square matrix with spectral radius ρ and ergodicity coefficient τ .
 - (a) P is irreducible and aperiodic.
 - (b) ρ is positive and is a simple eigenvalue of P ; in particular, the index of P at ρ is 1.
 - (c) There exist positive right and left eigenvectors of P corresponding to ρ , in particular, ρ is a distinguished eigenvalue of both P and P^T .
 - (d) ρ is the only distinguished eigenvalue of P .
 - (e) ρ is the only eigenvalue λ of P with $|\lambda| = \rho$.
 - (f) If $\mathbf{x} \in \mathbb{R}^n$ satisfies $\mathbf{x} \geq 0$ and either $(\rho I - P)\mathbf{x} \geq 0$ or $(\rho I - P)\mathbf{x} \leq 0$, then $(\rho I - P)\mathbf{x} = 0$.
 - (g) If $\mathbf{v}$ and $\mathbf{w}$ are positive right and left eigenvectors of P corresponding to ρ (note that $\mathbf{w}$ is a row vector), then $\lim_{m \rightarrow \infty} (\frac{P}{\rho})^m = \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}}$ and the convergence is geometric with rate $\frac{\tau}{\rho}$.
 - (h) $Q \equiv \rho I - P$ has a group inverse; further, if $\mathbf{v}$ and $\mathbf{w}$ are positive right and left eigenvectors of P corresponding to ρ , then $Q + \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}}$ is nonsingular, $Q^\# = (Q + \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}})^{-1}(I - \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}})$, and $\frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}} = I - QQ^\#$.
 - (i) $\lim_{m \rightarrow \infty} \sum_{t=0}^{m-1} (\frac{P}{\rho})^t - m \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}} = (\rho I - P)^\#$ and the convergence is geometric with rate $\frac{\tau}{\rho}$.
2. (*Characterizing Irreducibility*) Let P be a nonnegative $n \times n$ matrix with spectral radius ρ . The following are equivalent:
 - (a) P is irreducible.
 - (b) $\sum_{s=0}^{n-1} P^s > 0$.
 - (c) $(I + P)^{n-1} > 0$.
 - (d) The digraph of P is strongly connected, i.e., P has a single access equivalence class.
 - (e) Every eigenvector of P corresponding to ρ is a scalar multiple of a positive vector.
 - (f) For some $\mu > \rho$, $\mu I - P$ is nonsingular and $(\mu I - P)^{-1} > 0$.
 - (g) For every $\mu > \rho$, $\mu I - P$ is nonsingular and $(\mu I - P)^{-1} > 0$.
3. (*Characterizing Aperiodicity*) Let P be an irreducible nonnegative $n \times n$ matrix. The following are equivalent:
 - (a) P is aperiodic.
 - (b) $P^m > 0$ for some m . (See Section 29.6.)
 - (c) $P^m > 0$ for all $m \geq n$.
4. (*The Period*) Let P be an irreducible nonnegative $n \times n$ matrix with period q .
 - (a) q is the greatest common divisor of $\{m : m \text{ is a positive integer and } (P^m)_{ii} > 0\}$ for any one, or equivalently all, $i \in \{1, \dots, n\}$.

(b) There exists a partition $C_1, \dots, C_q$ of $\{1, \dots, n\}$ such that:

- i. For $s, t = 1, \dots, q$, $P[C_s, C_t] \neq 0$ if and only if $t = s + 1$ (with $q + 1$ identified with 1); in particular, P is permutation similar to a block rectangular matrix having a representation

$$\begin{bmatrix} 0 & P[C_1, C_2] & 0 & \dots & 0 \\ 0 & 0 & P[C_2, C_3] & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & P[C_{q-1}, C_q] \\ P[C_q, C_1] & 0 & 0 & \dots & 0 \end{bmatrix}.$$

- ii. $P^q[C_s]$ is irreducible for $s = 1, \dots, q$ and $P^q[C_s, C_t] = 0$ for $s, t = 1, \dots, n$ with $s \neq t$; in particular, P^q is permutation similar to a block diagonal matrix having irreducible blocks on the diagonal.

5. (*Spectral Properties — The Perron–Frobenius Theorem*) [Fro12] Let P be an irreducible nonnegative square matrix with spectral radius ρ and period q .

- (a) ρ is positive and is a simple eigenvalue of P ; in particular, the index of P at ρ is 1.
 (b) There exist positive right and left eigenvectors of P corresponding to ρ ; in particular, ρ is a distinguished eigenvalue of both P and P^T .
 (c) ρ is the only distinguished eigenvalue of P and of P^T .
 (d) If $\mathbf{x} \in \mathbb{R}^n$ satisfies $\mathbf{x} \geq 0$ and either $(\rho I - P)\mathbf{x} \geq 0$ or $(\rho I - P)\mathbf{x} \leq 0$, then $(\rho I - P)\mathbf{x} = 0$.
 (e) The eigenvalues of P with modulus ρ are $\{\rho e^{(2\pi i)k/q} : k = 0, \dots, q-1\}$ (here, i is the complex root of -1) and each of these eigenvalues is simple. In particular, if P is aperiodic ($q = 1$), then every eigenvalue $\lambda \neq \rho$ of P satisfies $|\lambda| < \rho$.
 (f) $Q \equiv \rho I - P$ has a group inverse; further, if $\mathbf{v}$ and $\mathbf{w}$ are positive right and left eigenvectors of P corresponding to ρ , then $Q + \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}}$ is nonsingular, $Q^\# = (Q + \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}})^{-1}(I - \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}})$, and $\frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}} = I - QQ^\#$.

6. (*Convergence Properties of Powers*) Let P be an irreducible nonnegative square matrix with spectral radius ρ , index v , period q , and ergodicity coefficient τ . Also, let $\mathbf{v}$ and $\mathbf{w}$ be positive right and left eigenvectors of P corresponding to ρ and let $P^* \equiv \frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}}$.

- (a) $\lim_{m \rightarrow \infty} (\frac{P}{\rho})^m = P^*(C, 1)$.
 (b) $\lim_{m \rightarrow \infty} \frac{1}{q} \sum_{t=m}^{m+q-1} (\frac{P}{\rho})^t = P^*$ and the convergence is geometric with rate $\frac{\tau}{\rho} < 1$. In particular, if P is aperiodic ($q = 1$), then $\lim_{m \rightarrow \infty} (\frac{P}{\rho})^m = P^*$ and the convergence is geometric with rate $\frac{\tau}{\rho} < 1$.
 (c) For each $k = 0, \dots, q-1$, $\lim_{m \rightarrow \infty} (\frac{P}{\rho})^{mq+k}$ exists and the convergence of these sequences to their limit is geometric with rate $(\frac{\tau}{\rho})^q < 1$.
 (d) $\lim_{m \rightarrow \infty} \sum_{t=0}^{m-1} (\frac{P}{\rho})^t - mP^* = (I - \rho^{-1}P)^\#(C, 1)$; further, if P is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\frac{\tau}{\rho} < 1$.

7. (*Bounds on the Perron Value*) Let P be an irreducible nonnegative $n \times n$ matrix with spectral radius ρ , let μ be a nonnegative scalar, and let $\diamond \in \{<, \leq, =, \geq, >\}$. The following are equivalent:

- (a) $\rho \diamond \mu$.
 (b) There exists a vector $\mathbf{u} \gtrsim 0$ in $\mathbb{R}^n$ with $P\mathbf{u} \diamond \mu\mathbf{u}$.
 (c) There exists a vector $\mathbf{u} > 0$ in $\mathbb{R}^n$ with $P\mathbf{u} \diamond \mu\mathbf{u}$.

In particular,

$$\begin{aligned} \rho &= \max_{\mathbf{x} \gtrsim 0} \min_{\{i: x_i > 0\}} \frac{(P\mathbf{x})_i}{x_i} = \min_{\mathbf{x} \gtrsim 0} \max_{\{i: x_i > 0\}} \frac{(P\mathbf{x})_i}{x_i} \\ &= \max_{\mathbf{x} > 0} \min_i \frac{(P\mathbf{x})_i}{x_i} = \min_{\mathbf{x} > 0} \max_i \frac{(P\mathbf{x})_i}{x_i}. \end{aligned}$$

Since $\rho(P^T) = \rho(P)$, the above properties (and characterizations) of ρ can be expressed by applying the above conditions to P^T .

Consider the sets $\Omega(P) \equiv \{\mu \geq 0 : \exists \mathbf{x} \succeq 0, P\mathbf{x} \geq \mu\mathbf{x}\}$, $\Omega_1(P) \equiv \{\mu \geq 0 : \exists \mathbf{x} > 0, P\mathbf{x} \geq \mu\mathbf{x}\}$, $\Sigma(P) \equiv \{\mu \geq 0 : \exists \mathbf{x} \succeq 0, P\mathbf{x} \leq \mu\mathbf{x}\}$, $\Sigma_1(P) \equiv \{\mu \geq 0 : \exists \mathbf{x} > 0, P\mathbf{x} \leq \mu\mathbf{x}\}$; these sets were named the Collatz–Wielandt sets in [BS75], giving credit to ideas used in [Col42], [Wie50]. The above properties (and characterizations) of ρ can be expressed through maximal/minimal elements of the Collatz–Wielandt sets of P and P^T . (For further details see Chapter 26.)

8. (*Bounds on the Spectral Radius*) Let A be a complex $n \times n$ matrix and let P be an irreducible nonnegative $n \times n$ matrix such that $|A| \leq P$.

- (a) $\rho(A) \leq \rho(P)$.
- (b) [Wie50], [Sch96] $\rho(A) = \rho(P)$ if and only if there exist a complex number μ with $|\mu| = 1$ and a complex diagonal matrix D with $|D| = I$ such that $A = \mu D^{-1} P D$; in particular, in this case $|A| = P$.
- (c) If A is real and μ and $\diamond \in \{<, \leq\}$ satisfy the condition stated in 7b or 7c, then $\rho(A) < \mu$.
- (d) If A is real and μ and $\diamond \in \{\leq, =\}$ satisfy the condition stated in 7b or 7c, then $\rho(A) \leq \mu$.

9. (*Functional Inequalities*) Consider the function $\rho(\cdot)$ mapping irreducible, nonnegative $n \times n$ matrices to their spectral radius.

- (a) $\rho(\cdot)$ is strictly increasing in each element (of the domain matrices), i.e., if A and B are irreducible, nonnegative $n \times n$ matrices with $A \succeq B \geq 0$, then $\rho(A) > \rho(B)$.
- (b) [Coh78] $\rho(\cdot)$ is (jointly) convex in the diagonal elements, i.e., if A and D are $n \times n$ matrices, with D diagonal, A and $A + D$ nonnegative and irreducible and if $0 < \alpha < 1$, then $\rho[\alpha A + (1 - \alpha)(A + D)] \leq \alpha \rho(A) + (1 - \alpha)\rho(A + D)$.

For further functional inequalities that concern the spectral radius see Fact 8 of Section 9.3.

10. (*Taylor Expansion of the Perron Value*) [HRR92] The function $\rho(\cdot)$ mapping irreducible nonnegative $n \times n$ matrices $X = [x_{ij}]$ to their spectral radius is differentiable of all orders and has a converging Taylor expansion. In particular, if P is an irreducible nonnegative $n \times n$ matrix with spectral radius ρ and corresponding positive right and left eigenvectors $\mathbf{v} = [v_i]$ and $\mathbf{w} = [w_j]$, normalized so that $\mathbf{w}\mathbf{v} = 1$, and if F is an $n \times n$ matrix with $P + \epsilon F \geq 0$ for all sufficiently small positive ϵ , then $\rho(P + \epsilon F) = \sum_{k=0}^{\infty} \rho_k \epsilon^k$ with $\rho_0 = \rho$, $\rho_1 = \mathbf{w}F\mathbf{v}$, $\rho_2 = \mathbf{w}F(\rho I - P)^{\#}F\mathbf{v}$, $\rho_3 = \mathbf{w}F(\rho I - P)^{\#}(\mathbf{w}F\mathbf{v}I - F)(\rho I - P)^{\#}F\mathbf{v}$; in particular, $\frac{\partial \rho(X)}{\partial x_{ij}}|_{X=P} = w_i v_j$.

An algorithm that iteratively generates all coefficients of the above Taylor expansion is available; see [HRR92].

11. (*Bounds on the Ergodicity Coefficient*) [RT85] Let P be an irreducible nonnegative $n \times n$ matrix with spectral radius ρ , corresponding positive right eigenvector $\mathbf{v}$, and ergodicity coefficient τ ; let D be a diagonal $n \times n$ matrix with positive diagonal elements; and let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Then

$$\tau \leq \max_{\mathbf{x} \in \mathbb{R}^n, \|\mathbf{x}\| \leq 1, \mathbf{x}^T D^{-1} P D \mathbf{v} = 0} \|\mathbf{x}^T D^{-1} P D\|.$$

Examples:

1. We illustrate Fact 1 using the matrix

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{8} & \frac{1}{8} \end{bmatrix}.$$

The eigenvalues of P are $1, \frac{1}{48}(-3 - \sqrt{33}), \frac{1}{48}(-3 + \sqrt{33})$, so $\rho(A) = 1$. Also, $\mathbf{v} = [1, 1, 1]^T$ and $\mathbf{w} = [57, 18, 32]$ are positive right and left eigenvectors, respectively, corresponding to eigenvalue

1 and

$$\frac{\mathbf{v}\mathbf{w}}{\mathbf{w}\mathbf{v}} = \begin{bmatrix} \frac{57}{107} & \frac{18}{107} & \frac{32}{107} \\ \frac{57}{107} & \frac{18}{107} & \frac{32}{107} \\ \frac{57}{107} & \frac{18}{107} & \frac{32}{107} \end{bmatrix}.$$

2. We illustrate parts of Facts 5 and 6 using the matrix

$$P \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The spectral radius of P is 1 with corresponding right and left eigenvectors $\mathbf{v} = (1, 1)^T$ and $\mathbf{w} = (1, 1)$, respectively, the period of P is 2, and $(I - P)^\# = \frac{I-P}{4}$. Evidently,

$$P^m = \begin{cases} I & \text{if } m \text{ is even} \\ P & \text{if } m \text{ is odd.} \end{cases}$$

In particular,

$$\lim_{m \rightarrow \infty} P^{2m} = I \quad \text{and} \quad \lim_{m \rightarrow \infty} P^{2m+1} = P$$

and

$$\frac{1}{2}[P^m + P^{m+1}] = \frac{I + P}{2} = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} = \mathbf{v}(\mathbf{w}\mathbf{v})^{-1}\mathbf{w} \quad \text{for each } m = 0, 1, \dots,$$

assuring that, trivially,

$$\lim_{m \rightarrow \infty} \frac{1}{2} \sum_{t=m}^{m+1} P^t = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} = \mathbf{v}(\mathbf{w}\mathbf{v})^{-1}\mathbf{w}.$$

In this example, $\tau(P)$ is 0 (as the maximum over the empty set) and the convergence of the above sequences is geometric with rate 0. Finally,

$$\sum_{t=0}^{m-1} P^t = \begin{cases} \frac{m(I+P)}{2} & \text{if } m \text{ is even} \\ \frac{m(I+P)}{2} + \frac{I-P}{2} & \text{if } m \text{ is odd,} \end{cases}$$

implying that

$$\lim_{m \rightarrow \infty} P^m = \frac{(I + P)}{2} = \mathbf{v}(\mathbf{w}\mathbf{v})^{-1}\mathbf{w} \quad (\text{C},1)$$

and

$$\lim_{m \rightarrow \infty} \sum_{t=0}^{m-1} P^t - m \left(\frac{I + P}{2} \right) = \frac{I - P}{4} \quad (\text{C},1).$$

3. We illustrate parts of Fact 10 using the matrix P of Example 2 and $F \equiv \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $\rho(P + \epsilon F) =$

$$\sqrt{1 + \epsilon} = 1 + \frac{1}{2}\epsilon + \sum_{k=2}^{\infty} \epsilon^k \frac{(-1)^{k+1}}{k2^{k-2}} \binom{2k-3}{k-2}.$$

4. [RT85, Theorem 4.1] and [Hof67] With $\|\cdot\|$ as the 1-norm on $\mathbb{R}^n$ and $d_1, \dots, d_n$ as the (positive) diagonal elements of D , the bound in Fact 11 on the coefficient of ergodicity $\tau(P)$ of P becomes

$$\max_{r,s=1,\dots,n, r \neq s} \frac{1}{d_s v_r + d_r v_s} \left(\sum_{k=1}^n d_k |v_s P_{rk} - v_r P_{sk}| \right).$$

With $D = I$, a relaxation of this bound on $\tau(P)$ yields the expression

$$\leq \min \left\{ \rho - \sum_{j=1}^n \min_i \left(\frac{P_{ij} v_j}{v_i} \right), \sum_{j=1}^n \max_i \left(\frac{P_{ij} v_j}{v_i} \right) - \rho \right\}.$$

5. [RT85, Theorem 4.3] For a positive vector $\mathbf{u} \in \mathbb{R}^n$, consider the function $M^{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}$ defined for $\mathbf{a} \in \mathbb{R}^n$ by

$$M^{\mathbf{u}}(\mathbf{a}) = \max\{\mathbf{x}^T \mathbf{a} : \mathbf{x} \in \mathbb{R}^n, \|\mathbf{x}\| \leq 1, \mathbf{x}^T \mathbf{u} = 0\}.$$

This function has a simple explicit representation obtained by sorting the ratios $\frac{a_i}{u_j}$, i.e., identifying a permutation $j(1), \dots, j(n)$ of $1, \dots, n$ such that

$$\frac{a_{j(1)}}{u_{j(1)}} \leq \frac{a_{j(2)}}{u_{j(2)}} \leq \dots \leq \frac{a_{j(n)}}{u_{j(n)}}.$$

With k^* as the smallest integer in $\{1, \dots, n\}$ such that $2 \sum_{p=1}^{k^*} u_{j(p)} > \sum_{t=1}^n u_t$ and

$$\mu \equiv 1 + \left(\sum_{t=1}^n u_t - 2 \sum_{p=1}^{k^*} u_{j(p)} \right),$$

we have that

$$M^{\mathbf{u}}(\mathbf{a}) = \sum_{p=1}^{k^*-1} a_{j(p)} + \mu a_{j(k^*)} - \sum_{p=k^*+1}^n a_{j(p)}.$$

With $\|\cdot\|$ as the ∞ -norm on $\mathbb{R}^n$ and $(D^{-1}PD)_1, \dots, (D^{-1}PD)_n$ as the columns of $D^{-1}PD$, the bound in Fact 11 on the coefficient of ergodicity $\tau(P)$ of P becomes

$$\max_{r=1, \dots, n} M^{D^{-1}\mathbf{w}}[(D^{-1}PD)_r].$$

9.3 Reducible Matrices

Definitions:

For a nonnegative $n \times n$ matrix P with spectral radius ρ :

A **basic class** of P is an access equivalence class B of P with $\rho(P[B]) = \rho$.

The **period** of an access equivalence class C of P (also known as the **index of imprimitivity** of C) is the period of the (irreducible) matrix $P[C]$.

The **period** of P (also known as the **index of imprimitivity** of P) is the least common multiple of the periods of its basic classes.

P is **aperiodic** if its period is 1.

The **index** of P , denoted v_P , is $v_P(\rho)$.

The **co-index** of P , denoted $\bar{v}_P$, is $\max\{v_P(\lambda) : \lambda \in \sigma(P), |\lambda| = \rho \text{ and } \lambda \neq \rho\}$ (with the maximum over the empty set defined as 0).

The **basic reduced digraph** of P , denoted $R^*(P)$, is the digraph whose vertex-set is the set of basic classes of P and whose arcs are the pairs (B, B') of distinct basic classes of P for which there exists a simple walk in $R[\Gamma(P)]$ from B to B' .

The **height** of a basic class is the largest number of vertices on a simple walk in $R^*(P)$ which ends at B .

The **principal submatrix** of P at a distinguished eigenvalue λ , denoted $P[\lambda]$, is the principal submatrix of P corresponding to a set of vertices of $\Gamma(P)$ having no access to a vertex of an access equivalence class C that satisfies $\rho(P[C]) > \lambda$.

P is **convergent** or **transient** if $\lim_{m \rightarrow \infty} P^m = 0$.

P is **semiconvergent** if $\lim_{m \rightarrow \infty} P^m$ exists.

P is **weakly expanding** if $P\mathbf{u} \geq \mathbf{u}$ for some $\mathbf{u} > 0$.

P is **expanding** if for some $P\mathbf{u} > \mathbf{u}$ for some $\mathbf{u} > 0$.

An $n \times n$ **matrix polynomial of degree d** in the (integer) variable m is a polynomial in m with coefficients that are $n \times n$ matrices (expressible as $S(m) = \sum_{t=0}^d m^t B_t$ with $B_1, \dots, B_d$ as $n \times n$ matrices and $B_d \neq 0$).

Facts:

Facts requiring proofs for which no specific reference is given can be found in [BP94, Chap. 2].

1. The set of basic classes of a nonnegative matrix is always nonempty.
2. (*Spectral Properties of the Perron Value*) Let P be a nonnegative $n \times n$ matrix with spectral radius ρ and index ν .
 - (a) [Fro12] ρ is an eigenvalue of P .
 - (b) [Fro12] There exist semipositive right and left eigenvectors of P corresponding to ρ , i.e., ρ is a distinguished eigenvalue of both P and P^T .
 - (c) [Rot75] ν is the largest number of vertices on a simple walk in $R^*(P)$.
 - (d) [Rot75] For each basic class B having height h , there exists a generalized eigenvector $\mathbf{v}^B$ in $N_\rho^h(P)$, with $(\mathbf{v}^B)_i > 0$ if $i \mapsto B$ and $(\mathbf{v}^B)_i = 0$ otherwise.
 - (e) [Rot75] The dimension of $N_\rho^\nu(P)$ is the number of basic classes of P . Further, if $B_1, \dots, B_p$ are the basic classes of P and $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_p}$ are generalized eigenvectors of P at ρ that satisfy the conclusions of Fact 2(d) with respect to $B_1, \dots, B_p$, respectively, then $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_p}$ form a basis of $N_\rho^\nu(P)$.
 - (f) [RiSc78, Sch86] If $B_1, \dots, B_p$ is an enumeration of the basic classes of P with nondecreasing heights (in particular, $s < t$ assures that we do not have $B_t \mapsto B_s$), then there exist generalized eigenvectors $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_p}$ of P at ρ that satisfy the assumptions and conclusions of Fact 2(e) and a nonnegative $p \times p$ upper triangular matrix M with all diagonal elements equal to ρ , such that

$$P[\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_p}] = [\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_p}]M$$

(in particular, $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_p}$ is a basis of $N_\rho^\nu(P)$). Relationships between the matrix M and the Jordan Canonical Form of P are beyond the scope of the current review; see [Sch56], [Sch86], [HS89], [HS91a], [HS91b], [HRS89], and [NS94].

- (g) [Vic85], [Sch86], [Tam04] If $B_1, \dots, B_r$ are the basic classes of P having height 1 and $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_r}$ are generalized eigenvectors of P at ρ that satisfy the conclusions of Fact 2(d) with respect to $B_1, \dots, B_r$, respectively, then $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_r}$ are linearly independent, nonnegative eigenvectors of P at ρ that span the cone $(\mathbb{R}_0^+)^n \cap N_\rho^1(P)$; that is, each vector in the cone $(\mathbb{R}_0^+)^n \cap N_\rho^1(P)$ is a linear combination with nonnegative coefficients of $\mathbf{v}^{B_1}, \dots, \mathbf{v}^{B_r}$ (in fact, the sets $\{\alpha \mathbf{v}^{B_s} : \alpha \geq 0\}$ for $s = 1, \dots, r$ are the extreme rays of the cone $(\mathbb{R}_0^+)^n \cap N_\rho^1(P)$).
3. (*Spectral Properties of Eigenvalues $\lambda \neq \rho(P)$ with $|\lambda| = \rho(P)$*) Let P be a nonnegative $n \times n$ matrix with spectral radius ρ , index ν , co-index $\bar{\nu}$, period q , and coefficient of ergodicity τ .
 - (a) [Rot81a] The following are equivalent:
 - i. $\{\lambda \in \sigma(P) \setminus \{\rho\} : |\lambda| = \rho\} = \emptyset$.
 - ii. $\bar{\nu} = 0$.
 - iii. P is aperiodic ($q = 1$).

- (b) [Rot81a] If $\lambda \in \sigma(P) \setminus \{\rho\}$ and $|\lambda| = \rho$, then $(\frac{\lambda}{\rho})^h = 1$ for some $h \in \{2, \dots, n\}$; further, $q = \min\{h = 2, \dots, n : (\frac{\lambda}{\rho})^h = 1 \text{ for each } \lambda \in \sigma(P) \setminus \{\rho\} \text{ with } |\lambda| = \rho\} \leq n$ (here the minimum over the empty set is taken to be 1).
- (c) [Rot80] If $\lambda \in \sigma(P) \setminus \{\rho\}$ and $|\lambda| = \rho$, then $v_P(\lambda)$ is bounded by the largest number of vertices on a simple walk in $R^*(P)$ with each vertex corresponding to a (basic) access equivalence class C that has $\lambda \in \sigma(P[C])$; in particular, $\bar{v} \leq v$.
4. (*Distinguished Eigenvalues*) Let P be a nonnegative $n \times n$ matrix.

- (a) [Vic85] λ is a distinguished eigenvalue of P if and only if there is a final set C with $\rho(P[C]) = \lambda$.

It is noted that the set of distinguished eigenvalues of P and P^T need not coincide (and the above characterization of distinguished eigenvalues is not invariant of the application of the transpose operator). (See Example 1 below.)

- (b) [HS88b] If λ is a distinguished eigenvalue, $v_P(\lambda)$ is the largest number of vertices on a simple walk in $R^*(P[\lambda])$.
- (c) [HS88b] If $\mu > 0$, then $\mu \leq \min\{\lambda : \lambda \text{ is a distinguished eigenvalue of } P\}$ if and only if there exists a vector $\mathbf{u} > 0$ with $P\mathbf{u} \geq \mu\mathbf{u}$.

(For additional characterizations of the minimal distinguished eigenvalue, see the concluding remarks of Facts 12(h) and 12(i).)

Additional properties of distinguished eigenvalues λ of P that depend on $P[\lambda]$ can be found in [HS88b] and [Tam04].

5. (*Convergence Properties of Powers*) Let P be a nonnegative $n \times n$ matrix with positive spectral radius ρ , index v , co-index $\bar{v}$, period q , and coefficient of ergodicity τ (for the case where $\rho = 0$, see Fact 12(j) below).
- (a) [Rot81a] There exists an $n \times n$ matrix polynomial $S(m)$ of degree $v - 1$ in the (integer) variable m such that $\lim_{m \rightarrow \infty} [(\frac{P}{\rho})^m - S(m)] = 0$ (C, p) for every $p \geq \bar{v}$; further, if P is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\frac{\tau}{\rho} < 1$.
- (b) [Rot81a] There exist matrix polynomials $S^0(m), \dots, S^{q-1}(m)$ of degree $v - 1$ in the (integer) variable m , such that for each $k = 0, \dots, q - 1$, $\lim_{m \rightarrow \infty} [(\frac{P}{\rho})^{mq+k} - S^k(m)] = 0$ and the convergence of these sequences to their limit is geometric with rate $(\frac{\tau}{\rho})^q < 1$.
- (c) [Rot81a] There exists a matrix polynomial $T(m)$ of degree v in the (integer) variable m with $\lim_{m \rightarrow \infty} [\sum_{s=0}^{m-1} (\frac{P}{\rho})^s - T(m)] = 0$ (C, p) for every $p \geq \bar{v}$; further, if P is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\frac{\tau}{\rho} < 1$.
- (d) [FrSc80] The limit of $\frac{p^m}{\rho^m m^{v-1}} [I + \frac{P}{\rho} + \dots + (\frac{P}{\rho})^{q-1}]$ exists and is semipositive.
- (e) [Rot81b] Let $\mathbf{x} = [x_i]$ be a nonnegative vector in $\mathbb{R}^n$ and let $i \in \langle n \rangle$. With $K(i, \mathbf{x}) \equiv \{j \in \langle n \rangle : j \mapsto i\} \cap \{j \in \langle n \rangle : u \mapsto j \text{ for some } u \in \langle n \rangle \text{ with } x_u > 0\}$,

$$r(i|\mathbf{x}, P) \equiv \inf\{\alpha > 0 : \lim_{m \rightarrow \infty} \alpha^{-m} (P^m \mathbf{x})_i = 0\} = \rho(P[K(i, \mathbf{x})])$$

and if $r \equiv r(i|\mathbf{x}, P) > 0$,

$$k(i|\mathbf{x}, P) \equiv \inf\{k = 0, 1, \dots : \lim_{m \rightarrow \infty} m^{-k} r^{-m} (P^m \mathbf{x})_i = 0\} = v_{P[K(i, \mathbf{x})]}(r).$$

Explicit expressions for the polynomials mentioned in Facts 5(a) to 5(d) in terms of characteristics of the underlying matrix P are available in Fact 12(a)ii for the case where $v = 1$ and in [Rot81a] for the general case. In fact, [Rot81a] provides (explicit) polynomial approximations of additional high-order partial sums of normalized powers of nonnegative matrices.

6. (*Bounds on the Perron Value*) Let P be a nonnegative $n \times n$ matrix with spectral radius ρ and let μ be a nonnegative scalar.

- (a) For $\diamond \in \{<, \leq, =, \geq, >\}$,

$$[P\mathbf{u} \diamond \mu\mathbf{u} \text{ for some vector } \mathbf{u} > 0] \Rightarrow [\rho \diamond \mu];$$

further, the inverse implication holds for $\diamond$ as $<$, implying that

$$\rho = \max_{\mathbf{x} \geq 0} \min_{\{i: x_i > 0\}} \frac{(A\mathbf{x})_i}{x_i}.$$

- (b) For $\diamond \in \{\leq, \leq, =, \geq, \geq\}$,

$$[\rho \diamond \mu] \Rightarrow [P\mathbf{u} \diamond \mu\mathbf{u} \text{ for some vector } \mathbf{u} \geq 0];$$

further, the inverse implication holds for $\diamond$ as $\geq$.

- (c) $\rho < \mu$ if and only if $P\mathbf{u} < \rho\mathbf{u}$ for some vector $\mathbf{u} \geq 0$.

Since $\rho(P^T) = \rho(P)$, the above properties (and characterizations) of ρ can be expressed by applying the above conditions to P^T . (See Example 3 below.)

Some of the above results can be expressed in terms of the Collatz–Wielandt sets. (See Fact 7 of Section 9.2 and Chapter 26.)

7. (*Bounds on the Spectral Radius*) Let P be a nonnegative $n \times n$ matrix and let A be a complex $n \times n$ matrix such that $|A| \leq P$. Then $\rho(A) \leq \rho(P)$.
8. (*Functional Inequalities*) Consider the function $\rho(\cdot)$ mapping nonnegative $n \times n$ matrices to their spectral radius.
 - (a) $\rho(\cdot)$ is nondecreasing in each element (of the domain matrices); that is, if A and B are nonnegative $n \times n$ matrices with $A \geq B \geq 0$, then $\rho(A) \geq \rho(B)$.
 - (b) [Coh78] $\rho(\cdot)$ is (jointly) convex in the diagonal elements; that is, if A and D are $n \times n$ matrices, with D diagonal, A and $A + D$ nonnegative, and if $0 < \alpha < 1$, then $\rho[\alpha A + (1 - \alpha)(A + D)] \leq \alpha \rho(A) + (1 - \alpha)\rho(A + D)$.
 - (c) [EJD88] If $A = [a_{ij}]$ and $B = [b_{ij}]$ are nonnegative $n \times n$ matrices, $0 < \alpha < 1$ and $C = [c_{ij}]$ with $c_{ij} = a_{ij}^\alpha b_{ij}^{1-\alpha}$ for each $i, j = 1, \dots, n$, then $\rho(C) \leq \rho(A)^\alpha \rho(B)^{1-\alpha}$.

Further functional inequalities about $\rho(\cdot)$ can be found in [EJD88] and [EHP90].

9. (*Resolvent Expansions*) Let P be a nonnegative square matrix with spectral radius ρ and let $\mu > \rho$. Then $\mu I - P$ is invertible and

$$(\mu I - P)^{-1} = \sum_{t=0}^{\infty} \frac{P^t}{\mu^{t+1}} \geq \frac{I}{\mu} + \frac{P}{\mu^2} \geq \frac{I}{\mu} \geq 0$$

(the invertibility of $\mu I - P$ and the power series expansion of its inverse do not require nonnegativity of P).

For explicit expansions of the resolvent about the spectral radius, that is, for explicit power series representations of $[(z + \rho)I - P]^{-1}$ with $|z|$ positive and sufficiently small, see [Rot81c], and [HNR90] (the latter uses such expansions to prove Perron–Frobenius-type spectral results for nonnegative matrices).

10. (*Puiseux Expansions of the Perron Value*) [ERS95] The function $\rho(\cdot)$ mapping irreducible nonnegative $n \times n$ matrices $X = [x_{ij}]$ to their spectral radius has a converging Puiseux (fractional power series) expansion at each point; i.e., if P is a nonnegative $n \times n$ matrix and if F is an $n \times n$ matrix with $P + \epsilon F \geq 0$ for all sufficiently small positive ϵ , then $\rho(P + \epsilon F)$ has a representation $\sum_{k=0}^{\infty} \rho_k \epsilon^{k/q}$ with $\rho_0 = \rho(P)$ and q as a positive integer.
11. (*Bounds on the Ergodicity Coefficient*) [RT85, extension of Theorem 3.1] Let P be a nonnegative $n \times n$ matrix with spectral radius ρ , corresponding semipositive right eigenvector $\mathbf{v}$, and ergodicity

coefficient τ , let D be a diagonal $n \times n$ matrix with positive diagonal elements, and let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Then

$$\tau \leq \max_{\mathbf{x} \in \mathbb{R}^n, \|\mathbf{x}\| \leq 1, \mathbf{x}^T D^{-1} P D = 0} \|\mathbf{x}^T D^{-1} P D\|.$$

12. (*Special Cases*) Let P be a nonnegative $n \times n$ matrix with spectral radius ρ , index ν , and period q .

(a) (*Index 1*) Suppose $\nu = 1$.

- i. $\rho I - P$ has a group inverse.
- ii. [Rot81a] With $P^\star \equiv I - (\rho I - P)(\rho I - P)^\#$, all of the convergence properties stated in Fact 6 of Section 9.2 apply.
- iii. If $\rho > 0$, then $\frac{P^m}{\rho^m}$ is bounded in m (element-wise).
- iv. $\rho = 0$ if and only if $P = 0$.

(b) (*Positive eigenvector*) The following are equivalent:

- i. P has a positive right eigenvector corresponding to ρ .
- ii. The final classes of P are precisely its basic classes.
- iii. There is no vector $\mathbf{w}$ satisfying $\mathbf{w}^T P \leq \rho \mathbf{w}^T$.

Further, when the above conditions hold:

- i. $\nu = 1$ and the conclusions of Fact 12(a) hold.
- ii. If P satisfies the above conditions and $P \neq 0$, then $\rho > 0$ and there exists a diagonal matrix D having positive diagonal elements such that $S \equiv \frac{1}{\rho} D^{-1} P D$ is stochastic (that is, $S \geq 0$ and $S\mathbf{1} = \mathbf{1}$; see Chapter 4).

(c) [Sch53] There exists a vector $\mathbf{x} > 0$ with $P\mathbf{x} \leq \rho\mathbf{x}$ if and only if every basic class of P is final.

(d) (*Positive generalized eigenvector*) [Rot75], [Sch86], [HS88a] The following are equivalent:

- i. P has a positive right generalized eigenvector at ρ .
- ii. Each final class of P is basic.
- iii. $P\mathbf{u} \geq \rho\mathbf{u}$ for some $\mathbf{u} > 0$.
- iv. Every vector $\mathbf{w} \geq 0$ with $\mathbf{w}^T P \leq \rho \mathbf{w}^T$ must satisfy $\mathbf{w}^T P = \rho \mathbf{w}^T$.
- v. ρ is the only distinguished eigenvalue of P .

(e) (*Convergent/Transient*) The following are equivalent:

- i. P is convergent.
- ii. $\rho < 1$.
- iii. $I - P$ is invertible and $(I - P)^{-1} \geq 0$.
- iv. There exists a positive vector $\mathbf{u} \in R^n$ with $P\mathbf{u} < \mathbf{u}$.

Further, when the above conditions hold, $(I - P)^{-1} = \sum_{t=0}^{\infty} P^t \geq I$.

(f) (*Semiconvergent*) The following are equivalent:

- i. P is semiconvergent.
- ii. Either $\rho < 1$ or $\rho = \nu = 1$ and 1 is the only eigenvalue λ of P with $|\lambda| = 1$.

(g) (*Bounded*) P^m is bounded in m (element-wise) if and only if either $\rho < 1$ or $\rho = 1$ and $\nu=1$.

(h) (*Weakly Expanding*) [HS88a], [TW89] [DR05] The following are equivalent:

- i. P is weakly expanding.
- ii. There is no vector $\mathbf{w} \in R^n$ with $\mathbf{w} \geq 0$ and $\mathbf{w}^T P \leq \mathbf{w}^T$.
- iii. Every distinguished eigenvalue λ of P satisfies $\lambda \geq 1$.

- iv. Every final class C of P has $\rho(P[C]) \geq 1$.
- v. If C is a final set of P , then $\rho(P[C]) \geq 1$.

Given $\mu > 0$, the application of the above equivalence to $\frac{P}{\mu}$ yields characterizations of instances where each distinguished eigenvalue of P is bigger than or equal to μ .

(i) (*Expanding*) [HS88a], [TW89] [DR05] The following are equivalent:

- i. P is expanding.
- ii. There exists a vector $\mathbf{u} \in R^n$ with $\mathbf{u} \geq 0$ and $P\mathbf{u} > \mathbf{u}$.
- iii. There is no vector $\mathbf{w} \in R^n$ with $\mathbf{w} \succeq 0$ and $w^T P \leq w^T$.
- iv. Every distinguished eigenvalue λ of P satisfies $\lambda > 1$.
- v. Every final class C of P has $\rho(P[C]) > 1$.
- vi. If C is a final set of P , then $\rho(P[C]) > 1$.

Given $\mu > 0$, the application of the above equivalence to $\frac{P}{\mu}$ yields characterizations of instances where each distinguished eigenvalue of P is bigger than μ .

(j) (*Nilpotent*) The following are equivalent conditions:

- i. P is nilpotent; that is, $P^m = 0$ for some positive integer m .
- ii. P is permutation similar to an upper triangular matrix all of whose diagonal elements are 0.
- iii. $\rho = 0$.
- iv. $P^n = 0$.
- v. $P^\nu = 0$.

(k) (*Symmetric*) Suppose P is symmetric.

- i. $\rho = \max_{\mathbf{u} \succeq 0} \frac{\mathbf{u}^T P \mathbf{u}}{\mathbf{u}^T \mathbf{u}}$.
- ii. $\rho = \frac{\mathbf{u}^T P \mathbf{u}}{\mathbf{u}^T \mathbf{u}}$ for $\mathbf{u} \succeq 0$ if and only if $\mathbf{u}$ is an eigenvector of P corresponding to ρ .
- iii. [CHR97, Theorem 1] For $\mathbf{u}, \mathbf{w} \succeq 0$ with $w_i = \sqrt{u_i(Pu)_i}$ for $i = 1, \dots, n$, $\frac{\mathbf{u}^T P \mathbf{u}}{\mathbf{u}^T \mathbf{u}} \leq \frac{\mathbf{w}^T P \mathbf{w}}{\mathbf{w}^T \mathbf{w}}$ and equality holds if and only if $\mathbf{u}[\mathcal{S}]$ is an eigenvector of $P[\mathcal{S}]$ corresponding to ρ , where $\mathcal{S} \equiv \{i : u_i > 0\}$.

Examples:

1. We illustrate parts of Fact 2 using the matrix

$$P = \begin{bmatrix} 2 & 2 & 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The eigenvalues of P are 2, 1, and 0; so, $\rho(P) = 2 \in \sigma(P)$ as is implied by Fact 2(a). The vectors $\mathbf{v} = [1, 0, 0, 0, 0, 0]^T$ and $\mathbf{w} = [0, 0, 0, 1, 1, 1]$ are semipositive right and left eigenvectors corresponding to the eigenvalue 2; their existence is implied by Fact 2(b).

The basic classes are $B_1 = \{1\}$, $B_1 = \{2\}$ and $B_3 = \{4, 5\}$. The digraph corresponding to P , its reduced digraph, and the basic reduced digraph of P are illustrated in Figure 9.1. From Figure 9.1(c), the largest number of vertices in a simple walk in the basic reduced digraph of P is 2 (going from B_1 to either B_2 or B_3); hence, Fact 2(c) implies that $\nu_P(2) = 2$. The height of basic class B_1 is 1 and the height of basic classes B_2 and B_3 is 2. Semipositive generalized eigenvectors of P at (the eigenvalue)

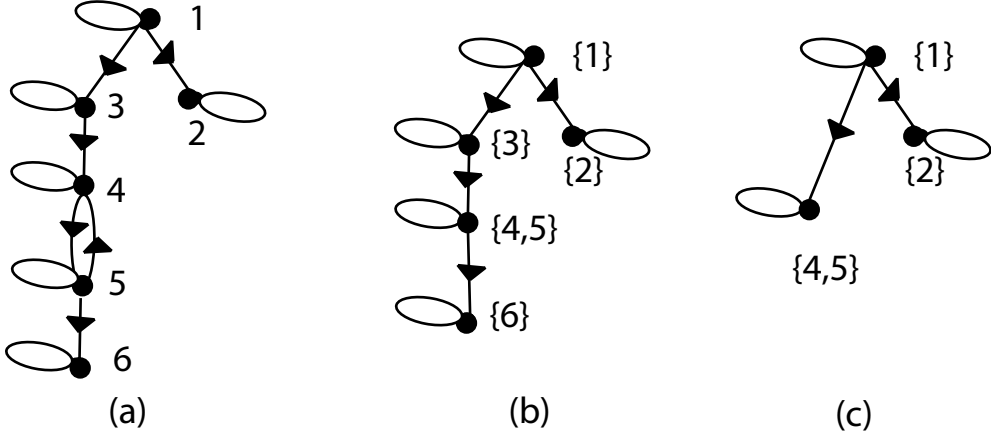


FIGURE 9.1 (a) The digraph $\Gamma(P)$, (b) reduced digraph $R[\Gamma(P)]$, and (c) basic reduced digraph $R^*(P)$.

2 that satisfy the assumptions of Fact 2(f) are $\mathbf{u}^{B_1} = [1, 0, 0, 0, 0, 0]^T$, $\mathbf{u}^{B_2} = [1, 1, 0, 0, 0, 0]^T$, and $\mathbf{u}^{B_3} = [1, 0, 2, 1, 1, 0]^T$. The implied equality

$$P[\mathbf{u}^{B_1}, \dots, \mathbf{u}^{B_p}] = [\mathbf{u}^{B_1}, \dots, \mathbf{u}^{B_p}]M$$

of Fact 2(f) holds as

$$\begin{bmatrix} 2 & 2 & 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

In particular, Fact 2(e) implies that $\mathbf{u}^{B_1}, \mathbf{u}^{B_2}, \mathbf{u}^{B_3}$ form a basis of $N_{\rho(P)}^{v(P)} = N_2^2$. We note that while there is only a single basic class of height 1, $\dim[N_{\rho}^1(P)] = 2$ and $\mathbf{u}^{B_1}, 2\mathbf{u}^{B_2} - \mathbf{u}^{B_3} = [-1, 2, -2, -1, -1, 0]^T$ form a basis of $N_{\rho}^1(P)$. Still, Fact 2(g) assures that $(\mathbb{R}_0^+)^n \cap N_{\rho}^1(P)$ is the cone $\{\alpha \mathbf{u}^{B_1} : \alpha \geq 0\}$ (consisting of its single ray).

Fact 4(a) and Figure 9.1 imply that the distinguished eigenvalues of P are 1 and 2, while 2 is the only distinguished eigenvalue of P^T .

2. Let $H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; properties of H were demonstrated in Example 2 of section 9.2. We will demonstrate Facts 2(c), 5(b), and 5(a) on the matrix

$$P \equiv \begin{bmatrix} H & I \\ 0 & H \end{bmatrix}.$$

The spectral radius of P is 1 and its basic classes of P are $B_1 = \{1, 2\}$ and $B_2 = \{3, 4\}$ with B_1 having access to B_2 . Thus, the index of 1 with respect to P , as the largest number of vertices on a walk of the marked reduced graph of P , is 2 (Fact 2(c)). Also, as the period of each of the two basic

classes of P is 2, the period of P is 2. To verify the convergence properties of P , note that

$$P^m = \begin{cases} \begin{bmatrix} I & mH \\ 0 & I \end{bmatrix} & \text{if } m \text{ is even} \\ \begin{bmatrix} H & mI \\ 0 & H \end{bmatrix} & \text{if } m \text{ is odd,} \end{cases}$$

immediately providing matrix-polynomials $S^0(m)$ and $S^1(m)$ of degree 1 such that $\lim_{m \rightarrow \infty} P^{2m} - S^0(m) = 0$ and $\lim_{m \rightarrow \infty} P^{2m+1} - S^1(m) = 0$. In this example, $\tau(P)$ is 0 (as the maximum over the empty set) and the convergence of the above sequences is geometric with rate 0.

The above representation of P^m shows that

$$P^m = \begin{bmatrix} H^m & mH^{m+1} \\ 0 & H^m \end{bmatrix}$$

and Example 2 of section 9.2 shows that

$$\lim_{m \rightarrow \infty} H^m = \frac{I + H}{2} = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} \quad (\text{C},1).$$

We next consider the upper-right blocks of P^m . We observe that

$$\begin{aligned} \frac{1}{m} \sum_{t=0}^{m-1} P^t[B_1, B_2] &= \begin{cases} \frac{mI}{4} + \frac{(m-2)H}{4} & \text{if } m \text{ is even} \\ \frac{(m-1)^2 I}{4m} + \frac{(m^2-1)H}{4m} & \text{if } m \text{ is odd,} \end{cases} \\ &= \begin{cases} \frac{m(I+H)}{4} - \frac{H}{2} & \text{if } m \text{ is even} \\ \frac{m(I+H)}{4} - \frac{I}{2} + \frac{I-H}{4m} & \text{if } m \text{ is odd,} \end{cases} \end{aligned}$$

implying that

$$\lim_{m \rightarrow \infty} \frac{1}{m} \sum_{t=0}^{m-1} P^t[B_1, B_2] - m \left(\frac{I+H}{4} \right) + \frac{I+H}{4} = 0 \quad (\text{C},1).$$

As $m-1 = \frac{1}{m} \sum_{t=0}^{m-1} t$ for each $m = 1, 2, \dots$, the above shows that

$$\lim_{m \rightarrow \infty} \frac{1}{m} \sum_{t=0}^{m-1} \left\{ P^t[B_1, B_2] - t \left(\frac{I+H}{4} \right) \right\} = 0 \quad (\text{C},1),$$

and, therefore (recalling that (C,1)-convergence implies (C,2)-convergence),

$$\lim_{m \rightarrow \infty} \left\{ P^m - \begin{bmatrix} .5 & .5 & -.25m & -.25m \\ .5 & .5 & -.25m & -.25m \\ 0 & 0 & .5 & .5 \\ 0 & 0 & .5 & .5 \end{bmatrix} \right\} = 0 \quad (\text{C},2).$$

3. Fact 6 implies many equivalencies, in particular, as the spectral radius of a matrix equals that of its transpose. For example, for a nonnegative $n \times n$ matrix P with spectral radius ρ and nonnegative scalar μ , the following are equivalent:

- (a) $\rho < \mu$.
- (b) $P\mathbf{u} < \mu\mathbf{u}$ for some vector $\mathbf{u} > 0$.
- (c) $\mathbf{w}^T P < \mu\mathbf{w}^T$ for some vector $\mathbf{w} > 0$.

- (d) $P\mathbf{u} < \rho\mathbf{u}$ for some vector $\mathbf{u} \geq 0$.
- (e) $\mathbf{w}^T P < \rho\mathbf{w}^T$ for some vector $\mathbf{w} \geq 0$.
- (f) There is no vector $\mathbf{u} \geq 0$ satisfying $P\mathbf{u} \geq \mu\mathbf{u}$.
- (g) There is no vector $\mathbf{w} \geq 0$ satisfying $\mathbf{w}^T P \geq \mu\mathbf{w}^T$.

9.4 Stochastic and Substochastic Matrices

(For more information about stochastic matrices see Chapter 54 (including examples).)

Definitions:

A square $n \times n$ matrix $P = [p_{ij}]$ is **stochastic** if it is nonnegative and $P\mathbf{1} = \mathbf{1}$ where $\mathbf{1} = [1, \dots, 1]^T \in R^n$. (Stochastic matrices are sometimes referred to as **row-stochastic**, while **column-stochastic** matrices are matrices whose transpose is (row-)stochastic.)

A square $n \times n$ matrix P is **doubly stochastic** if both P and its transpose are stochastic. The set of doubly stochastic matrices of order n is denoted Ω_n .

A square $n \times n$ matrix P is **substochastic** if it is nonnegative and $P\mathbf{1} \leq \mathbf{1}$.

A transient substochastic matrix is also called **stopping**.

An **ergodic class** of a stochastic matrix P is a basic class of P .

A **transient class** of a stochastic matrix P is an access equivalence class of P which is not ergodic.

A **state** of an $n \times n$ stochastic matrix P is an index $i \in \{1, \dots, n\}$. Such a state is **ergodic** or **transient** depending on whether it belongs to an ergodic class or to a transient class.

A **stationary distribution** of a stochastic matrix P is a nonnegative vector π that satisfies $\pi^T \mathbf{1} = 1$ and $\pi^T P = \pi^T$.

Facts:

Facts requiring proofs for which no specific reference is given follow directly from facts in Sections 9.2 and 9.3 and/or can be found in [BP94, Chap. 8].

1. Let $P = [p_{ij}]$ be an $n \times n$ stochastic matrix.
 - (a) $\rho(P) = 1$, $\mathbf{1} \in R^n$ is a right eigenvector of P corresponding to 1 and the stationary distributions of P are nonnegative eigenvectors of P corresponding to 1.
 - (b) $\nu_P(1) = 1$.
 - (c) $I - P$ has a group inverse.
 - (d) The height of every ergodic class is 1.
 - (e) The final classes of P are precisely its ergodic classes.
 - (f)
 - i. For every ergodic class C , P has a unique stationary distribution π^C of P with $(\pi^C)_i > 0$ if $i \in C$ and $(\pi^C)_i = 0$ otherwise.
 - ii. If $C^1, \dots, C^p$ are the ergodic classes of P , then the corresponding stationary distributions $\pi^{C^1}, \dots, \pi^{C^p}$ (according to Fact 1(f)i above) form a basis of the set of left eigenvectors of P corresponding to the eigenvalue 1; further, every stationary distribution of P is a convex combination of these vectors.

(g)

- i. Let T and R be the sets of transient and ergodic states of P , respectively. The matrix $I - P[T]$ is nonsingular and for each ergodic class C of P , the vector $\mathbf{u}^C$ given by

$$(\mathbf{u}^C)[K] = \begin{cases} e & \text{if } K = C \\ 0 & \text{if } K = R \setminus C \\ (I - P[T])^{-1} P[T, C]e & \text{if } K = T \end{cases}$$

is a right eigenvector of P corresponding to the eigenvalue 1; in particular, $(\mathbf{u}^C)_i > 0$ if i has access to C and $(\mathbf{u}^C)_i = 0$ if i does not have access to C .

- ii. If $C^1, \dots, C^p$ are the ergodic classes of P , then the corresponding vectors $\mathbf{u}^{C^1}, \dots, \mathbf{u}^{C^p}$ (referred to in Fact 1(g)i above) form a basis of the set of right eigenvectors of P corresponding to the eigenvalue 1; further, $\sum_{i=1}^p \mathbf{u}^{C^i} = e$.
- (h) Let $C^1, \dots, C^p$ be the ergodic classes of P , $\pi^{C^1}, \dots, \pi^{C^p}$ the corresponding stationary distributions (referred to in Fact 1(f)i above), and $\mathbf{u}^{C^1}, \dots, \mathbf{u}^{C^p}$ the corresponding eigenvectors referred to in Fact 1(g)i above. Then the matrix

$$P^* = \begin{bmatrix} \mathbf{u}^{C^1}, \dots, \mathbf{u}^{C^p} \end{bmatrix} \begin{bmatrix} \pi^{C^1} \\ \vdots \\ \pi^{C^p} \end{bmatrix}$$

is stochastic and satisfies $P^*[\langle n \rangle, C] = 0$ if C is a transient class of P , $P^*[i, C] \neq 0$ if C is an ergodic class and i has access to C , and $P^*[i, C] = 0$ if C is an ergodic class and i does not have access to C .

- (i) The matrix P^* from Fact 1(h) above has the representation $I - (I - P)^\#(I - P)$; further, $I - P + P^*$ is nonsingular and $(I - P)^\# = (I - P + P^*)^{-1}(I - P^*)$.
- (j) With P^* as the matrix from Fact 1(h) above, $\lim_{m \rightarrow \infty} P^m = P^*$ (C,1); further, when P is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\tau(P) < 1$.
- (k) With P^* as the matrix from Fact 1(h) above, $\lim_{m \rightarrow \infty} \sum_{t=0}^{m-1} P^t - mP^* = (I - P)^\#$ (C,1); further, when P is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\tau(P) < 1$.
- (l) With D a diagonal $n \times n$ matrix with positive diagonal elements and $\|\cdot\|$ a norm on $\mathbb{R}^n$,

$$\tau(P) \leq \max_{\mathbf{x} \in \mathbb{R}^n, \|\mathbf{x}\| \leq 1, \mathbf{x}^T D^{-1} \mathbf{1} = 0} \|\mathbf{x}^T D^{-1} P D\|.$$

In particular, with $\|\cdot\|$ as the 1-norm on $\mathbb{R}^n$ and $D = I$, the above bound specializes to

$$\tau(P) \leq \max_{r,s=1,\dots,n, r \neq s} \sum_{k=1}^n \frac{|p_{rk} - p_{sk}|}{2} \leq \min \left\{ 1 - \sum_{k=1}^n \min_r p_{rk}, \sum_{k=1}^n \max_r p_{rk} - 1 \right\}$$

(cf. Fact 11 of section 9.3 and Example 4 of section 9.2).

- (m) For every $0 < \alpha < 1$, $P_\alpha \equiv (1 - \alpha)I + \alpha P$ is an aperiodic stochastic matrix whose ergodic classes, transient classes, stationary distributions, and the vectors of Fact 1(g)i coincide with those of P . In particular, with P^* and P_α^* as the matrices from Fact 1(h) corresponding to P and P_α , respectively, $\lim_{m \rightarrow \infty} (P_\alpha)^m = P_\alpha^* = P^*$.

2. Let P be an irreducible stochastic matrix with coefficient of ergodicity τ .
 - (a) P has a unique stationary distribution, say π . Also, up to scalar multiple, $\mathbf{1}$ is a unique right eigenvector of P corresponding to the eigenvalue 1.
 - (b) With π as the unique stationary distribution of P , the matrix P^* from Fact 1(h) above equals $\mathbf{1}\pi$.
3. A doubly stochastic matrix is a convex combination of permutation matrices (in fact, the $n \times n$ permutation matrices are the extreme points of the set Ω_n of $n \times n$ doubly stochastic matrices).
4. Let P be an $n \times n$ substochastic matrix.
 - (a) $\rho(P) \leq 1$.
 - (b) $\nu_P(1) \leq 1$.
 - (c) $I - P$ has a group inverse.
 - (d) The matrix $P^* \equiv I - (I - P)^\#(I - P)$ is substochastic; further, $I - P + P^*$ is nonsingular and $(I - P)^\# = (I - P + P^*)^{-1}(I - P^*)$.
 - (e) With P^* as in Fact 4(d), $\lim_{m \rightarrow \infty} P^m = P^*(C, 1)$; further, when every access equivalence class C with $\rho(P[C]) = 1$ is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\max\{|\lambda| : \lambda \in \sigma(P) \text{ and } |\lambda| \neq 1\} < 1$.
 - (f) With P^* as the matrix from Fact 4(d) above, $\lim_{m \rightarrow \infty} \sum_{t=0}^{m-1} P^t - mP^* = (I - P)^\#(C, 1)$; further, when every access equivalence class C with $\rho(P[C]) = 1$ is aperiodic, this limit holds as a regular limit and the convergence is geometric with rate $\max\{|\lambda| : \lambda \in \sigma(P) \text{ and } |\lambda| \neq 1\} < 1$.
 - (g) The following are equivalent:
 - i. P is stopping.
 - ii. $\rho(P) < 1$.
 - iii. $I - P$ is invertible.
 - iv. There exists a positive vector $\mathbf{u} \in R^n$ with $P\mathbf{u} < \mathbf{u}$.

Further, when the above conditions hold, $(I - P)^{-1} = \sum_{t=0}^{\infty} P^t \geq 0$.

9.5 M-Matrices

Definitions:

An $n \times n$ real matrix $A = [a_{ij}]$ is a **Z-matrix** if its off-diagonal elements are nonpositive, i.e., if $a_{ij} \leq 0$ for all $i, j = 1, \dots, n$ with $i \neq j$.

An **M_0 -matrix** is a Z-matrix A that can be written as $A = sI - P$ with P as a nonnegative matrix and with s as a scalar satisfying $s \geq \rho(P)$.

An **M-matrix** A is a Z-matrix A that can be written as $A = sI - P$ with P as a nonnegative matrix and with s as a scalar satisfying $s > \rho(P)$.

A square real matrix A is an **inverse M-matrix** if it is nonsingular and its inverse is an M-matrix.

A square real matrix A is **inverse-nonnegative** if it is nonsingular and $A^{-1} \geq 0$ (the property is sometimes referred to as **inverse-positivity**).

A square real matrix A has a **convergent regular splitting** if A has a representation $A = M - N$ such that $N \geq 0$, M invertible with $M^{-1} \geq 0$ and $M^{-1}N$ is convergent.

A square complex matrix A is **positive stable** if the real part of each eigenvalue of A is positive; A is **nonnegative stable** if the real part of each eigenvalue of A is nonnegative.

An $n \times n$ complex matrix $A = [a_{ij}]$ is **strictly diagonally dominant (diagonally dominant)** if $|a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$ ($|a_{ii}| \geq \sum_{j=1, j \neq i}^n |a_{ij}|$) for $i = 1, \dots, n$.

An $n \times n$ M-matrix A **satisfies property C** if there exists a representation of A of the form $A = sI - P$ with $s > 0$, $P \geq 0$ and $\frac{P}{s}$ semiconvergent.

Facts:

Facts requiring proofs for which no specific reference is given follow directly from results about nonnegative matrices stated in Sections 9.2 and 9.3 and/or can be found in [BP94, Chap. 6].

1. Let A be an $n \times n$ real matrix with $n \geq 2$. The following are equivalent:
 - (a) A is an M -matrix; that is, A is a Z -matrix that can be written as $sI - P$ with P nonnegative and $s > \rho(P)$.
 - (b) A is a nonsingular M_0 -matrix.
 - (c) For each nonnegative diagonal matrix D , $A + D$ is inverse-nonnegative.
 - (d) For each $\mu \geq 0$, $A + \mu I$ is inverse-nonnegative.
 - (e) Each principal submatrix of A is inverse-nonnegative.
 - (f) Each principal submatrix of A of orders $1, \dots, n$ is inverse-nonnegative.
2. Let $A = [a_{ij}]$ be an $n \times n$ Z -matrix. The following are equivalent:*
 - (a) A is an M -matrix.
 - (b) Every real eigenvalue of A is positive.
 - (c) $A + D$ is nonsingular for each nonnegative diagonal matrix D .
 - (d) All of the principal minors of A are positive.
 - (e) For each $k = 1, \dots, n$, the sum of all the $k \times k$ principal minors of A is positive.
 - (f) There exist lower and upper triangular matrices L and U , respectively, with positive diagonal elements such that $A = LU$.
 - (g) A is permutation similar to a matrix satisfying condition 2(f).
 - (h) A is positive stable.
 - (i) There exists a diagonal matrix D with positive diagonal elements such that $AD + DA^T$ is positive definite.
 - (j) There exists a vector $\mathbf{x} > 0$ with $A\mathbf{x} > 0$.
 - (k) There exists a vector $\mathbf{x} > 0$ with $A\mathbf{x} \succeq 0$ and $\sum_{j=1}^i a_{ij}x_j > 0$ for $i = 1, \dots, n$.
 - (l) A is permutation similar to a matrix satisfying condition 2(k).
 - (m) There exists a vector $\mathbf{x} > 0$ such that $A\mathbf{x} \succeq 0$ and the matrix $\hat{A} = [\hat{a}_{ij}]$ defined by

$$\hat{a}_{ij} = \begin{cases} 1 & \text{if either } a_{ij} \neq 0 \text{ or } (A\mathbf{x})_i \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

is irreducible.

- (n) All the diagonal elements of A are positive and there exists a diagonal matrix D such that AD is strictly diagonally dominant.
- (o) A is inverse-nonnegative.
- (p) Every representation of A of the form $A = M - N$ with $N \geq 0$ and M inverse-positive must have $M^{-1}N$ convergent (i.e., $\rho(M^{-1}N) < 1$).
- (q) For each vector $\mathbf{y} \geq 0$, the set $\{\mathbf{x} \geq 0 : A^T \mathbf{x} \leq \mathbf{y}\}$ is bounded and A is nonsingular.

*Each of the 17 conditions that are listed in Fact 2 is a representative of a set of conditions that are known to be equivalent for all matrices (not just Z -matrices); see [BP94, Theorem 6.2.3]. For additional characterizations of M -matrices, see [FiSc83].

3. Let A be an irreducible $n \times n$ Z -matrix with $n \geq 2$. The following are equivalent:
 - (a) A is an M -matrix.
 - (b) A is nonsingular and $A^{-1} > 0$.
 - (c) $A\mathbf{x} \geq 0$ for some $\mathbf{x} > 0$.
4. Let $A = [a_{ij}]$ be an $n \times n$ M -matrix and let $B = [b_{ij}]$ be an $n \times n$ Z -matrix with $B \geq A$. Then:
 - (a) B is an M -matrix.
 - (b) $\det B \geq \det A$.
 - (c) $A^{-1} \geq B^{-1}$.
 - (d) $\det A \leq a_{11} \dots a_{nn}$.
5. If P is an inverse M -matrix, then $P \geq 0$ and $\Gamma(P)$ is transitive; that is, if (v, u) and (u, w) are arcs of $\Gamma(P)$, then so is (v, w) .
6. Let A be an $n \times n$ real matrix with $n \geq 2$. The following are equivalent:
 - (a) A is a nonsingular M_0 -matrix.
 - (b) For each diagonal matrix D with positive diagonal elements, $A + D$ is inverse-nonnegative.
 - (c) For each $\mu > 0$, $A + \mu I$ is inverse-nonnegative.
7. Let A be an $n \times n$ Z -matrix. The following are equivalent:*
 - (a) A is an M_0 -matrix.
 - (b) Every real eigenvalue of A is nonnegative.
 - (c) $A + D$ is nonsingular for each diagonal matrix D having positive diagonal elements.
 - (d) For each $k = 1, \dots, n$, the sum of all the $k \times k$ principal minors of A is nonnegative.
 - (e) A is permutation similar to a matrix having a representation LU with L and U as lower and upper triangular matrices having positive diagonal elements.
 - (f) A is nonnegative stable.
 - (g) There exists a nonnegative matrix Y satisfying $YA^{k+1} = A^k$ for some $k \geq 1$.
 - (h) A has a representation of the form $A = M - N$ with M inverse-nonnegative, $N \geq 0$ and $B \equiv M^{-1}N$ satisfying $\cap_{k=0}^{\infty} \text{range}(B^k) = \cap_{k=0}^{\infty} \text{range}(A^k)$ and $\rho(B) \leq 1$.
 - (i) A has a representation of the form $A = M - N$ with M inverse-nonnegative, $M^{-1}N \geq 0$ and $B \equiv M^{-1}N$ satisfying $\cap_{k=0}^{\infty} \text{range}(B^k) = \cap_{k=0}^{\infty} \text{range}(A^k)$ and $\rho(B) \leq 1$.
8. Let A be an M_0 -matrix.
 - (a) A satisfies property C if and only if $v_A(0) \leq 1$.
 - (b) A is permutation similar to a matrix having a representation LU with L as a lower triangular M -matrix and U as an upper triangular M_0 matrix.
9. [BP94, Theorem 8.4.2] If P is substochastic (see Section 9.4), then $I - P$ is an M_0 -matrix satisfying property C.
10. Let A be an irreducible $n \times n$ singular M_0 -matrix.
 - (a) A has rank $n - 1$.
 - (b) There exists a vector $\mathbf{x} > 0$ such that $A\mathbf{x} = 0$.

*Each of the 9 conditions that are listed in Fact 7 is a representative of a set of conditions that are known to be equivalent for all matrices (not just Z -matrices); see [BP94, Theorem 6.4.6]. For additional characterizations of M -matrices, see [FiSc83].

- (c) A has property C .
- (d) Each principal submatrix of A other than A itself is an M -matrix.
- (e) $[Ax \geq 0] \Rightarrow [Ax = 0]$.

9.6 Scaling of Nonnegative Matrices

A scaling of a (usually nonnegative) matrix is the outcome of its pre- and post-multiplication by diagonal matrices having positive diagonal elements. Scaling problems concern the search for scalings of given matrices such that specified properties are satisfied. Such problems are characterized by:

- (a) The class of matrices to be scaled.
- (b) Restrictions on the pre- and post-multiplying diagonal matrices to be used.
- (c) The target property.

Classes of matrices under (a) may refer to arbitrary rectangular matrices, square matrices, symmetric matrices, positive semidefinite matrices, etc. For possible properties of pre- and post-multiplying diagonal matrices under (b) see the following Definition subsection. Finally, examples for target properties under (c) include:

- i. The specification of the row- and/or column-sums; for example, being stochastic or being doubly stochastic. See the following Facts subsection.
- ii. The specification of the row- and/or column-maxima.
- iii. (For a square matrix) being line-symmetric, that is, having each row-sum equal to the corresponding column-sum.
- iv. Being optimal within a prescribed class of scalings under some objective function. One example of such optimality is to minimize the maximal element of a scalings of the form XAX^{-1} . Also, in numerical analysis, preconditioning a matrix may involve its replacement with a scaling that has a low ratio of largest to smallest element; so, a potential target property is to be a minimizer of this ratio among all scalings of the underlying matrix.

Typical questions that are considered when addressing scaling problems include:

- (a) Characterizing existence of a scaling that satisfies the target property (precisely or approximately).
- (b) Computing a scaling of a given matrix that satisfies the target property (precisely or approximately) or verifying that none exists.
- (c) Determining complexity bounds for corresponding computation.

Early references that address scaling problems include [Kru37], which describes a heuristic for finding a doubly stochastic scaling of a positive square matrix, and Sinkhorn's [Sin64] pioneering paper, which provides a formal analysis of that problem. The subject has been intensively studied and an aspiration to provide a comprehensive survey of the rich literature is beyond the scope of the current review; consequently, we address only scaling problems where the target is to achieve, precisely or approximately, prescribed row- and column-sums.

Definitions:

Let $A = [a_{ij}]$ be an $m \times n$ matrix.

A **scaling** (sometimes referred to as an **equivalence-scaling** or a **DAE-scaling**) of A is any matrix of the form DAE where D and E are square diagonal matrices having positive diagonal elements; such a scaling is a **row-scaling** of A if $E = I$ and it is a **normalized-scaling** if $\det(D) = \det(E) = 1$.

If $m = n$, a scaling DAE of A is a **similarity-scaling** (sometimes referred to as a DAD^{-1} **scaling**) of A if $E = D^{-1}$, and DAE is a **symmetric-scaling** (sometimes referred to as a DAD **scaling**) of A if $E = D$.

The **support** (or **sparsity pattern**) of A , denoted $\text{Struct}(A)$, is the set of indices ij with $a_{ij} \neq 0$; naturally, this definition applies to vectors.

Facts:

1. (*Prescribed-Line-Sum Scalings*) [RoSc89] Let $A = [a_{ij}] \in \mathbb{R}^{m \times n}$ be a nonnegative matrix, and let $\mathbf{r} = [r_i] \in \mathbb{R}^m$ and $\mathbf{c} = [c_j] \in \mathbb{R}^n$ be positive vectors.

(a) The following are equivalent:

- i. There exists a scaling B of A with $B\mathbf{1} = \mathbf{r}$ and $\mathbf{1}^T B = \mathbf{c}^T$.
- ii. There exists nonnegative $m \times n$ matrix B having the same support as A with $B\mathbf{1} = \mathbf{r}$ and $\mathbf{1}^T B = \mathbf{c}^T$.
- iii. For every $I \subseteq \{1, \dots, m\}$ and $J \subseteq \{1, \dots, n\}$ for which $A[I^c, J] = 0$,

$$\sum_{i \in I} r_i \geq \sum_{j \in J} c_j$$

and equality holds if and only if $A[I, J^c] = 0$.

- iv. $\mathbf{1}^T \mathbf{r} = \mathbf{1}^T \mathbf{r}$ and the following (geometric) optimization problem has an optimal solution:

$$\min \mathbf{x}^T A \mathbf{y}$$

$$\text{subject to : } \mathbf{x} = [x_i] \in \mathbb{R}^m, \mathbf{y} = [y_j] \in \mathbb{R}^n$$

$$\mathbf{x} \geq 0, \mathbf{y} \geq 0$$

$$\prod_{i=1}^m (x_i)^{r_i} = \prod_{j=1}^n (y_j)^{c_j} = 1.$$

A standard algorithm for approximating a scaling of a matrix to one that has prescribed row- and column-sums (when one exists) is to iteratively scale rows and columns separately so as to achieve corresponding line-sums.

- (b) Suppose $\mathbf{1}^T \mathbf{r} = \mathbf{1}^T \mathbf{r}$ and $\bar{\mathbf{x}} = [\bar{x}_i]$ and $\bar{\mathbf{y}} = [\bar{y}_j]$ form an optimal solution of the optimization problem of Fact 1(d). Let $\bar{\lambda} \equiv \frac{\bar{\mathbf{x}}^T A \bar{\mathbf{y}}}{\mathbf{1}^T \mathbf{r}}$ and let $\bar{X} \in \mathbb{R}^{m \times m}$ and $\bar{Y} \in \mathbb{R}^{n \times n}$ be the diagonal matrices having diagonal elements $\bar{X}_{ii} = \frac{\bar{x}_i}{\bar{\lambda}}$ and $\bar{Y}_{jj} = \bar{y}_j$. Then $B \equiv \bar{X} A \bar{Y}$ is a scaling of A satisfying $B\mathbf{1} = \mathbf{r}$ and $\mathbf{1}^T B = \mathbf{c}^T$.
- (c) Suppose $\bar{X} \in \mathbb{R}^{m \times m}$ and $\bar{Y} \in \mathbb{R}^{n \times n}$ are diagonal matrices such that $B \equiv \bar{X} A \bar{Y}$ is a scaling of A satisfying $B\mathbf{1} = \mathbf{r}$ and $\mathbf{1}^T B = \mathbf{c}^T$. Then $\mathbf{1}^T \mathbf{r} = \mathbf{1}^T \mathbf{r}$ and with

$$\bar{\lambda} \equiv \prod_{i=1}^m (\bar{X}_{ii})^{-r_i / \mathbf{1}^T \mathbf{r}}$$

and

$$\bar{\mu} \equiv \prod_{j=1}^n (\bar{Y}_{jj})^{-c_j / \mathbf{1}^T \mathbf{c}},$$

the vectors $\bar{\mathbf{x}} = [\bar{x}_i] \in \mathbb{R}^m$ and $\bar{\mathbf{y}} = [\bar{y}_j] \in \mathbb{R}^n$ with $\bar{x}_i = \bar{\lambda} X_{ii}$ for $i = 1, \dots, m$ and $\bar{y}_j = \bar{\mu} Y_{jj}$ for $j = 1, \dots, n$ are optimal for the optimization problem of Fact 1(d).

2. (*Approximate Prescribed-Line-Sum Scalings*) [RoSc89] Let $A = [a_{ij}] \in \mathbb{R}^{m \times n}$ be a nonnegative matrix, and let $\mathbf{r} = [r_i] \in \mathbb{R}^m$ and $\mathbf{c} = [c_j] \in \mathbb{R}^n$ be positive vectors.

(a) The following are equivalent:

- i. For every $\epsilon > 0$ there exists a scaling B of A with $\|B\mathbf{1} - \mathbf{r}\|_1 \leq \epsilon$ and $\|\mathbf{1}^T B - \mathbf{c}^T\|_1 \leq \epsilon$.

- ii. There exists nonnegative $m \times n$ matrix $A' = [a'_{ij}]$ with $\text{Struct}(A') \subseteq \text{Struct}(A)$ and $a'_{ij} = a_{ij}$ for each $ij \in \text{Struct}(A')$ such that A' has a scaling B satisfying $B\mathbf{1} = \mathbf{r}$ and $\mathbf{1}^T B = \mathbf{c}^T$.
- iii. For every $\epsilon > 0$ there exists a matrix B having the same support as A and satisfying $\|B\mathbf{1} - \mathbf{r}\|_1 \leq \epsilon$ and $\|\mathbf{1}^T B - \mathbf{c}^T\|_1 \leq \epsilon$.
- iv. There exists a matrix B satisfying $\text{Struct}(B) \subseteq \text{Struct}(A)$, $B\mathbf{1} = \mathbf{r}$ and $\mathbf{1}^T B = \mathbf{c}^T$.
- v. For every $I \subseteq \{1, \dots, m\}$ and $J \subseteq \{1, \dots, n\}$ for which $A[I^c, J] = 0$,

$$\sum_{i \in I} r_i \geq \sum_{j \in J} c_j.$$

- vi. $\mathbf{1}^T \mathbf{r} = \mathbf{1}^T \mathbf{c}$ and the objective of the optimization problem of Fact 2(a)iii is bounded away from zero.

See [NR99] for a reduction of the problem of finding a scaling of A that satisfies $\|B\mathbf{1} - \mathbf{r}\|_1 \leq \epsilon$ and $\|\mathbf{1}^T B - \mathbf{c}^T\|_1 \leq \epsilon$ for a given $\epsilon > 0$ to the approximate solution of geometric program that is similar to the one in Fact 1(a)iv and for the description of an (ellipsoid) algorithm that solves the latter with complexity bound of $O(1)(m+n)^4 \ln[2 + \frac{mn\sqrt{m^3+n^3} \ln(mn\beta)}{\epsilon^3}]$, where β is the ratio between the largest and smallest positive entries of A .

9.7 Miscellaneous Topics

In this subsection, we mention several important topics about nonnegative matrices that are not covered in detail in the current section due to size constraint; some relevant material appears in other sections.

9.7.1 Nonnegative Factorization and Completely Positive Matrices

A *nonnegative factorization* of a nonnegative matrix $A \in R^{m \times n}$ is a representation $A = LR$ of A with L and R as nonnegative matrices. The *nonnegative rank* of A is the smallest number of columns of L (rows of R) in such a factorization.

A square matrix A is *doubly nonnegative* if it is nonnegative and positive semidefinite. Such a matrix A is *completely positive* if it has a nonnegative factorization $A = BB^T$; the *CP-rank* of A is then the smallest number of columns of a matrix B in such a factorization.

Facts about nonnegative factorizations and completely positive matrices can be found in [CR93], [BSM03], and [CP05].

9.7.2 The Inverse Eigenvalue Problem

The *inverse eigenvalue problem* concerns the identification of necessary conditions and sufficient conditions for a finite set of complex numbers to be the spectrum of a nonnegative matrix.

Facts about the inverse eigenvalue problem can be found in [BP94, Sections 4.2 and 11.2] and Chapter 20.

9.7.3 Nonhomogenous Products of Matrices

A *nonhomogenous product* of nonnegative matrices is the finite matrix product of nonnegative matrices $P_1 P_2 \dots P_m$, generalizing powers of matrices where the multiplicands are equal (i.e., $P_1 = P_2 = \dots = P_m$); the study of such products focuses on the case where the multiplicands are taken from a prescribed set.

Facts about Perron–Frobenius type properties of nonhomogenous products of matrices can be found in [Sen81], and [Har02].

9.7.4 Operators Determined by Sets of Nonnegative Matrices in Product Form

A finite set of nonnegative $n \times n$ matrices $\{P_\delta : \delta \in \Delta\}$ is said to be in *product form* if there exists finite sets of row vectors $\Delta_1, \dots, \Delta_n$ such that $\Delta = \prod_{i=1}^n \Delta_i$ and for each $\delta = (\delta_1, \dots, \delta_n) \in \Delta$, P_δ is the matrix whose rows are, respectively, $\delta_1, \dots, \delta_n$. Such a family determines the operators $P_\Delta^{\max}$ and $P_\Delta^{\min}$ on $\mathbb{R}^n$ with $P_\Delta^{\max}x = \max_{\delta \in \Delta} P_\delta x$ and $P_\Delta^{\min}x = \min_{\delta \in \Delta} P_\delta x$ for each $x \in \mathbb{R}^n$.

Facts about Perron–Frobenius-type properties of the operators corresponding to families of matrices in product form can be found in [Zij82], [Zij84], and [RW82].

9.7.5 Max Algebra over Nonnegative Matrices

Matrix operations under the max algebra are executed with the max operator replacing (real) addition and (real) addition replacing (real) multiplication.

Perron–Frobenius-type results and scaling results are available for nonnegative matrices when considered as operators under the max algebra; see [RSS94], [Bap98], [But03], [BS05], and Chapter 25.

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References

- [Bap98] R.B. Bapat, A max version of the Perron–Frobenius Theorem, *Lin. Alg. Appl.*, 3:18, 275–276, 1998.
- [BS75] G.P. Barker and H. Schneider, Algebraic Perron–Frobenius Theory, *Lin. Alg. Appl.*, 11:219–233, 1975.
- [BNS89] A. Berman, M. Neumann, and R.J. Stern, *Nonnegative Matrices in Dynamic Systems*, John Wiley & Sons, New York, 1989.
- [BP94] A. Berman and R.J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*, Academic, 1979, 2nd ed., SIAM, 1994.
- [BSM03] A. Berman and N. Shaked-Monderer, *Completely Positive Matrices*, World Scientific, Singapore, 2003.
- [But03] P. Butkovic, Max-algebra: the linear algebra of combinatorics? *Lin. Alg. Appl.*, 367:313–335, 2003.
- [BS05] P. Butkovic and H. Schneider, Applications of max algebra to diagonal scaling of matrices, *ELA*, 13:262–273, 2005.
- [CP05] M. Chu and R. Plemmons, Nonnegative matrix factorization and applications, *Bull. Lin. Alg. Soc. — Image*, 34:2–7, 2005.
- [Coh78] J.E. Cohen, Derivatives of the spectral radius as a function of non-negative matrix elements, *Mathematical Proceedings of the Cambridge Philosophical Society*, 83:183–190, 1978.
- [CR93] J.E. Cohen and U.G. Rothblum, Nonnegative ranks, decompositions and factorizations of non-negative matrices, *Lin. Alg. Appl.*, 190:149–168, 1993.
- [Col42] L. Collatz, *Einschliessungssatz für die charakteristischen Zahlen von Matrizen*, *Math. Z.*, 48:221–226, 1942.
- [CHR97] D. Coppersmith, A.J. Hoffman, and U.G. Rothblum, Inequalities of Rayleigh quotients and bounds on the spectral radius of nonnegative symmetric matrices, *Lin. Alg. Appl.*, 263:201–220, 1997.
- [DR05] E.V. Denardo and U.G. Rothblum, Totally expanding multiplicative systems, *Lin. Alg. Appl.*, 406:142–158, 2005.
- [ERS95] B.C. Eaves, U.G. Rothblum, and H. Schneider, Perron–Frobenius theory over real closed fields and fractional power series expansions, *Lin. Alg. Appl.*, 220:123–150, 1995.
- [EHP90] L. Elsner, D. Hershkowitz, and A. Pinkus, Functional inequalities for spectral radii of nonnegative matrices, *Lin. Alg. Appl.*, 129:103–130, 1990.

- [EJD88] L. Elsner, C. Johnson, and D. da Silva, The Perron root of a weighted geometric mean of nonnegative matrices, *Lin. Multilin. Alg.*, 24:1–13, 1988.
- [FiSc83] M. Fiedler and H. Schneider, Analytic functions of M-matrices and generalizations, *Lin. Multilin. Alg.*, 13:185–201, 1983.
- [FrSc80] S. Friedland and H. Schneider, The growth of powers of a nonnegative matrix, *SIAM J. Alg. Disc. Meth.*, 1:185–200, 1980.
- [Fro08] G.F. Frobenius, *Über Matrizen aus positiven Elementen*, S.-B. Preuss. Akad. Wiss. Berlin, 471–476, 1908.
- [Fro09] G.F. Frobenius, *Über Matrizen aus positiven Elementen*, II, S.-B. Preuss. Akad. Wiss. Berlin, 514–518, 1909.
- [Fro12] G.F. Frobenius, *Über Matrizen aus nicht negativen Elementen*, Sitzungsber. Kön. Preuss. Akad. Wiss. Berlin, 456–457, 1912.
- [Gan59] F.R. Gantmacher, *The Theory of Matrices*, Vol. II, Chelsea Publications, London, 1958.
- [Har02] D.J. Hartfiel, *Nonhomogeneous Matrix Products*, World Scientific, River Edge, NJ, 2002.
- [HJ85] R.A. Horn and C.R. Johnson, *Matrix Analysis*, Cambridge University Press, Cambridge, 1985.
- [HNR90] R.E. Hartwig, M. Neumann, and N.J. Rose, An algebraic-analytic approach to nonnegative basis, *Lin. Alg. Appl.*, 133:77–88, 1990.
- [HRR92] M. Haviv, Y. Ritov, and U.G. Rothblum, Taylor expansions of eigenvalues of perturbed matrices with applications to spectral radii of nonnegative matrices, *Lin. Alg. Appl.*, 168:159–188, 1992.
- [HS88a] D. Hershkowitz and H. Schneider, Solutions of Z-matrix equations, *Lin. Alg. Appl.*, 106:25–38, 1988.
- [HS88b] D. Hershkowitz and H. Schneider, On the generalized nullspace of M-matrices and Z-matrices, *Lin. Alg. Appl.*, 106:5–23, 1988.
- [HRS89] D. Hershkowitz, U.G. Rothblum, and H. Schneider, The combinatorial structure of the generalized nullspace of a block triangular matrix, *Lin. Alg. Appl.*, 116:9–26, 1989.
- [HS89] D. Hershkowitz and H. Schneider, Height bases, level bases, and the equality of the height and the level characteristic of an M-matrix, *Lin. Multilin. Alg.*, 25:149–171, 1989.
- [HS91a] D. Hershkowitz and H. Schneider, Combinatorial bases, derived Jordan and the equality of the height and level characteristics of an M-matrix, *Lin. Multilin. Alg.*, 29:21–42, 1991.
- [HS91b] D. Hershkowitz and H. Schneider, On the existence of matrices with prescribed height and level characteristics, *Israel Math J.*, 75:105–117, 1991.
- [Hof67] A.J. Hoffman, Three observations on nonnegative matrices, *J. Res. Nat. Bur. Standards-B. Math and Math. Phys.*, 71B:39–41, 1967.
- [Kru37] J. Kruithof, *Telefoonverkeersrekening*, *De Ingenieur*, 52(8):E15–E25, 1937.
- [LT85] P. Lancaster and M. Tismenetsky, *The Theory of Matrices*, 2nd ed., Academic Press, New York, 1985.
- [Mac00] C.R. MacCluer, The many proofs and applications of Perron's theorem, *SIAM Rev.*, 42:487–498, 2000.
- [Min88] H. Minc, *Nonnegative Matrices*, John Wiley & Sons, New York, 1988.
- [NR99] A. Nemirovski and U.G. Rothblum, On complexity of matrix scaling, *Lin. Alg. Appl.*, 302-303:435–460, 1999.
- [NS94] M. Neumann and H. Schneider, Algorithms for computing bases for the Perron eigenspace with prescribed nonnegativity and combinatorial properties, *SIAM J. Matrix Anal. Appl.*, 15:578–591, 1994.
- [Per07a] O. Perron, *Grundlagen für eine Theorie des Jacobischen Kettenbruchalgorithmus*, *Math. Ann.*, 63:11–76, 1907.
- [Per07b] O. Perron, *Zür Theorie der über Matrizen*, *Math. Ann.*, 64:248–263, 1907.
- [RiSc78] D. Richman and H. Schneider, On the singular graph and the Weyr characteristic of an M-matrix, *Aequationes Math.*, 17:208–234, 1978.
- [Rot75] U.G. Rothblum, Algebraic eigenspaces of nonnegative matrices, *Lin. Alg. Appl.*, 12:281–292, 1975.
- [Rot80] U.G. Rothblum, Bounds on the indices of the spectral-circle eigenvalues of a nonnegative matrix, *Lin. Alg. Appl.*, 29:445–450, 1980.

- [Rot81a] U.G. Rothblum, Expansions of sums of matrix powers, *SIAM Rev.*, 23:143–164, 1981.
- [Rot81b] U.G. Rothblum, Sensitive growth analysis of multiplicative systems I: the stationary dynamic approach, *SIAM J. Alg. Disc. Meth.*, 2:25–34, 1981.
- [Rot81c] U.G. Rothblum, Resolvent expansions of matrices and applications, *Lin. Alg. Appl.*, 38:33–49, 1981.
- [RoSc89] U.G. Rothblum and H. Schneider, Scalings of matrices which have prespecified row-sums and column-sums via optimization, *Lin. Alg. Appl.*, 114/115:737–764, 1989.
- [RSS94] U.G. Rothblum, H. Schneider, and M.H. Schneider, Scalings of matrices which have prespecified row-maxima and column-maxima, *SIAM J. Matrix Anal.*, 15:1–15, 1994.
- [RT85] U.G. Rothblum and C.P. Tan, Upper bounds on the maximum modulus of subdominant eigenvalues of nonnegative matrices, *Lin. Alg. Appl.*, 66:45–86, 1985.
- [RW82] U.G. Rothblum and P. Whittle, Growth optimality for branching Markov decision chains, *Math. Op. Res.*, 7:582–601, 1982.
- [Sch53] H. Schneider, An inequality for latent roots applied to determinants with dominant principal diagonal, *J. London Math. Soc.*, 28:8–20, 1953.
- [Sch56] H. Schneider, The elementary divisors associated with 0 of a singular M-matrix, *Proc. Edinburgh Math. Soc.*, 10:108–122, 1956.
- [Sch86] H. Schneider, The influence of the marked reduced graph of a nonnegative matrix on the Jordan form and on related properties: a survey, *Lin. Alg. Appl.*, 84:161–189, 1986.
- [Sch96] H. Schneider, Commentary on “Unzerlegbare, nicht negative Matrizen,” in *Helmut Wielandt’s “Mathematical Works,”* Vol. 2, B. Huppert and H. Schneider, Eds., Walter de Gruyter Berlin, 1996.
- [Sen81] E. Seneta, *Non-negative matrices and Markov chains*, Springer Verlag, New York, 1981.
- [Sin64] R. Sinkhorn, A relationship between arbitrary positive and stochastic matrices, *Ann. Math. Stat.*, 35:876–879, 1964.
- [Tam04] B.S. Tam, The Perron generalized eigenspace and the spectral cone of a cone-preserving map, *Lin. Alg. Appl.*, 393:375–429, 2004.
- [TW89] B.S. Tam and S.F. Wu, On the Collatz-Wielandt sets associated with a cone-preserving map, *Lin. Alg. Appl.*, 125:77–95, 1989.
- [Var62] R.S. Varga, *Matrix Iterative Analysis*, Prentice-Hall, Upper Saddle River, NJ, 1962, 2nd ed., Springer, New York, 2000.
- [Vic85] H.D. Victory, Jr., On nonnegative solutions to matrix equations, *SIAM J. Alg. Dis. Meth.*, 6: 406–412, 1985.
- [Wie50] H. Wielandt, *Unzerlegbare, nicht negative Matrizen*, *Mathematische Zeitschrift*, 52:642–648, 1950.
- [Zij82] W.H.M. Zijm, *Nonnegative Matrices in Dynamic Programming*, Ph.D. dissertation, Mathematisch Centrum, Amsterdam, 1982.
- [Zij84] W.H.M. Zijm, Generalized eigenvectors and sets of nonnegative matrices, *Lin. Alg. Appl.*, 59:91–113, 1984.

10

Partitioned Matrices

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10.1 Submatrices and Block Matrices

Definitions:

Let $A \in F^{m \times n}$. Then the row indices of A are $\{1, \dots, m\}$, and the column indices of A are $\{1, \dots, n\}$. Let α, β be nonempty sets of indices with $\alpha \subseteq \{1, \dots, m\}$ and $\beta \subseteq \{1, \dots, n\}$.

A **submatrix** $A[\alpha, \beta]$ is a matrix whose rows have indices α among the row indices of A , and whose columns have indices β among the column indices of A . $A(\alpha, \beta) = A[\alpha^c, \beta^c]$, where α^c is the complement of α .

A **principal submatrix** is a submatrix $A[\alpha, \alpha]$, denoted more compactly as $A[\alpha]$.

Let the set $\{1, \dots, m\}$ be partitioned into the subsets $\alpha_1, \dots, \alpha_r$ in the usual sense of partitioning a set (so that $\alpha_i \cap \alpha_j = \emptyset$, for all $i \neq j$, $1 \leq i, j \leq r$, and $\alpha_1 \cup \dots \cup \alpha_r = \{1, \dots, m\}$), and let $\{1, \dots, n\}$ be partitioned into the subsets $\beta_1, \dots, \beta_s$.

The matrix $A \in F^{m \times n}$ is said to be **partitioned** into the submatrices $A[\alpha_i, \beta_j]$, $1 \leq i \leq r$, $1 \leq j \leq s$.

A **block matrix** is a matrix that is partitioned into submatrices $A[\alpha_i, \beta_j]$ with the row indices $\{1, \dots, m\}$ and column indices $\{1, \dots, n\}$ partitioned into subsets sequentially, i.e., $\alpha_1 = \{1, \dots, i_1\}$, $\alpha_2 = \{i_1 + 1, \dots, i_2\}$, etc.

Each entry of a block matrix, which is a submatrix $A[\alpha_i, \beta_j]$, is called a **block**, and we will sometimes write $A = [A_{ij}]$ to label the blocks, where $A_{ij} = A[\alpha_i, \beta_j]$.

If the block matrix $A \in F^{m \times p}$ is partitioned with α_i s and β_j s, $1 \leq i \leq r$, $1 \leq j \leq s$, and the block matrix $B \in F^{p \times n}$ is partitioned with β_j s and γ_k s, $1 \leq j \leq s$, $1 \leq k \leq t$, then the partitions of A and B are said to be **conformal** (or sometimes **conformable**).

Facts:

The following facts can be found in [HJ85]. This information is also available in many other standard references such as [LT85] or [Mey00].

1. Two block matrices $A = [A_{ij}]$ and $B = [B_{ij}]$ in $F^{m \times n}$, which are both partitioned with the same α_i s and β_j s, $1 \leq i \leq r$, $1 \leq j \leq s$, may be added block-wise, as with the usual matrix addition, so that the (i, j) block entry of $A + B$ is $(A + B)_{ij} = A_{ij} + B_{ij}$.

2. If the block matrix $A \in F^{m \times p}$ and the block matrix $B \in F^{p \times n}$ have conformal partitions, then we can think of A and B as having entries, which are blocks, so that we can then multiply A and B block-wise to form the $m \times n$ block matrix $C = AB$. Then $C_{ik} = \sum_{j=1}^s A_{ij}B_{jk}$, and the matrix C will be partitioned with the α_i s and γ_k s, $1 \leq i \leq r$, $1 \leq k \leq t$, where A is partitioned with α_i s and β_j s, $1 \leq i \leq r$, $1 \leq j \leq s$, and B is partitioned with β_j s and γ_k s, $1 \leq j \leq s$, $1 \leq k \leq t$.
3. With addition and multiplication of block matrices described as in Facts 1 and 2 the usual properties of associativity of addition and multiplication of block matrices hold, as does distributivity, and commutativity of addition. The additive identity $\mathbf{0}$ and multiplicative identity I are the same under block addition and multiplication, as with the usual matrix addition and multiplication. The additive identity $\mathbf{0}$ has zero matrices as blocks; the multiplicative identity I has multiplicative identity submatrices as diagonal blocks and zero matrices as off-diagonal blocks.
4. If the partitions of A and B are conformal, the partitions of B and A are not necessarily conformal, even if BA is defined.
5. Let $A \in F^{n \times n}$ be a block matrix of the form $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & \mathbf{0} \end{bmatrix}$, where A_{12} is $k \times k$, and A_{21} is $(n-k) \times (n-k)$. Then $\det(A) = (-1)^{(n+1)k} \det(A_{12}) \det(A_{21})$.

Examples:

1. Let the block matrix $A \in \mathbb{C}^{n \times n}$ given by $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ be Hermitian. Then A_{11} and A_{22} are Hermitian, and $A_{21} = A_{12}^*$.
2. If $A = [a_{ij}]$, then $A[\{i\}, \{j\}]$ is the 1×1 matrix whose entry is a_{ij} . The submatrix $A(\{i\}, \{j\})$ is the submatrix of A obtained by deleting row i and column j of A .

$$\text{Let } A = \begin{bmatrix} 1 & -2 & 5 & 3 & -1 \\ -3 & 0 & 1 & 6 & 1 \\ 2 & 7 & 4 & 5 & -7 \end{bmatrix}.$$

3. Then $A[\{2\}, \{3\}] = [a_{23}] = [1]$ and $A(\{2\}, \{3\}) = \begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & 7 & 5 & -7 \end{bmatrix}$.
4. Let $\alpha = \{1, 3\}$ and $\beta = \{1, 2, 4\}$. Then the submatrix $A[\alpha, \beta] = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 7 & 5 \end{bmatrix}$, and the principal submatrix $A[\alpha] = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$.
5. Let $\alpha_1 = \{1\}$, $\alpha_2 = \{2, 3\}$ and let $\beta_1 = \{1, 2\}$, $\beta_2 = \{3\}$, $\beta_3 = \{4, 5\}$. Then the block matrix, with (i, j) block entry $A_{ij} = A[\alpha_i, \beta_j]$, $1 \leq i \leq 2$, $1 \leq j \leq 3$, is

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix} = \left[\begin{array}{ccc|ccc} 1 & -2 & 5 & 3 & -1 & \\ - & - & - & - & - & \\ -3 & 0 & 1 & 6 & 1 & \\ 2 & 7 & 4 & 5 & -7 & \end{array} \right].$$

6. Let $B = \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \end{bmatrix} = \left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 6 & -2 & \\ - & - & - & - & - & \\ -4 & 0 & 5 & 3 & 7 & \\ 1 & 1 & -2 & 2 & 6 & \end{array} \right]$. Then the matrices A (of this example) and B are partitioned with the same α_i s and β_j s, so they can be added block-wise

$$\begin{aligned}
 & \text{as } \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \end{bmatrix} \\
 &= \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} & A_{13} + B_{13} \\ A_{21} + B_{21} & A_{22} + B_{22} & A_{23} + B_{23} \end{bmatrix} \\
 &= \left[\begin{array}{cc|c|cc} 3 & -3 & 5 & 9 & -3 \\ \hline -7 & 0 & 6 & 9 & 8 \\ \hline 3 & 8 & 2 & 7 & -1 \end{array} \right].
 \end{aligned}$$

7. The block matrices $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{13} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} \\ B_{21} \\ B_{31} \end{bmatrix}$ have conformal partitions if the β_j

index sets, which form the submatrices $A_{ij} = A[\alpha_i, \beta_j]$ of A , are the same as the β_j index sets, which form the submatrices $B_{jk} = B[\beta_j, \gamma_k]$ of B . For instance, the matrix

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix} = \left[\begin{array}{cc|c|cc} 1 & -2 & 5 & 3 & -1 \\ \hline -3 & 0 & 1 & 6 & 1 \\ \hline 2 & 7 & 4 & 5 & -7 \end{array} \right]$$

$$\text{and the matrix } B = \begin{bmatrix} B_{11} \\ B_{21} \\ B_{31} \end{bmatrix} = \left[\begin{array}{cc} 4 & -1 \\ 3 & 9 \\ \hline 5 & 2 \\ \hline 2 & -8 \\ 7 & -1 \end{array} \right] \text{ have conformal partitions, so } A \text{ and } B \text{ can be}$$

multiplied block-wise to form the 3×2 matrix

$$\begin{aligned}
 AB &= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} \\ A_{21}B_{11} + A_{22}B_{21} + A_{23}B_{31} \end{bmatrix} \\
 &= \left[\begin{array}{cc} [1 \ -2] \begin{bmatrix} 4 & -1 \\ 3 & 9 \end{bmatrix} + 5[5 \ 2] + [3 \ -1] \begin{bmatrix} 2 & -8 \\ 7 & -1 \end{bmatrix} \\ \hline [-3 \ 0] \begin{bmatrix} 4 & -1 \\ 3 & 9 \end{bmatrix} + [1] \begin{bmatrix} 5 & 2 \end{bmatrix} + \begin{bmatrix} 6 & 1 \\ 5 & -7 \end{bmatrix} \begin{bmatrix} 2 & -8 \\ 7 & -1 \end{bmatrix} \end{array} \right] \\
 &= \left[\begin{array}{cc} [-2 \ -19] + [25 \ 10] + [-1 \ -23] \\ \hline [-12 \ 3] + [5 \ 2] + [-39 \ -33] \end{array} \right] = \begin{bmatrix} 22 & -32 \\ 12 & -44 \\ 10 & 36 \end{bmatrix}.
 \end{aligned}$$

8. Let $A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array} \right]$ and $B = \left[\begin{array}{c|c} 7 & 8 \\ 9 & 0 \\ \hline 1 & 2 \end{array} \right]$. Then A and B have conformal partitions. BA

is defined, but B and A do not have conformal partitions.

10.2 Block Diagonal and Block Triangular Matrices

Definitions:

A matrix $A = [a_{ij}] \in F^{n \times n}$ is **diagonal** if $a_{ij} = 0$, for all $i \neq j$, $1 \leq i, j \leq n$.

A diagonal matrix $A = [a_{ij}] \in F^{n \times n}$ is said to be **scalar** if $a_{ii} = a$, for all i , $1 \leq i \leq n$, and some scalar $a \in F$, i.e., $A = aI_n$.

A matrix $A \in F^{n \times n}$ is **block diagonal** if A as a block matrix is partitioned into submatrices $A_{ij} \in F^{n_i \times n_j}$, so that $A = [A_{ij}]$, $\sum_{i=1}^k n_i = n$, and $A_{ij} = \mathbf{0}$, for all $i \neq j$, $1 \leq i, j \leq k$. Thus, $A =$

$$\begin{bmatrix} A_{11} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & A_{22} & \cdots & \mathbf{0} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & A_{kk} \end{bmatrix}.$$
 This block diagonal matrix A is denoted $A = \text{diag}(A_{11}, \dots, A_{kk})$, where A_{ii} is $n_i \times n_i$, (or sometimes denoted $A = A_{11} \oplus \cdots \oplus A_{kk}$, and called the **direct sum** of $A_{11}, \dots, A_{kk}$).

A matrix $A = [a_{ij}] \in F^{n \times n}$ is **upper triangular** if $a_{ij} = 0$, for all $i > j$, $1 \leq i, j \leq n$.

An upper triangular matrix $A = [a_{ij}] \in F^{n \times n}$ is **strictly upper triangular** if $a_{ij} = 0$ for all $i \geq j$, $1 \leq i, j \leq n$.

A matrix $A \in F^{n \times n}$ is **lower triangular** if $a_{ij} = 0$ for all $i < j$, $1 \leq i, j \leq n$, i.e., if A^T is upper triangular.

A matrix $A \in F^{n \times n}$ is **strictly lower triangular** if A^T is strictly upper triangular.

A matrix is **triangular** if it is upper or lower triangular.

A matrix $A \in F^{n \times n}$ is **block upper triangular**, if A as a block matrix is partitioned into the submatrices $A_{ij} \in F^{n_i \times n_j}$, so that $A = [A_{ij}]$, $\sum_{i=1}^k n_i = n$, and $A_{ij} = \mathbf{0}$, for all $i > j$, $1 \leq i, j \leq k$, i.e., considering

the A_{ij} blocks as the entries of A , A is upper triangular. Thus, $A =$

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1k} \\ \mathbf{0} & A_{22} & \cdots & A_{2k} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & A_{kk} \end{bmatrix},$$

where each

A_{ij} is $n_i \times n_j$, and $\sum_{i=1}^k n_i = n$. The matrix A is **strictly block upper triangular** if $A_{ij} = \mathbf{0}$, for all $i \geq j$, $1 \leq i, j \leq k$.

A matrix $A \in F^{n \times n}$ is **block lower triangular** if A^T is block upper triangular.

A matrix $A \in F^{n \times n}$ is **strictly block lower triangular** if A^T is strictly block upper triangular.

A matrix $A = [a_{ij}] \in F^{n \times n}$ is **upper Hessenberg** if $a_{ij} = 0$, for all $i - 2 \geq j$, $1 \leq i, j \leq n$, i.e., A has

the form $A =$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n-1} & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n-1} & a_{2n} \\ 0 & a_{32} & a_{33} & \cdots & a_{3n-1} & a_{3n} \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ 0 & 0 & \cdots & a_{n-1n-2} & a_{n-1n-1} & a_{n-1n} \\ 0 & 0 & \cdots & 0 & a_{nn-1} & a_{nn} \end{bmatrix}.$$

A matrix $A = [a_{ij}] \in F^{n \times n}$ is **lower Hessenberg** if A^T is upper Hessenberg.

Facts:

The following facts can be found in [HJ85]. This information is also available in many other standard references such as [LT85] or [Mey00].

1. Let $D, D' \in F^{n \times n}$ be any diagonal matrices. Then $D + D'$ and DD' are diagonal, and $DD' = D'D$. If $D = \text{diag}(d_1, \dots, d_n)$ is nonsingular, then $D^{-1} = \text{diag}(1/d_1, \dots, 1/d_n)$.
2. Let $D \in F^{n \times n}$ be a matrix such that $DA = AD$ for all $A \in F^{n \times n}$. Then D is a scalar matrix.
3. If $A \in F^{n \times n}$ is a block diagonal matrix, so that $A = \text{diag}(A_{11}, \dots, A_{kk})$, then $\text{tr}(A) = \sum_{i=1}^k \text{tr}(A_{ii})$, $\det(A) = \prod_{i=1}^k \det(A_{ii})$, $\text{rank}(A) = \sum_{i=1}^k \text{rank}(A_{ii})$, and $\sigma(A) = \sigma(A_{11}) \cup \dots \cup \sigma(A_{kk})$.
4. Let $A \in F^{n \times n}$ be a block diagonal matrix, so that $A = \text{diag}(A_{11}, A_{22}, \dots, A_{kk})$. Then A is nonsingular if and only if A_{ii} is nonsingular for each i , $1 \leq i \leq k$. Moreover, $A^{-1} = \text{diag}(A_{11}^{-1}, A_{22}^{-1}, \dots, A_{kk}^{-1})$.
5. See Chapter 4.3 for information on diagonalizability of matrices.
6. Let $A \in F^{n \times n}$ be a block diagonal matrix, so that $A = \text{diag}(A_{11}, \dots, A_{kk})$. Then A is diagonalizable if and only if A_{ii} is diagonalizable for each i , $1 \leq i \leq k$.
7. If $A, B \in F^{n \times n}$ are upper (lower) triangular matrices, then $A + B$ and AB are upper (lower) triangular. If the upper (lower) triangular matrix $A = [a_{ij}]$ is nonsingular, then A^{-1} is upper (lower) triangular with diagonal entries $1/a_{11}, \dots, 1/a_{nn}$.

8. Let $A \in F^{n \times n}$ be block upper triangular, so that $A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1k} \\ \mathbf{0} & A_{22} & \cdots & A_{2k} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & A_{kk} \end{bmatrix}$. Then $\text{tr}(A) = \sum_{i=1}^k \text{tr}(A_{ii})$, $\det(A) = \prod_{i=1}^k \det(A_{ii})$, $\text{rank}(A) \geq \sum_{i=1}^k \text{rank}(A_{ii})$, and $\sigma(A) = \sigma(A_{11}) \cup \dots \cup \sigma(A_{kk})$.
9. Let $A = (A_{ij}) \in F^{n \times n}$ be a block triangular matrix (either upper or lower triangular). Then A is nonsingular if and only if A_{ii} is nonsingular for each i , $1 \leq i \leq k$. Moreover, the $n_i \times n_i$ diagonal block entries of A^{-1} are A_{ii}^{-1} , for each i , $1 \leq i \leq k$.
10. *Schur's Triangularization Theorem:* Let $A \in \mathbb{C}^{n \times n}$. Then there is a unitary matrix $U \in \mathbb{C}^{n \times n}$ so that U^*AU is upper triangular. The diagonal entries of U^*AU are the eigenvalues of A .
11. Let $A \in \mathbb{R}^{n \times n}$. Then there is an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ so that V^TAV is of upper Hessenberg

form $\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1k} \\ \mathbf{0} & A_{22} & \cdots & A_{2k} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & A_{kk} \end{bmatrix}$, where each A_{ii} , $1 \leq i \leq k$, is 1×1 or 2×2 . Moreover, when A_{ii} is

1×1 , the entry of A_{ii} is an eigenvalue of A , whereas when A_{ii} is 2×2 , then A_{ii} has two eigenvalues which are nonreal complex conjugates of each other, and are eigenvalues of A .

12. (For more information on unitary triangularization, see Chapter 7.2.)
13. Let $A = [A_{ij}] \in F^{n \times n}$ with $|\sigma(A)| = n$, where $\lambda_1, \dots, \lambda_k \in \sigma(A)$ are the distinct eigenvalues of A . Then there is a nonsingular matrix $T \in F^{n \times n}$ so that $T^{-1}AT = \text{diag}(A_{11}, \dots, A_{kk})$, where each $A_{ii} \in F^{n_i \times n_i}$ is upper triangular with all diagonal entries of A_{ii} equal to λ_i , for $1 \leq i \leq k$ (and $\sum_{i=1}^k n_i = n$).

14. Let $A \in F^{n \times n}$ be a block upper triangular matrix, of the form $A = \begin{bmatrix} A_{11} & A_{12} \\ \mathbf{0} & A_{22} \end{bmatrix}$, where A_{ij} is

$n_i \times n_j$, $1 \leq i, j \leq 2$, and $\sum_{i=1}^2 n_i = n$. (Note that any block upper triangular matrix can be said to have this form.) Let $\mathbf{x}$ be an eigenvector of A_{11} , with corresponding eigenvalue λ , so that $A_{11}\mathbf{x} = \lambda\mathbf{x}$, where $\mathbf{x}$ is a (column) vector with n_1 components. Then the (column) vector with n

components $\begin{bmatrix} \mathbf{x} \\ \mathbf{0} \end{bmatrix}$ is an eigenvector of A with eigenvalue λ . Let $\mathbf{y}$ be a left eigenvector of A_{22} , with

corresponding eigenvalue μ , so that $\mathbf{y}A_{22} = \mathbf{y}\mu$, where $\mathbf{y}$ is a row vector with n_2 components. Then the (row) vector with n components $[\mathbf{0} \quad \mathbf{y}]$ is a left eigenvector of A with eigenvalue μ .

Examples:

1. The matrix $A = \begin{bmatrix} 1 & 3 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 6 & 8 \\ 0 & 0 & 0 & 7 & 9 \end{bmatrix}$ is a block diagonal matrix, and the trace, determinant, and

rank can be calculated block-wise, where $A_{11} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $A_{22} = 5$, and $A_{33} = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$, as $\text{tr}(A) = 25 = \sum_{i=1}^3 \text{tr}(A_{ii})$, $\det(A) = (-2)(5)(-2) = \prod_{i=1}^3 \det(A_{ii})$, and $\text{rank}(A) = 5 = \sum_{i=1}^3 \text{rank}(A_{ii})$.

2. Let $A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix} \in F^{3 \times 3}$, an upper triangular matrix. If a, d, f are nonzero, then $A^{-1} = \begin{bmatrix} \frac{1}{a} & -\frac{b}{ad} & \frac{be-cd}{adf} \\ 0 & \frac{1}{d} & -\frac{e}{df} \\ 0 & 0 & \frac{1}{f} \end{bmatrix}$.

3. The matrix $B = \begin{bmatrix} 1 & 3 & 0 & 0 & 0 \\ 2 & 4 & 5 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 8 \end{bmatrix}$ is not block diagonal. However, B is block upper triangular,

with $B_{11} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $B_{22} = 0$, $B_{33} = \begin{bmatrix} 0 & 7 \\ 0 & 8 \end{bmatrix}$, $B_{12} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$, $B_{13} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, and $B_{23} = \begin{bmatrix} 6 & 0 \end{bmatrix}$.

Notice that $4 = \text{rank}(B) \geq \sum_{i=1}^3 \text{rank}(B_{ii}) = 2 + 0 + 1 = 3$.

4. The 4×4 matrix $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 5 & 0 \\ 6 & 7 & 8 & 9 \\ 10 & 11 & 12 & 13 \end{bmatrix}$ is not lower triangular, but is lower Hessenberg.

10.3 Schur Complements

In this subsection, the square matrix $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ is partitioned as a block matrix, where A_{11} is nonsingular.

Definitions:

The **Schur complement** of A_{11} in A is the matrix $A_{22} - A_{21}A_{11}^{-1}A_{12}$, sometimes denoted A/A_{11} .

Facts:

- [Zha99] $\begin{bmatrix} I & \mathbf{0} \\ -A_{21}A_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} I & -A_{11}^{-1}A_{12} \\ \mathbf{0} & I \end{bmatrix} = \begin{bmatrix} A_{11} & \mathbf{0} \\ \mathbf{0} & A/A_{11} \end{bmatrix}$.
- [Zha99] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where A_{11} is nonsingular. Then $\det(A) = \det(A_{11})\det(A/A_{11})$.

Also, $\text{rank}(A) = \text{rank}(A_{11}) + \text{rank}(A/A_{11})$.

3. [Zha99] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$. Then A is nonsingular if and only if both A_{11} and the Schur complement of A_{11} in A are nonsingular.
4. [HJ85] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where A_{11} , A_{22} , A/A_{11} , A/A_{22} , and A are nonsingular. Then $A^{-1} = \begin{bmatrix} (A/A_{22})^{-1} & -A_{11}^{-1}A_{12}(A/A_{11})^{-1} \\ -A_{22}^{-1}A_{21}(A/A_{22})^{-1} & (A/A_{11})^{-1} \end{bmatrix}$.
5. [Zha99] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where A_{11} , A_{22} , A/A_{11} , and A/A_{22} are nonsingular. An equation relating the Schur complements of A_{11} in A and A_{22} in A is $(A/A_{22})^{-1} = A_{11}^{-1} + A_{11}^{-1}A_{12}(A/A_{11})^{-1}A_{21}A_{11}^{-1}$.
6. [LT85] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where the $k \times k$ matrix A_{11} is nonsingular. Then $\text{rank}(A) = k$ if and only if $A/A_{11} = \mathbf{0}$.
7. [Hay68] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{bmatrix}$ be Hermitian, where A_{11} is nonsingular. Then the inertia of A is $\text{in}(A) = \text{in}(A_{11}) + \text{in}(A/A_{11})$.
8. [Hay68] Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{bmatrix}$ be Hermitian, where A_{11} is nonsingular. Then A is positive (semi)definite if and only if both A_{11} and A/A_{11} are positive (semi)definite.

Examples:

1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$. Then with $A_{11} = 1$, we have

$$\begin{aligned} \begin{bmatrix} 1 & \mathbf{0} \\ -\begin{bmatrix} 4 \\ 7 \end{bmatrix} & I_2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix} \begin{bmatrix} 1 & -[2 \ 3] \\ \mathbf{0} & I_2 \end{bmatrix} &= \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \begin{bmatrix} 5 & 6 \\ 8 & 10 \end{bmatrix} \end{bmatrix} - \begin{bmatrix} 4 \\ 7 \end{bmatrix} 1^{-1} [2 \ 3] \\ &= \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \begin{bmatrix} -3 & -6 \\ -6 & -11 \end{bmatrix} \end{bmatrix}. \end{aligned}$$

2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$. With $A_{11} = 1$, and $A_{22} = \begin{bmatrix} 5 & 6 \\ 8 & 10 \end{bmatrix}$, then $A/A_{11} = \begin{bmatrix} -3 & -6 \\ -6 & -11 \end{bmatrix}$, $A/A_{22} =$

$$1 - [2 \ 3] \begin{bmatrix} 5 & 6 \\ 8 & 10 \end{bmatrix}^{-1} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = -\frac{3}{2}, \text{ and}$$

$$\begin{aligned} A^{-1} &= \begin{bmatrix} (-\frac{3}{2})^{-1} & -[2 \ 3] \begin{bmatrix} -3 & -6 \\ -6 & -11 \end{bmatrix}^{-1} \\ -\begin{bmatrix} 5 & 6 \\ 8 & 10 \end{bmatrix}^{-1} \begin{bmatrix} 4 \\ 7 \end{bmatrix} (-\frac{3}{2})^{-1} & \begin{bmatrix} -3 & -6 \\ -6 & -11 \end{bmatrix}^{-1} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{2}{3} & \frac{1}{3}[-4 \ 3] \\ \begin{bmatrix} -\frac{2}{3} \\ 1 \end{bmatrix} & -\frac{1}{3} \begin{bmatrix} -11 & 6 \\ 6 & -3 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} & -\frac{4}{3} & 1 \\ -\frac{2}{3} & \frac{11}{3} & -2 \\ 1 & -2 & 1 \end{bmatrix}. \end{aligned}$$

3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$.

Then, from Fact 5,

$$\begin{aligned} (A/A_{22})^{-1} &= \left(-\frac{3}{2}\right)^{-1} = 1^{-1} + 1^{-1} \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -3 & -6 \\ -6 & -11 \end{bmatrix}^{-1} \begin{bmatrix} 4 \\ 7 \end{bmatrix} 1^{-1}, \\ &= A_{11}^{-1} + A_{11}^{-1} A_{12} (A/A_{11})^{-1} A_{21} A_{11}^{-1}. \end{aligned}$$

10.4 Kronecker Products

Definitions:

Let $A \in F^{m \times n}$ and $B \in F^{p \times q}$. Then the **Kronecker product** (sometimes called the **tensor product**) of A

and B , denoted $A \otimes B$, is the $mp \times nq$ partitioned matrix $A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{bmatrix}$.

Let $A \in F^{m \times n}$ and let the j th column of A , namely, $A[\{1, \dots, m\}, \{j\}]$ be denoted a_j , for $1 \leq j \leq n$.

The column vector with mn components, denoted $\text{vec}(A)$, defined by $\text{vec}(A) = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in F^{mn}$, is the

vec-function of A , i.e., $\text{vec}(A)$ is formed by stacking the columns of A on top of each other in their natural order.

Facts:

All of the following facts except those for which a specific reference is given can be found in [LT85].

1. Let $A \in F^{m \times n}$ and $B \in F^{p \times q}$. If $a \in F$, then $a(A \otimes B) = (aA) \otimes B = A \otimes (aB)$.
2. Let $A, B \in F^{m \times n}$ and $C \in F^{p \times q}$. Then $(A + B) \otimes C = A \otimes C + B \otimes C$.
3. Let $A \in F^{m \times n}$ and $B, C \in F^{p \times q}$. Then $A \otimes (B + C) = A \otimes B + A \otimes C$.
4. Let $A \in F^{m \times n}$, $B \in F^{p \times q}$, and $C \in F^{r \times s}$. Then $A \otimes (B \otimes C) = (A \otimes B) \otimes C$.
5. Let $A \in F^{m \times n}$ and $B \in F^{p \times q}$. Then $(A \otimes B)^T = A^T \otimes B^T$.
6. [MM64] Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$. Then $\overline{(A \otimes B)} = \overline{A} \otimes \overline{B}$.
7. [MM64] Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$. Then $(A \otimes B)^* = A^* \otimes B^*$.
8. Let $A \in F^{m \times n}$, $B \in F^{p \times q}$, $C \in F^{n \times r}$, and $D \in F^{q \times s}$. Then $(A \otimes B)(C \otimes D) = AC \otimes BD$.
9. Let $A \in F^{m \times n}$ and $B \in F^{p \times q}$. Then $A \otimes B = (A \otimes I_p)(I_n \otimes B) = (I_m \otimes B)(A \otimes I_q)$.
10. If $A \in F^{m \times m}$ and $B \in F^{n \times n}$ are nonsingular, then $A \otimes B$ is nonsingular and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.
11. Let $A_1, A_2, \dots, A_k \in F^{m \times m}$, and $B_1, B_2, \dots, B_k \in F^{n \times n}$. Then $(A_1 \otimes B_1)(A_2 \otimes B_2) \cdots (A_k \otimes B_k) = (A_1 A_2 \cdots A_k) \otimes (B_1 B_2 \cdots B_k)$.
12. Let $A \in F^{m \times m}$ and $B \in F^{n \times n}$. Then $(A \otimes B)^k = A^k \otimes B^k$.

13. If $A \in F^{m \times m}$ and $B \in F^{n \times n}$, then there is an $mn \times mn$ permutation matrix P so that $P^T(A \otimes B)P = B \otimes A$.
14. Let $A, B \in F^{m \times n}$. Then $\text{vec}(aA + bB) = a \text{vec}(A) + b \text{vec}(B)$, for any $a, b \in F$.
15. If $A \in F^{m \times n}$, $B \in F^{p \times q}$, and $X \in F^{n \times p}$, then $\text{vec}(AXB) = (B^T \otimes A)\text{vec}(X)$.
16. If $A \in F^{m \times m}$ and $B \in F^{n \times n}$, then $\det(A \otimes B) = (\det(A))^n (\det(B))^m$, $\text{tr}(A \otimes B) = (\text{tr}(A))(\text{tr}(B))$, and $\text{rank}(A \otimes B) = (\text{rank}(A))(\text{rank}(B))$.
17. Let $A \in F^{m \times m}$ and $B \in F^{n \times n}$, with $\sigma(A) = \{\lambda_1, \dots, \lambda_m\}$ and $\sigma(B) = \{\mu_1, \dots, \mu_n\}$. Then $A \otimes B \in F^{mn \times mn}$ has eigenvalues $\{\lambda_s \mu_t | 1 \leq s \leq m, 1 \leq t \leq n\}$. Moreover, if the right eigenvectors of A are denoted $\mathbf{x}_i$, and the right eigenvectors of B are denoted $\mathbf{y}_j$, so that $A\mathbf{x}_i = \lambda_i \mathbf{x}_i$ and $B\mathbf{y}_j = \mu_j \mathbf{y}_j$, then $(A \otimes B)(\mathbf{x}_i \otimes \mathbf{y}_j) = \lambda_i \mu_j (\mathbf{x}_i \otimes \mathbf{y}_j)$.
18. Let $A \in F^{m \times m}$ and $B \in F^{n \times n}$, with $\sigma(A) = \{\lambda_1, \dots, \lambda_m\}$ and $\sigma(B) = \{\mu_1, \dots, \mu_n\}$. Then $(I_n \otimes A) + (B \otimes I_m)$ has eigenvalues $\{\lambda_s + \mu_t | 1 \leq s \leq m, 1 \leq t \leq n\}$.
19. Let $p(x, y) \in F[x, y]$ so that $p(x, y) = \sum_{i,j=1}^k a_{ij} x^i y^j$, where $a_{ij} \in F$, $1 \leq i \leq k$, $1 \leq j \leq k$. Let $A \in F^{m \times m}$ and $B \in F^{n \times n}$. Define $p(A; B)$ to be the $mn \times mn$ matrix $p(A; B) = \sum_{i,j=1}^k a_{ij}(A^i \otimes B^j)$. If $\sigma(A) = \{\lambda_1, \dots, \lambda_m\}$ and $\sigma(B) = \{\mu_1, \dots, \mu_n\}$, then $\sigma(p(A; B)) = \{p(\lambda_s, \mu_t) | 1 \leq s \leq m, 1 \leq t \leq n\}$.
20. Let $A_1, A_2 \in F^{m \times m}$, $B_1, B_2 \in F^{n \times n}$. If A_1 and A_2 are similar, and B_1 and B_2 are similar, then $A_1 \otimes B_1$ is similar to $A_2 \otimes B_2$.
21. If $A \in F^{m \times n}$, $B \in F^{p \times q}$, and $X \in F^{n \times p}$, then $\text{vec}(AX) = (I_p \otimes A)\text{vec}(X)$, $\text{vec}(XB) = (B^T \otimes I_n)\text{vec}(X)$, and $\text{vec}(AX + XB) = [(I_p \otimes A) + (B^T \otimes I_n)]\text{vec}(X)$.
22. If $A \in F^{m \times n}$, $B \in F^{p \times q}$, $C \in F^{m \times q}$, and $X \in F^{n \times p}$, then the equation $AXB = C$ can be written in the form $(B^T \otimes A)\text{vec}(X) = \text{vec}(C)$.
23. Let $A \in F^{m \times m}$, $B \in F^{n \times n}$, $C \in F^{m \times n}$, and $X \in F^{m \times n}$. Then the equation $AX + XB = C$ can be written in the form $[(I_n \otimes A) + (B^T \otimes I_m)]\text{vec}(X) = \text{vec}(C)$.
24. Let $A \in \mathbb{C}^{m \times m}$ and $B \in \mathbb{C}^{n \times n}$ be Hermitian. Then $A \otimes B$ is Hermitian.
25. Let $A \in \mathbb{C}^{m \times m}$ and $B \in \mathbb{C}^{n \times n}$ be positive definite. Then $A \otimes B$ is positive definite.

Examples:

1. Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Then $A \otimes B = \begin{bmatrix} 1 & 2 & 3 & -1 & -2 & -3 \\ 4 & 5 & 6 & -4 & -5 & -6 \\ 7 & 8 & 9 & -7 & -8 & -9 \\ 0 & 0 & 0 & 2 & 4 & 6 \\ 0 & 0 & 0 & 8 & 10 & 12 \\ 0 & 0 & 0 & 14 & 16 & 18 \end{bmatrix}$.
2. Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$. Then $\text{vec}(A) = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}$.
3. Let $A \in F^{m \times m}$ and $B \in F^{n \times n}$. If A is upper (lower) triangular, then $A \otimes B$ is block upper (lower) triangular. If A is diagonal then $A \otimes B$ is block diagonal. If both A and B are upper (lower) triangular, then $A \otimes B$ is (upper) triangular. If both A and B are diagonal, then $A \otimes B$ is diagonal.
4. Let $A \in F^{m \times n}$ and $B \in F^{p \times q}$. If $A \otimes B = \mathbf{0}$, then $A = \mathbf{0}$ or $B = \mathbf{0}$.
5. Let $A \in F^{m \times n}$. Then $A \otimes I_n = \begin{bmatrix} a_{11}I_n & a_{12}I_n & \cdots & a_{1n}I_n \\ a_{21}I_n & a_{22}I_n & \cdots & a_{2n}I_n \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}I_n & a_{m2}I_n & \cdots & a_{mn}I_n \end{bmatrix} \in F^{mn \times n^2}$. Let $B \in F^{p \times p}$. Then $I_n \otimes B = \text{diag}(B, B, \dots, B) \in F^{np \times np}$, and $I_m \otimes I_n = I_{mn}$.

References

- [Hay68] E. Haynsworth, Determination of the Inertia of a Partitioned Matrix, *Lin. Alg. Appl.*, 1:73–81 (1968).
- [HJ85] R. A. Horn and C. R. Johnson, *Matrix Analysis*, Cambridge University Press, Combridge, 1985.
- [LT85] P. Lancaster and M. Tismenetsky, *The Theory of Matrices, with Applications*, 2nd ed., Academic Press, San Diego, 1985.
- [MM64] M. Marcus and H. Minc, *A Survey of Matrix Theory and Matrix Inequalities*, Prindle, Weber, & Schmidt, Boston, 1964.
- [Mey00] C. Meyer, *Matrix Analysis and Applied Linear Algebra*, SIAM, 2000.
- [Zha99] F. Zhang, *Matrix Theory*, Springer-Verlag, New York, 1999.

Advanced Linear Algebra

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Functions of Matrices

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Matrix functions are used in many areas of linear algebra and arise in numerous applications in science and engineering. The most common matrix function is the matrix inverse; it is not treated specifically in this chapter, but is covered in Section 1.1 and Section 38.2. This chapter is concerned with general matrix functions as well as specific cases such as matrix square roots, trigonometric functions, and the exponential and logarithmic functions.

The specific functions just mentioned can all be defined via power series or as the solution of nonlinear systems. For example, $\cos(A) = I - A^2/2! + A^4/4! - \dots$. However, a general theory exists from which a number of properties possessed by all matrix functions can be deduced and which suggests computational methods. This chapter treats general theory, then specific functions, and finally outlines computational methods.

11.1 General Theory

Definitions:

A function of a matrix can be defined in several ways, of which the following three are the most generally useful.

- *Jordan canonical form definition.* Let $A \in \mathbb{C}^{n \times n}$ have the Jordan canonical form $Z^{-1}AZ = J_A = \text{diag}(J_1(\lambda_1), J_2(\lambda_2), \dots, J_p(\lambda_p))$, where Z is nonsingular,

$$J_k(\lambda_k) = \begin{bmatrix} \lambda_k & 1 & & \\ & \lambda_k & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_k \end{bmatrix} \in \mathbb{C}^{m_k \times m_k}, \quad (11.1)$$